

A Hardy-Gauge Contour Method for Density One of Zeta Zeros on the Critical Line

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Abstract

We construct a finite-dimensional Hermitian contour form from an analytic Hardy gauge and a near-lattice packet family. Its zero-side residue representation converts off-critical-line stationary geometry into negative inertia. On the arithmetic side, the same form admits an exact right-edge decomposition whose principal term is the pullback of a completed lag operator through the common packet/carrier analysis map. A direct Schur estimate gives a uniform spectral lower bound for this lag operator on the mean-zero fibres. The stationary edge contribution has rank $o(N_T)$, while the regular remainder is $o(N_T)$ in the relative Hilbert–Schmidt metric. A finite-dimensional inertia argument then implies that all but $o(N_T)$ zeros in a dyadic interval lie on the critical line, and hence that $N_0(T)/N(T) \rightarrow 1$.

1 Main theorem and notation

Let

$$N_T := N(T, 2T)$$

denote the number, with multiplicity, of nontrivial zeros of ζ with imaginary part in $[T, 2T]$, and let $N_{0,T}$ denote the number of those zeros on the critical line. Thus $N_T \asymp T \log T$. Put

$$L := \log \frac{T}{2\pi}.$$

Theorem 1.1 (Density one on the critical line). *As $T \rightarrow \infty$,*

$$\boxed{\frac{N_{0,T}}{N_T} = 1 - o(1)}.$$

Consequently, by dyadic summation,

$$\boxed{\frac{N_0(T)}{N(T)} \rightarrow 1}.$$

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Hardy's theorem established infinitely many zeros on the critical line, and Selberg proved that a positive proportion lie there [5, 58, 9]. The modern quantitative line begins with Levinson's mollifier method and its later refinements [8, 60, 3, 2, 4, 55, 56]. Closely related work on simple and distinct zeros, pair correlation, and finite-dimensional positivity includes [42, 10, 51, 31, 18, 22, 11, 13, 40, 41, 1]. In particular, Alpöge and Furman obtained an unconditional proportion exceeding two thirds for zeros that are both simple and on the critical line by combining a finite compression of a Hermitian zero form with inertia.

A second strand concerns the size and statistics of ζ , its derivatives, and the Hardy Z -function. Mean-value and discrete-moment methods were developed through the work of Gonek, Conrey–Ghosh, Conrey–Ghosh–Gonek, Ng, Garunkštis–Steuding, Hughes, Milinovich, and later authors [39, 28, 29, 24, 30, 38, 52, 43, 49, 50, 44, 45, 54, 7]. These results are especially close to the present zero-side use of stationary points of the Hardy gauge and to the right-edge expansion through Z'_1/Z_1 .

The arithmetic side also intersects the extensive literature on twisted moments, Dirichlet-polynomial mean values, mollifiers, and ratios of L -functions [12, 36, 35, 14, 15, 16, 25, 26, 27, 34, 32, 33, 59, 6]. Adjacent work on zero gaps, derivative zeros, and critical-line geometry includes [23, 17, 20, 19, 37, 21, 46, 47, 48, 53, 57]. The argument below uses a different assembly: the same Hardy-gauge contour matrix is represented on the zero side and on the packetized right edge, its principal lag operator is given a uniform spectral floor, and the remaining terms are separated into low-rank and relative Hilbert–Schmidt classes.

The proof is organized around the normalized contour matrix introduced in Section 2, whose zero-side and arithmetic-side representations are developed in the subsequent sections.

2 The invariant Hardy-gauge contour matrix

2.1 The analytic Hardy gauge

Fix once and for all

$$c := 2.$$

For horizontal heights $T_1 < T_2$ avoiding the zeros relevant to the contour, let

$$\mathcal{R}_T^{\text{sym}} := \{\sigma + it : -1 \leq \sigma \leq 2, T_1 \leq t \leq T_2\}. \quad (1)$$

The existence of admissible $T_1 \in [T, T+1]$ and $T_2 \in [2T, 2T+1]$ is proved in Proposition 3.10. On a simply connected neighbourhood of this symmetric high rectangle choose a holomorphic branch

$$q(s) := \chi(1-s)^{1/2}.$$

Fix its sign at one point of the critical line so that the resulting Hardy gauge is real there, and put

$$\mathcal{Z}(s) := q(s)\zeta(s).$$

This normalization determines the branch on the rectangle. Indeed the holomorphic function

$$q^\dagger(s) := \overline{q(1-\bar{s})}$$

satisfies $(q^\dagger)^2 = \chi(s) = q(s)^{-2}$; the chosen sign gives $q^\dagger = q^{-1}$ at the base point and hence everywhere. Using the zeta functional equation therefore yields the exact reflection law

$$\boxed{\mathcal{Z}(1-\bar{s}) = \overline{\mathcal{Z}(s)}}. \quad (2)$$

In particular $F(z) = \mathcal{Z}(1/2 + iz)$ satisfies $F(\bar{z}) = \overline{F(z)}$ and is real on the real axis. The factor q is analytic and nonvanishing, so $\mathcal{Z}$ and ζ have exactly the same zeros with the same multiplicities. Define

$$f(s) := -\frac{1}{2} \frac{\chi'}{\chi}(s), \quad Z_1(s) := \zeta'(s) + f(s)\zeta(s).$$

Since $q'/q = f$, differentiation gives the exact identity

$$\boxed{\mathcal{Z}'(s) = q(s)Z_1(s).} \quad (3)$$

Thus the stationary zeros of the Hardy-gauge detector are precisely the zeros of the same Z_1 which appears on the arithmetic side.

2.2 Curvature representation of the contour form

Let a be the physical shift and put

$$p := La.$$

The curvature source is generated before any right-edge expansion from one shifted Hardy product. Algebraically,

$$\mathcal{Z}(s)\mathcal{Z}''(s) - \mathcal{Z}'(s)^2 = \frac{1}{2} \partial_a^2 (\mathcal{Z}(s+a)\mathcal{Z}(s-a)) \Big|_{a=0}. \quad (4)$$

Passing from a to p introduces the positive scalar factor L^2 . We divide the raw matrix by this factor once and for all. Since the rescaling is positive, it does not change inertia.

Proposition 2.1 (Exact normalized same-form identity). *Let*

$$C_{\mathcal{Z}}(s) := \mathcal{Z}(s)\mathcal{Z}''(s) - \mathcal{Z}'(s)^2, \quad C_{\mathcal{Z}}^{(p)}(s) := L^{-2}C_{\mathcal{Z}}(s).$$

Equivalently,

$$C_{\mathcal{Z}}^{(p)}(s) = \frac{1}{2} \partial_p^2 (\mathcal{Z}(s+p/L)\mathcal{Z}(s-p/L)) \Big|_{p=0}.$$

Let $\{\psi_\nu\}_{\nu \in \mathcal{I}_T}$ be the smooth entire packet family specified below and write

$$B_v(z) := \sum_{\nu \in \mathcal{I}_T} v_\nu \psi_\nu(z), \quad B_v^\#(z) := \overline{B_v(\bar{z})}.$$

With $z(s) := -i(s - \frac{1}{2})$ define the holomorphic polarization

$$\boxed{\mathcal{B}_{v,w}(s) := B_w(z(s))B_v^\#(z(s)).} \quad (5)$$

Then the normalized invariant contour form is exactly

$$\boxed{\mathcal{A}_T(v, w) = \frac{1}{2\pi i} \oint_{\partial \mathcal{R}_T^{\text{sym}}} \frac{\mathcal{Z}''}{\mathcal{Z}'}(s) C_{\mathcal{Z}}^{(p)}(s) \mathcal{B}_{v,w}(s) ds = \frac{1}{2\pi i} \oint_{\partial \mathcal{R}_T^{\text{sym}}} \frac{Z_1'}{Z_1}(s) C_{\mathcal{Z}}^{(p)}(s) \mathcal{B}_{v,w}(s) ds.} \quad (6)$$

After straightening the critical line by $s = 1/2 + iz$, the same form is exactly (9). Thus the zero-side stationary residue detector and the arithmetic Z_1'/Z_1 source are literally the same Hermitian contour matrix.

Proof. Since (3) gives

$$\frac{\mathcal{Z}''}{\mathcal{Z}'} = f + \frac{Z_1'}{Z_1},$$

and f , $C_{\mathcal{Z}}^{(p)}$ and $\mathcal{B}_{v,w}$ are holomorphic on the high rectangle, the integral of $f C_{\mathcal{Z}}^{(p)} \mathcal{B}_{v,w}$ around the closed contour is zero. This proves the second equality in (6).

For the first equality, expand

$$\frac{\mathcal{Z}''}{\mathcal{Z}'} C_{\mathcal{Z}}^{(p)} = L^{-2} \left(\frac{\mathcal{Z} \mathcal{Z}''^2}{\mathcal{Z}'} - \mathcal{Z}' \mathcal{Z}'' \right).$$

The second term, after multiplication by the holomorphic packet polarization, is holomorphic and hence has zero closed-contour integral. Put $F(z) = \mathcal{Z}(1/2 + iz)$. Then

$$\mathcal{Z}' = -iF', \quad \mathcal{Z}'' = -F'', \quad ds = i dz,$$

and therefore the remaining term transforms to

$$\frac{1}{2\pi i} \oint L^{-2} \left(-F(z) \frac{F''(z)^2}{F'(z)} \right) B_w(z) B_v^\#(z) dz,$$

which is exactly (9). Finally, (4) shows that this same form is generated from one shifted Hardy product before any contour movement or stationary reduction. $\square$

With the packet notation above, the normalized Hermitian contour form is

$$\mathcal{A}_T(v, w) := \frac{1}{2\pi i} \int_{\partial \mathcal{R}_T^{\text{sym}}} \mathcal{K}(s; B_v, \overline{B_w}) ds, \quad (7)$$

where $\mathcal{K}$ is the exact Hardy-gauge curvature integrand obtained from (4), before Dirichlet expansion, stationary phase, Perron inversion, shell analysis, or arithmetic contraction. Define the finite matrix A_T by

$$\langle A_T v, w \rangle = \mathcal{A}_T(v, w).$$

The reflection law (2) together with (5) gives

$$\mathcal{A}_T(w, v) = \overline{\mathcal{A}_T(v, w)},$$

so A_T is Hermitian. It is the invariant Hermitian object throughout the argument. The positive form $\mathfrak{G}_T$ introduced later is not a second invariant object; it is the fixed reference metric used for relative operator estimates.

Lemma 2.2 (Invariance under standard reductions). *Every later change of representation used below belongs to one of the following four classes.*

- (i) *An exact congruence $A \mapsto U^* A U$ with U invertible; inertia is preserved exactly.*
- (ii) *Restriction to a subspace of codimension d ; at most d negative directions are lost.*
- (iii) *Addition of a matrix of rank r ; negative inertia changes by at most r .*
- (iv) *Addition of a Hermitian remainder R relative to a positive reference Gram G , measured by*

$$\|G^{-1/2} R G^{-1/2}\|_{S_2}^2.$$

Proof. The first assertion is Sylvester's law of inertia. The second is the elementary dimension inequality for the intersection of a negative subspace with a codimension- d subspace. The third follows by applying the same argument to $\ker E$, whose codimension is at most r . The fourth is quantified in Lemma 14.1 below. $\square$

3 Zero side: stationary residues force negative inertia

Straighten the critical line by

$$F(z) := \mathcal{Z}\left(\frac{1}{2} + iz\right).$$

The symmetric rectangle (1) becomes

$$\Omega_T := \{z : T_1 \leq \Re z \leq T_2, |\Im z| \leq c - \tfrac{1}{2}\}. \quad (8)$$

Thus Ω_T is invariant under $z \mapsto \bar{z}$ and contains both members of every reflected off-critical pair in the height range. Moreover $F(\bar{z}) = \overline{F(z)}$ and F is real on $\mathbb{R}$. For a test function h put $h^\#(z) = \overline{h(\bar{z})}$. In zero coordinates the same normalized contour matrix has kernel

$$\mathcal{A}_F(h_1, h_2) = \frac{1}{2\pi i} \int_{\partial\Omega_T} L^{-2} \left(-F(z) \frac{F''(z)^2}{F'(z)} \right) h_2(z) h_1^\#(z) dz. \quad (9)$$

We write $A_F(T) := A_T$ for this original matrix. If H is a sufficiently small real-symmetric perturbation of F , $A_H(T)$ denotes the same contour construction with F replaced by H . The residue lemmas below are stated for such a generic H and are then applied to the perturbation constructed in Section 15.

Lemma 3.1 (Stationary residue blocks). *If $H'(c) = 0$ and $H''(c) \neq 0$, then*

$$\text{Res}_{z=c} L^{-2} \left(-H(z) \frac{H''(z)^2}{H'(z)} \right) = -L^{-2} H(c) H''(c) =: w_c.$$

For $c \in \mathbb{R}$, the contribution is a negative scalar exactly when $H(c)H''(c) > 0$. For a nonreal conjugate pair $\rho, \bar{\rho}$, the residue block is

$$\begin{pmatrix} 0 & \overline{w_\rho} \\ w_\rho & 0 \end{pmatrix},$$

which has exactly one negative eigenvalue whenever $w_\rho \neq 0$.

Proof. Since

$$H'(z) = H''(c)(z - c) + O((z - c)^2),$$

the residue is $-L^{-2}H(c)H''(c)$. The positive scalar L^{-2} does not change its sign. The conjugate-pair block has characteristic polynomial $\lambda^2 - |w_\rho|^2$. $\square$

Let $B(H)$ denote the number of real stationary points with $H(c)H''(c) > 0$, and let $P(H')$ denote the number of nonreal conjugate pairs of stationary zeros.

3.1 Exact packet evaluation rank

Let $C_{\text{pkt}} > 0$ be a fixed constant, chosen first in the global parameter hierarchy of Section 13, before the remaining buffer and symbol exponents. Put

$$\boxed{L_{\text{pkt}} := L + C_{\text{pkt}} \log L.} \quad (10)$$

The central width- L band is the physical zeta band; the additional $C_{\text{pkt}} \log L$ is a two-sided reserve used for packet-dimension slack and the smooth transition caps. Choose a real C^∞ function ϕ_L

which equals $L_{\text{pkt}}^{-1/2}$ on $[-L_{\text{pkt}}/2, L_{\text{pkt}}/2]$, satisfies $|\phi_L| \leq L_{\text{pkt}}^{-1/2}$, and is supported in $[-L_{\text{pkt}}/2 - \Delta, L_{\text{pkt}}/2 + \Delta]$ for one fixed $\Delta > 0$. Put

$$\psi_L(z) := \int_{\mathbb{R}} \phi_L(u) e^{iuz} du, \quad \psi_{L,\tau}(z) := \psi_L(z - \tau).$$

The reserve supplies more than the $O(T)$ dimensional slack required by the Riemann–von Mangoldt secondary term and leaves fixed-width room for the smooth transition caps. By Proposition 6.1, no arithmetic outer translation has to be accommodated. Since $L_{\text{pkt}} = L + O(\log L)$, all normalized analytic scales below change only by $O(\log L/L)$, and on a fixed horizontal strip the larger exponential type costs only a fixed power of L . Repeated integration by parts gives, on every fixed horizontal strip and for every fixed M ,

$$|\psi_L(z)| \ll_M L_{\text{pkt}}^{1/2} (1 + |\Re z|)^{-M} e^{(L_{\text{pkt}}/2 + \Delta)|\Im z|}. \quad (11)$$

Lemma 3.2 (Local packet centres give surjective evaluation). *Let $z_1, \dots, z_m$ be distinct complex stationary nodes and let $I_1, \dots, I_m$ be arbitrary nonempty open intervals of admissible packet centres. There exist $\tau_j \in I_j$ such that*

$$\det(\psi_L(z_i - \tau_j))_{1 \leq i, j \leq m} \neq 0.$$

Hence the packet evaluation map onto the stationary nodes is surjective.

Proof. First fix distinct z_i . If

$$\sum_i c_i \psi_L(z_i - \tau) = 0$$

on a nonempty real interval in τ , analyticity makes the identity global. Fourier uniqueness then gives

$$\phi_L(u) \sum_i c_i e^{iuz_i} = 0$$

for almost every u . Since ϕ_L is nonzero on an open interval, the exponential polynomial $\sum_i c_i e^{iuz_i}$ vanishes identically. Its first m derivatives at one point form a Vandermonde system, so all $c_i = 0$. Thus the functions $\tau \mapsto \psi_L(z_i - \tau)$ are linearly independent on every interval. The determinant above is real analytic in $(\tau_1, \dots, \tau_m)$ and is not identically zero; a nonzero real-analytic function cannot vanish on the open box $I_1 \times \dots \times I_m$. $\square$

Lemma 3.3 (Near-lattice packet Riesz bounds). *For lattice centres*

$$\tau_j^0 = \tau_* + \frac{2\pi n_j}{L_{\text{pkt}}}$$

one has, for all large L ,

$$\boxed{2\pi \sum_j |c_j|^2 \leq \left\| \sum_j c_j \psi_{L, \tau_j^0} \right\|_2^2 \leq 6\pi \sum_j |c_j|^2.} \quad (12)$$

The same inequalities, with slightly weaker fixed constants, hold when every centre is moved by at most T^{-100} .

Proof. Plancherel gives

$$\left\| \sum_j c_j \psi_{L, \tau_j^0} \right\|_2^2 = 2\pi \int |\phi_L(u)|^2 \left| \sum_j c_j e^{-iu\tau_j^0} \right|^2 du.$$

On the flat interval $[-L_{\text{pkt}}/2, L_{\text{pkt}}/2]$, the exponentials $e^{-2\pi i n_j u / L_{\text{pkt}}}$ are orthogonal; since $|\phi_L|^2 = L_{\text{pkt}}^{-1}$ there, this contributes exactly $2\pi \sum_j |c_j|^2$ and proves the lower bound. The full support has length $L_{\text{pkt}} + 2\Delta < 2L_{\text{pkt}}$ for large T . By splitting it into at most three intervals of length at most L_{pkt} and using the same period orthogonality, together with $|\phi_L|^2 \leq L_{\text{pkt}}^{-1}$, its total contribution is at most $6\pi \sum_j |c_j|^2$.

For perturbed centres, the corresponding Gram entries depend continuously on the centres. On the support $|u| \ll L_{\text{pkt}} \asymp L$ a displacement T^{-100} changes one phase by $O(LT^{-100})$. Since the packet dimension is $O(TL)$, the crude matrix norm perturbation is $O(T^{-99}L^2) = o(1)$, and the fixed Riesz bounds persist. $\square$

Lemma 3.4 (Packet dimension has logarithmic reserve). *For the choice (10), the admissible near-lattice packet family has*

$$d_T = \frac{T}{2\pi}L + \frac{C_{\text{pkt}}T}{2\pi} \log L + O(L). \quad (13)$$

and therefore contains, for all large T , at least as many columns as there are zeros of F' in the retained dyadic rectangle even after the $O(T)$ terminal packet collars are removed. The reserve itself is only $O(T \log L) = o(N_T)$.

Proof. The Riemann–von Mangoldt formula gives

$$N(T, 2T) = \frac{T}{2\pi}L + \frac{(2\log 2 - 1)T}{2\pi} + O(\log T).$$

By Proposition 3.10 and $F' = iqZ_1$ up to a nonzero constant factor,

$$N(F'; T_1, T_2) = \frac{T}{2\pi}L + \frac{(2\log 2 - 1)T}{2\pi} + O((\log T)^C).$$

A lattice of spacing $2\pi/L_{\text{pkt}}$ on a physical interval of length $T + O(1)$ has

$$d_T = \frac{T}{2\pi}(L + C_{\text{pkt}} \log L) + O(L)$$

columns, proving (13). The terminal packet collars remove only $C_{\text{del}}T + o(T)$ columns for a fixed C_{del} . Since $C_{\text{pkt}} \log L \rightarrow \infty$, the reserve term eventually dominates both the linear Riemann–von Mangoldt secondary term and the collar loss for every fixed $C_{\text{pkt}} > 0$. No further lower bound on C_{pkt} is required. All later frame, mean, shell, and edge restrictions are applied after the zero-side evaluation map has been constructed and do not enter this surjectivity count. $\square$

By Lemmas 3.2, 3.3, and 3.4, the packet centres may therefore be chosen inside T^{-100} neighbourhoods of the lattice so that the evaluation map is surjective while the packet synthesis retains a uniform Riesz bound.

Lemma 3.5 (Coordinate restriction preserves the common proof estimates). *Let $J \subset \mathcal{I}_T$ be any subset of the admissible packet coordinates and let P_J be the corresponding coordinate projection. Replacing the packet space by $P_J \mathbb{C}^{\mathcal{I}_T}$ replaces every contour/source matrix by the compression*

$P_J A P_J^*$ and every positive reference Gram by $P_J G P_J^*$. The packet Riesz lower bound is unchanged, the exact contour identities remain valid, all edge ranks do not increase, and absolute Hilbert–Schmidt norms of compressed remainder matrices do not increase. Moreover the shell/frame trace-defect arguments remain valid with the retained coordinate count in place of the full count.

Proof. Every contour, stationary, shell, and source construction is linear in the packet synthesis coefficients, so coordinate deletion is literal matrix compression. A subsequence of a Riesz sequence inherits the same lower and upper Riesz constants. Rank is monotone under compression, and $\|P_J R P_J^*\|_{S_2} \leq \|R\|_{S_2}$. More importantly, if the common reference analysis map is W , then on the restricted packet space it is exactly $W_J := W P_J^*$ and

$$G_J = W_J^* W_J = P_J W^* W P_J^*.$$

Since the restriction is on the input space, the least singular value of W_J is no smaller than the least singular value of W on the original retained space. The primitive mixed-derivative bounds used to prove the relative S_2 estimates are uniform on the output fibres and are unchanged by deleting packet inputs. Hence the relative remainder criterion may be rerun with W_J and gives the same $o(N_T)$ asymptotic (with the smaller packet dimension). The trace-defect proofs are sums of nonnegative diagonal contributions over packet coordinate vectors, so deleting coordinates only removes summands and gives the same normalized estimate for the smaller dimension. Exact algebraic contour/source identities commute with this restriction because they hold entrywise. $\square$

Lemma 3.6 (Horizontal connector decay). *Delete packet centres within physical distance T/L of the two horizontal heights T_1, T_2 . This costs $O(T) = o(N_T)$ packet directions. On the remaining packet space, every fixed curvature derivative of a horizontal connector is $O_A(T^{-A})$ after arbitrary prescribed fixed A , up to fixed polylogarithmic local-source factors.*

Proof. The critical packet spacing is $\asymp L^{-1}$, so a total physical collar of length $O(T/L)$ contains $O(T)$ centres. On either horizontal connector the straightened packet variable has real separation at least T/L from every retained centre, while its imaginary part remains in a fixed strip. Hence (11), after any fixed number of curvature derivatives, gives

$$|\psi_{L,\tau}| \ll_M T^{C_M} (T/L)^{-M}.$$

All zeta, gamma, and fixed local-source factors on the horizontal strip have at most polynomial or polylogarithmic size. Choosing M after the fixed loss exponent gives $O_A(T^{-A})$. The same estimate holds for the adjoint orientation and for the two mixed derivatives used in the relative Hilbert–Schmidt criterion. $\square$

For a generic H with simple stationary nodes and no common zero of H, H' , the residue theorem gives an exact pullback

$$A_H = \mathcal{E}_H^* W_H \mathcal{E}_H,$$

where W_H is the direct sum of the real scalar residues and the 2×2 conjugate-pair blocks of Lemma 3.1. By Lemma 3.2, $\mathcal{E}_H$ is surjective. Restricting to a complement of $\ker \mathcal{E}_H$ and applying Sylvester’s law therefore gives

$$\boxed{n_-(A_H) = B(H) + P(H')}. \quad (14)$$

Lemma 3.7 (Real zero/stationary bookkeeping). *For a real analytic function with simple real zeros and stationary points,*

$$R(H') = R(H) + 2B(H) + O(1),$$

where R denotes the number of real zeros in the interval under consideration.

Proof. Between consecutive real zeros, Rolle's theorem forces at least one stationary point. On every zero-free component the stationary extrema alternate. Any additional stationary points therefore occur in pairs consisting of one good and one bad extremum. Only the two terminal components contribute the $O(1)$ error. $\square$

For the original gauge $\mathcal{Z}$ and its stationary source Z_1 , put

$$P(s) := -\frac{\zeta'}{\zeta}(s), \quad R(s) := \frac{Z_1(s)}{\zeta(s)} = f(s) - P(s).$$

The functional equation gives $R(s) = -R(1-s)$. At an m -fold zeta zero,

$$\zeta(s) = (s-\rho)^m u(s), \quad u(\rho) \neq 0,$$

so

$$Z_1(s) = m(s-\rho)^{m-1}u(\rho) + O((s-\rho)^m),$$

and $R = Z_1/\zeta$ has a simple pole at every zeta zero, independently of its multiplicity. Moreover, for $|t|$ large there are no zeros of Z_1 in $\Re s \geq 2$: there $Z_1 = \zeta(f-P)$, with $P = O(1)$ and $|f| \asymp \log |t|$. The functional equation (20) then excludes zeros in $\Re s \leq -1$. Thus every high zero of Z_1 relevant to the fixed disks below lies in $-1 < \Re s < 2$; the corresponding assertion for the nontrivial zeta zeros is standard from its Euler product and functional equation.

Lemma 3.8 (Local Jensen count). *Uniformly for $U \asymp T$,*

$$\boxed{N_{Z_1}([U, U+1]) + N_\zeta([U, U+1]) \ll \log U.} \quad (15)$$

Proof. Center a fixed disk at $s_U = 2 + iU$. Absolute convergence and Stirling give

$$\zeta(s_U) \asymp 1, \quad Z_1(s_U) = \zeta(s_U)\{f(s_U) - P(s_U)\}, \quad |Z_1(s_U)| \gg \log U.$$

On a fixed larger disk, the classical convexity bound for ζ and the Cauchy estimate for ζ' give polynomial growth $U^{O(1)}$ for both $(s-1)\zeta(s)$ and $Z_1(s)$. Jensen's formula therefore bounds the number of zeros in the concentric smaller disk by $O(\log U)$. A fixed number of such disks covers the strip segment of height one, proving (15). $\square$

Lemma 3.9 (Local logarithmic-derivative factorization). *Let $F_0 \in \{\zeta, Z_1\}$ and $U \asymp T$. Uniformly for $-1 \leq \sigma \leq 2$ and $|t-U| \leq 1/10$, one has*

$$\frac{F_0'}{F_0}(s) = \sum_{\substack{F_0(\rho)=0 \\ |\rho-(2+iU)| < 7/2}} \frac{1}{s-\rho} + O((\log T)^{C_0}), \quad (16)$$

whenever s is not a zero of F_0 . The sum contains $O(\log T)$ terms, with multiplicity.

Proof. Put $z_U = 2 + iU$. On the fixed disk $|s - z_U| \leq 5$, both ζ and Z_1 are holomorphic for large U and have size $T^{O(1)}$. At the centre, absolute convergence and Stirling give

$$|\zeta(z_U)| \asymp 1, \quad |Z_1(z_U)| \asymp \log T.$$

Jensen's formula therefore gives $O(\log T)$ zeros, with multiplicity, in $|s - z_U| \leq 4$ for either function.

Choose a radius $r_U \in [7/2, 4]$ whose distance from every modulus $|\rho - z_U|$ of these zeros is at least $c/\log T$. Such a radius exists because there are only $O(\log T)$ forbidden intervals and the ambient radial interval has fixed length. Factor all zeros in $|s - z_U| < r_U$:

$$F_0(s) = e^{g_U(s)} \prod_{|\rho - z_U| < r_U} (s - \rho)$$

on the simply connected disk $|s - z_U| < r_U$, after removing the zeros from the quotient. On $|s - z_U| = r_U$ the polynomial-growth upper bound for F_0 , the separation of the boundary from the zeros, and the $O(\log T)$ zero count give

$$\Re g_U(s) = \log |e^{g_U(s)}| \ll \log T \log \log T.$$

At z_U , the lower bounds above and $|z_U - \rho| \leq 4$ give $\Re g_U(z_U) \gg -\log T$. Applying Borel–Carathéodory to $g_U - g_U(z_U)$ and then Cauchy’s estimate on the fixed inner disk $|s - z_U| \leq 31/10$ yields

$$|g'_U(s)| \ll (\log T)^{C_0}.$$

The entire horizontal segment $-1 \leq \sigma \leq 2$, $|t - U| \leq 1/10$, lies in that inner disk. Logarithmic differentiation gives the zero sum over $|\rho - z_U| < r_U$. The additional zeros with $7/2 \leq |\rho - z_U| < r_U$ are a fixed positive distance from the inner disk and there are only $O(\log T)$ of them, so their contribution is absorbed into the error in (16). $\square$

Proposition 3.10 (Good-height stationary-count decrement). *There are heights*

$$T_1 \in [T, T + 1], \quad T_2 \in [2T, 2T + 1]$$

such that the boundary rectangle $-1 \leq \sigma \leq 2$, $T_1 \leq t \leq T_2$ avoids all zeros and poles of R , and already on this good rectangle

$$\boxed{N(Z_1; T_1, T_2) - N(\zeta; T_1, T_2) = O((\log T)^C) = o(N_T).} \quad (17)$$

Returning the two horizontal heights to T and $2T$ gives

$$\boxed{N(Z_1; T, 2T) - N(\zeta; T, 2T) = O((\log T)^C) = o(N_T).} \quad (18)$$

Equivalently, because q is nonvanishing, the same two estimates hold with (Z_1, ζ) replaced by (F', F) .

Proof. Fix $A > 4$ and set $\delta_T = (\log T)^{-A}$. By Lemma 3.8, each unit interval contains only $O(\log T)$ ordinates of zeros of ζZ_1 . The union of intervals of radius δ_T around them has total length

$$O(\delta_T \log T) = O((\log T)^{1-A}) < \frac{1}{2}.$$

Hence T_1, T_2 can be chosen as stated, with the two horizontal sides at distance at least δ_T from every zero or pole of R in the fixed strip.

On $\Re s = 2$, absolute convergence and Stirling give

$$R(2 + it) = \frac{1}{2} \log \frac{t}{2\pi} + O(1), \quad T \leq t \leq 2T + 1.$$

For large T this lies in a fixed acute sector around the positive real axis, so the change of argument on the right vertical side is $O(1)$. The symmetry $R(s) = -R(1 - s)$ gives the same bound on the left side.

Consider one good horizontal side, at height $U \in \{T_1, T_2\}$. Since

$$\frac{R'}{R} = \frac{Z_1'}{Z_1} - \frac{\zeta'}{\zeta},$$

Lemma 3.9 applies separately to Z_1 and ζ . The selected horizontal line stays at distance at least δ_T from every zero of either function in the fixed strip, and each local zero sum has only $O(\log T)$ terms. Therefore, uniformly for $-1 \leq \sigma \leq 2$,

$$\left| \frac{R'}{R}(\sigma + iU) \right| \ll (\log T)^{C_0} + \delta_T^{-1} \log T \ll (\log T)^{A+C_1}. \quad (19)$$

Integrating over the bounded horizontal length gives a polylogarithmic argument variation. Thus the total boundary variation of $\arg R$ is $O((\log T)^C)$.

The argument principle identifies this variation divided by 2π with the number of zeros of R minus its poles. Since the divisor of $R = Z_1/\zeta$ is exactly the divisor difference of Z_1 and ζ , this proves (17). Returning to $T, 2T$ changes either zero count by only $O(\log T)$ by Lemma 3.8, proving (18). $\square$

Section 15 transfers the generic zero-side inertia identity (14) back to the original Hardy-gauge form. All right-edge and remainder estimates are proved for the unperturbed form A_F ; the generic perturbation is used only in the final finite-dimensional inertia comparison.

4 Right-edge arithmetic source and exact HLP algebra

4.1 Reflected archimedean contribution

Differentiate the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$. Since $f(s) = -\frac{1}{2}\chi'/\chi(s)$ and $f(s) = f(1-s)$, one obtains

$$\boxed{Z_1(s) = -\chi(s)Z_1(1-s)}. \quad (20)$$

Taking logarithmic derivatives gives

$$\boxed{L_1(s) + L_1(1-s) = \frac{\chi'}{\chi}(s), \quad L_1 := \frac{Z_1'}{Z_1}.} \quad (21)$$

The two vertical sides are related by the Hermitian reflection $\iota(s) := 1 - \bar{s}$, which preserves the positive height and maps the left vertical side onto the right one. The holomorphic map $s \mapsto 1-s$ alone would reverse the sign of the height and is not used as a geometric contour substitution. By (2), the curvature and packet polarizations on the left are the adjoints of their values at $\iota(s)$. Because the left boundary has the opposite orientation, its pullback to the upward right side carries the scalar factor

$$-\overline{L_1(1-\bar{s})} = -L_1(1-s) = L_1(s) - \frac{\chi'}{\chi}(s),$$

where the last equality is (21). Hence the vertical contribution is exactly

$$\boxed{I_{\text{right}} + I_{\text{right}}^* + J_{\text{arch}}, \quad J_{\text{arch}} \text{ has scalar source } -\frac{\chi'}{\chi} = 2f.} \quad (22)$$

Here I_{right} contains the right-edge logarithmic derivative L_1 and all stationary arithmetic information. No L_1 contour is pushed through an interior stationary zero. The separate archimedean self

source is holomorphic in the high rectangle and may be moved by Cauchy's theorem. For the completed Hermitian representation we distribute J_{arch} equally between the two adjoint orientations; hence each completed right-edge orientation carries the scalar source $f + L_1$.

For bounded normalized curvature p , put $a = p/L$ and let $\mathcal{B}_{v,w}(s)$ denote the holomorphic packet-polarization multiplier. Set

$$\mathcal{S}_a(s) := \mathcal{Z}(s+a)\mathcal{Z}(s-a)\mathcal{B}_{v,w}(s), \quad \Phi_J(s; a) := f(s)\mathcal{S}_a(s) = -\frac{1}{2}\frac{\chi'}{\chi}(s)\mathcal{S}_a(s).$$

Thus Φ_J is exactly one half of the full self source J_{arch} in (22). The shifts $a = O(L^{-1})$ leave the zeta poles at bounded imaginary height, so Φ_J is holomorphic on and inside the right half-rectangle

$$\mathcal{R}_T^+ := \{\sigma + it : 1/2 \leq \sigma \leq c, T_1 \leq t \leq T_2\},$$

which is the right half of (1), uniformly for $|p| \leq P$. With the vertical sides oriented upward and the usual top/bottom orientations, Cauchy's theorem gives the exact identity

$$\boxed{\int_{\Gamma_{1/2}^+} \Phi_J(s; a) ds = \int_{\Gamma_c^+} \Phi_J(s; a) ds + R_{\text{hor}, T}(a),} \quad (23)$$

where

$$R_{\text{hor}, T}(a) = - \int_{\mathcal{H}_T^+} \Phi_J(s; a) ds - \int_{\mathcal{H}_T^-} \Phi_J(s; a) ds.$$

The reflected Hermitian orientation gives the adjoint identity. Summing the two orientations doubles $\Phi_J = f\mathcal{S}_a$ and therefore reconstructs the full self source $-\chi'/\chi = 2f$ in (22), with the corresponding doubled right-edge and horizontal terms. Because all fixed p -derivatives remain holomorphic in the same rectangle,

$$\partial_p^j \mathcal{J}_{\text{crit}, T}(p) = \partial_p^j \mathcal{J}_{\text{right}, T}(p) + \partial_p^j R_{\text{hor}, T}(p), \quad 0 \leq j \leq 2. \quad (24)$$

Thus the critical- J contribution is fixed unambiguously before the common curvature is taken.

We insert the packet multiplier and the common curvature (4) before deforming the contour to the right edge. On that edge,

$$\mathcal{Z}(s+a)\mathcal{Z}(s-a) = \Gamma_a(s)\zeta(s+a)\zeta(s-a),$$

where

$$\Gamma_0(s) = \chi(1-s), \quad \partial_a \Gamma_a|_{a=0} = 0, \quad \partial_a^2 \Gamma_a|_{a=0} = 2f'(s)\chi(1-s).$$

Since $f'(\sigma + it) = O(T^{-1})$ on fixed strips with $t \asymp T$, the last term is assigned from the start to the regular remainder.

Fix the sign convention

$$P(s) := -\frac{\zeta'}{\zeta}(s), \quad Q(s) := -P'(s) = \sum_{n \geq 1} \frac{\Lambda(n) \log n}{n^s} \quad (\Re s > 1). \quad (25)$$

Then the right-edge logarithmic-derivative orientation itself has scalar source

$$L_1(s) = \frac{Z_1'}{Z_1}(s) = -P(s) + \frac{f'(s) + Q(s)}{f(s) - P(s)}.$$

The full archimedean self term in (22) is $2f$; after the symmetric half-allocation described above, each completed right-edge orientation therefore carries $f + L_1$. After separating the f' term, the pointwise completed source is

$$\widehat{H}_{\text{ex}}(s) = f(s) - P(s) + \frac{Q(s)}{f(s) - P(s)}. \quad (26)$$

On a smooth height block with reference height τ_b , put

$$F_b := f(1/2 + i\tau_b) \in \mathbb{R} \quad (27)$$

(the reality follows from $|\chi(1/2 + it)| = 1$ and $\partial_t \log \chi = i\chi'/\chi$ on the critical line), and define the frozen source

$$H_{\text{ex}}(s; F_b) = F_b - P(s) + \frac{Q(s)}{F_b - P(s)}. \quad (28)$$

Thus the term F_b is the frozen half of the archimedean self contribution and is not part of the Dirichlet series for Z'_1/Z_1 . The completed two-master model is

$$H_{\text{mod}}(s; F_b) = F_b - 2P(s) + P(s - 2/L). \quad (29)$$

Therefore

$$\Delta H(s; F_b) = P(s) + \frac{Q(s)}{F_b - P(s)} - P(s - 2/L). \quad (30)$$

This is an exact algebraic identity between the two frozen sources on the absolute-convergence right edge. The error introduced by replacing the pointwise source (26) with (28) is estimated below uniformly over the HLP hierarchy.

Lemma 4.1 (Exact curvature Dirichlet source). *Put*

$$\mathcal{C}(s) := \zeta(s)^2 Q(s).$$

Then the Hardy-gauge curvature satisfies the exact identity

$$C_{\mathcal{J}}(s) = q(s)^2 (\mathcal{C}(s) + f'(s)\zeta(s)^2), \quad q(s)^2 = \chi(1 - s). \quad (31)$$

Moreover

$$\mathcal{C}(s) = \sum_{n \geq 1} \frac{c(n)}{n^s}, \quad c(n) = \frac{1}{2} \sum_{ab=n} \log^2(a/b) \geq 0. \quad (32)$$

Consequently, after the global normalization L^{-2} already fixed in Section 2, the principal completed right-edge integrand has exactly one functional-equation factor $\chi(1 - s)$ and scalar Dirichlet source $\mathcal{C}(s)H_{\text{ex}}(s; F_b)$. Before freezing the slow factor, the exact completed product separates as

$$q(s)^2 \left[\mathcal{C}(s) \widehat{H}_{\text{ex}}(s) + R_{f'}^{\text{src}}(s) \right], \quad (33)$$

where

$$R_{f'}^{\text{src}} = f' \zeta^2 \widehat{H}_{\text{ex}} + \frac{f'}{f - P} \mathcal{C} + \frac{(f')^2}{f - P} \zeta^2. \quad (34)$$

Thus every term containing at least one factor f' belongs exactly once to the regular f' class, while the principal source contains none.

Proof. Since $\mathcal{Z} = q\zeta$ and $q'/q = f$,

$$\mathcal{Z}'' = q(\zeta'' + 2f\zeta' + (f' + f^2)\zeta).$$

Therefore

$$\begin{aligned} C_{\mathcal{Z}} &= \mathcal{Z} \mathcal{Z}'' - (\mathcal{Z}')^2 \\ &= q^2(\zeta\zeta'' - (\zeta')^2 + f'\zeta^2). \end{aligned}$$

Because $(\zeta'/\zeta)' = Q$, the first two terms equal $\zeta^2 Q = \mathcal{C}$, proving (31). Moreover (26) gives the exact completed scalar factor

$$f + L_1 = \widehat{H}_{\text{ex}} + \frac{f'}{f - P}.$$

Multiplying this by $C_{\mathcal{Z}}/q^2 = \mathcal{C} + f'\zeta^2$ gives (33)–(34). Finally the coefficient of n^{-s} in $\zeta\zeta'' - (\zeta')^2$ is

$$\sum_{ab=n} ((\log b)^2 - \log a \log b) = \frac{1}{2} \sum_{ab=n} (\log a - \log b)^2,$$

which proves (32). $\square$

From this point the spectral contour variable and the Perron variable are kept separate. The letter $s = \sigma + it$ is reserved for the high spectral contour. After stationary phase, the resulting coefficient sequences are represented by an independent Perron/Mellin variable z . We denote their generating functions by $P(z)$, $Q(z)$, $H_{\text{ex}}(z; F_b)$ and $H_{\text{mod}}(z; F_b)$ using the same coefficient formulas as above. All Laurent expansions near 1 below are in z ; no deformation of the original high spectral contour to $s = 1$ is ever performed.

Lemma 4.2 (Algebraic locator for the translated model scale). *Write $z = 1 + u$ and assume $F_b = L/2 + O(1)$. As a meromorphic resummation of the right-edge geometric series, the frozen ratio $H_{\text{ex}} = F_b - P + Q/(F_b - P)$ has, in a fixed normalized neighbourhood $|u| \leq C/L$, poles at $u = 0$ and at one point $u = u_*(F_b)$, where*

$$\boxed{u_*(F_b) = F_b^{-1} + O(F_b^{-2}) = \frac{2}{L} + O(L^{-2})}. \quad (35)$$

Their scalar residues are -2 and $+1$. The model source H_{mod} has principal parts $-2/u + 1/(u - 2/L)$.

This lemma is only an algebraic locator for the translated scale. In the actual Perron argument the quotient $Q/(F_b - P)$ is not transported as one meromorphic object through the Perron plane. The exact principal source is defined by the absolutely convergent geometric expansion on the safe right edge and by complete levelwise Laurent extraction at $u = 0$; only the resulting leading residue family is resummed. Thus u_ does not appear as a separate residue term in the contour decomposition; its role is only to identify the translated scale represented by the resummed model.*

Proof. The Laurent expansion of the logarithmic derivative at the zeta pole is

$$P(1 + u) = \frac{1}{u} + p_0 + O(u), \quad Q(1 + u) = \frac{1}{u^2} + O(1).$$

Put $D(u) = F_b - P(1 + u)$. Then

$$uD(u) = F_b u - 1 - p_0 u + O(u^2).$$

On a circle $|u - F_b^{-1}| = C_1 F_b^{-2}$ with C_1 fixed sufficiently large, Rouché's theorem gives exactly one zero u_* of D and $u_* = F_b^{-1} + O(F_b^{-2})$. Since

$$D'(u) = -P'(1+u) = Q(1+u),$$

the quotient Q/D has residue exactly 1 at u_* . At $u = 0$ the Laurent expansions above give

$$-P(1+u) = -\frac{1}{u} + O(1), \quad \frac{Q(1+u)}{F_b - P(1+u)} = -\frac{1}{u} + O(1),$$

while F_b is analytic. Hence the frozen Perron source H_{ex} has residue -2 at $u = 0$. This proves the exact-source statement without identifying the independent Perron variable with the original spectral Z_1 -function. For $H_{\text{mod}}(z; F_b) = F_b - 2P(z) + P(z - 2/L)$ the principal parts are immediately

$$-\frac{2}{u} + \frac{1}{u - 2/L}.$$

□

5 The physical one- χ carrier and exact dilation covariance

A right-edge monomial with total Dirichlet ratio

$$\xi = \frac{m}{k}$$

occurs inside a one- χ integral. After a logarithmic packet displacement w , put $\eta = e^w \xi$. The principal oscillatory kernel is

$$K_w(\xi) = \frac{1}{2\pi} \int A(t) \chi(1 - c - it) \eta^{-c - it} dt.$$

Stirling gives

$$\chi(1 - c - it) \eta^{-c - it} = \left(\frac{t}{2\pi} \right)^{c-1/2} \eta^{-c} e^{i\Phi_\eta(t)} (1 + O(t^{-1})),$$

with

$$\Phi_\eta(t) = t \log \frac{t}{2\pi\eta} - t - \frac{\pi}{4}.$$

Hence

$$\Phi'_\eta(t) = \log \frac{t}{2\pi\eta}, \quad t_* = 2\pi\eta, \quad \Phi''_\eta(t_*) = \frac{1}{t_*}.$$

At the saddle,

$$\left(\frac{t_*}{2\pi} \right)^{c-1/2} \eta^{-c} = \eta^{-1/2},$$

whereas the Gaussian Hessian factor is

$$\sqrt{\frac{2\pi}{\Phi''_\eta(t_*)}} = \sqrt{2\pi t_*} = 2\pi\eta^{1/2}.$$

Thus the powers of η cancel exactly. The stationary phase $+\pi/4$ cancels the $-\pi/4$ in Φ , and

$$e^{i\Phi_\eta(t_*)} e^{i\pi/4} = e^{-2\pi i\eta} = e(-\eta).$$

Consequently

$$\boxed{K_w(\xi) = A(2\pi e^w \xi) e(-e^w \xi) + E_w(\xi).} \quad (36)$$

There is no order-one pair-dependent Hessian weight.

If $D_w F(\xi) := F(e^w \xi)$, the principal stationary map satisfies

$$\boxed{\mathcal{S}_w^{\text{pr}} = D_w \mathcal{S}_0^{\text{pr}}.} \quad (37)$$

A smooth height cutoff transforms by the same dilation:

$$\omega\left(\frac{2\pi\xi}{T}\right) \mapsto \omega\left(\frac{2\pi e^w \xi}{T}\right).$$

Thus the phase, the saddle cutoff, and the principal amplitude are all carried by the same exact dilation. The higher stationary coefficients, Stirling corrections, and nonstationary leakage belong to the regular classes estimated in Proposition 12.4. The packet centres in the two horizontal collars are treated separately by the coordinate restriction V_{end} of Lemma 3.6; they do not contribute an edge or regular source term.

On a smooth height block of scale $H = T/L^D$, write $t = t_* v$ on the stationary sector. There $t_* \asymp T$ uniformly in the shell and fixed local-source labels, and the rescaled saddle $v = 1$ remains in a fixed compact subset of the block after the smooth stationary/nonstationary partition. Consequently every fixed-order stationary expansion is uniform in the retained indices. Differentiating the slow block amplitude contributes only fixed combinations of

$$T^{-1}, \quad H^{-1} = L^D/T, \quad T/H^2 = L^{2D}/T,$$

and further fixed derivatives give the same type of T^{-1} -times-polylog bounds. These terms therefore remain in the regular symbol class after the fixed derivative orders used below.

Lemma 5.1 (Uniform smooth stationary symbol). *Fix integers $r, s \geq 0$ and a bounded curvature range $|p| \leq P$. After the universal carrier $e(-my)$ is extracted, every direct stationary one- χ source occurring in the ordinary Hardy term, the summed exact HLP hierarchy, the translated/coalescing faces, and the fixed Stirling/Hardy corrections has on a retained unit shell $I_j = [J_j, J_j + 1]$ the form*

$$c_m a_{m,T}(y; p) e(-my),$$

with

$$\boxed{\sup_{y \in I_j, |p| \leq P} |\partial_y^r \partial_p^s a_{m,T}(y; p)| \ll_{r,s,P} L^{C_{r,s}} J_j^{-r}.} \quad (38)$$

The constants are uniform over the fixed local-source labels and over the summed geometric levels.

Proof. Call a height factor g a symbol of order R if

$$\sup_{0 \leq q \leq R} t^q |g^{(q)}(t)| \ll_R L^C$$

on $t \asymp T$. This class is closed under finite products and, when a factor is bounded below by a negative power of L , under reciprocal. Stirling's formula shows that

$$\lambda(t), \quad f(t)/L, \quad e^{p\lambda(t)}, \quad e^{\pm 2\lambda(t)}, \quad t^{\pm 2/L}$$

and every fixed retained gamma correction are symbols, uniformly for bounded p . The f' term gains an additional factor T^{-1} .

For the exact HLP hierarchy, level k carries the coefficient $\alpha_k(m)f(t)^{-k}$ and is supported only when $2^k \leq m$ by Lemma 7.1. Choose a fixed constant $C_H > 2$ so large that, on the stationary strip and for all sufficiently large T ,

$$q_H := \sup_t \left| \frac{L}{C_H f(t)} \right| < 1.$$

Set

$$b_{k,m} := \alpha_k(m) \left(\frac{C_H}{L} \right)^k, \quad c_m := \sum_{k \geq 1} b_{k,m}.$$

The coefficients $b_{k,m}$ are nonnegative. If $c_m \neq 0$, the summed hierarchy can therefore be written as $c_m a_{m,T}(t)$ with

$$a_{m,T}(t) = \sum_{k \geq 1} \frac{b_{k,m}}{c_m} \left(\frac{L}{C_H f(t)} \right)^k.$$

The weights $b_{k,m}/c_m$ form a probability vector. Every fixed height derivative of the last factor is bounded by a fixed polynomial in k times q_H^k , because $t^r f^{(r)}(t)/f(t) = O(L^{-1})$ for $r \geq 1$. Hence $a_{m,T}$ is a height symbol uniformly in m , without using an ℓ^2 estimate pointwise. Separately, the factorial majorant gives

$$\left(\sum_{m \leq x} c_m^2 \right)^{1/2} \leq \sum_{k \geq 1} \left(\frac{C_H}{L} \right)^k \left(\sum_{m \leq x} \alpha_k(m)^2 \right)^{1/2} \ll (xL)^{1/2} \sum_{k \geq 1} \frac{C_1^k}{\sqrt{k!}},$$

so the coefficient sequence has the required quadratic bound. Fixed p -derivatives and translated factors contribute only the already controlled polylogarithmic symbols and factors $(\log m)/L = O(1)$. Thus the complete exact HLP amplitude remains in a fixed polylogarithmic height-symbol class.

For one stationary monomial scale $t = t_* v$. After the exact Hessian cancellation already proved above, the phase is $t_* \Psi(v)$ with $\Psi'(1) = 0$, $\Psi''(1) \neq 0$, and uniformly bounded higher derivatives near $v = 1$. Fixed-order stationary phase therefore produces a finite expansion whose coefficients are height symbols, with a remainder $O(T^{-M} L^{C_M})$ after any prescribed fixed order M . Finally,

$$\partial_y^r g(2\pi m y) = (2\pi m)^r g^{(r)}(2\pi m y).$$

On the shell $my \asymp T$ and $y \asymp J_j$, so

$$\frac{m}{T} \asymp \frac{1}{J_j}.$$

The height-symbol bound therefore gives

$$|\partial_y^r g(2\pi m y)| \ll L^C J_j^{-r}.$$

The same argument is uniform after the fixed number of p -derivatives. This proves (38). $\square$

6 The completed three- Z master and common-prefix geometry

The three principal Z -faces are not three independent sources. We use the completed three-face Cauchy master and diagonal Mellin representation of Heap–Li–Zhao [6]. At fixed height the three faces arise from one completed shifted product and one joint three-variable residue. Only after this completion is the diagonal Mellin variable introduced. In the general HLZ twisted formula the

auxiliary outer Dirichlet polynomial produces indices h, k and the diagonal relation $hm = kn$. For $(h, k) = 1$ this is

$$\boxed{m = k\ell, \quad n = h\ell.} \quad (39)$$

The invariant contour form of this paper, however, contains no such auxiliary outer Dirichlet polynomial: its additional test factor is the packet polarization $\mathcal{B}_{v,w}$, which is retained as a smooth holomorphic height/Mellin weight. Thus the actual source specialization used below is $h = k = 1$, and (39) reduces to $m = n = \ell$.

Proposition 6.1 (Trivial outer twist of the invariant packet source). *In the application of the HLZ diagonal calculation to the contour matrix A_T , the auxiliary outer Dirichlet polynomial is identically one. Hence its outer indices satisfy*

$$\boxed{g = h = k = 1.}$$

If this source is embedded in any finite auxiliary divisor range $d \leq Y$, its Hilbert-valued coefficient family has only the coordinate $d = 1$ nonzero. Consequently the outer packet translations are the identity, the common divisor map of Proposition 6.8 restricts to the identity, and the factor de/g equals 1. In particular the outer excess annulus (59) is empty for the invariant contour source.

Proof. The exact same-form identity (6) contains the shifted Hardy product and the packet polarization, but no factor of the form $Q_{\text{out}}(s)\overline{Q_{\text{out}}}(1-s)$. The HLP series in (26) belongs to the logarithmic-derivative source and is not an auxiliary outer test polynomial. In the HLZ twisted formula the indices g, h, k arise only after expanding such an outer Dirichlet polynomial and its adjoint. Therefore the present contour form is the specialization $Q_{\text{out}} \equiv 1$, giving $g = h = k = 1$. The packet polarization remains in the analytic weight before the finite shift residues, so no discrete outer index is created by it. This specialization removes only the auxiliary outer indices g, h, k ; it does not by itself contract the internal HLP radial carrier M . That carrier is contracted only in the low-level common principal functional of Lemma 6.11, while the direct translated and high-level terms retain it. The remaining assertions follow immediately from the definitions of the translation and common-divisor maps. $\square$

Remark 6.2 (Scope of the HLZ input). We use only the completed three-face residue formula and diagonal calculation of Heap–Li–Zhao [6], specialized as in Proposition 6.1; the packet polarization is not identified with their auxiliary Dirichlet polynomial. Because the packet factor is holomorphic in the shift variables, it commutes exactly with the finite Cauchy residues. At $h = k = 1$ the finite Euler factor below is identically one. The packet-stable contour error retains the superlogarithmic/arbitrary-log saving after the fixed packet derivatives. All packet-uniform stationary and sideband estimates used below are proved directly rather than imported from the scalar moment theorem.

Lemma 6.3 (Packet-stable HLZ diagonalization). *Let $H_0(u, \mathbf{z})$ be the auxiliary entire factor in the diagonal lemma of Heap–Li–Zhao, normalized by $H_0(0, \mathbf{z}) = 1$ and vanishing at the two moving divisors*

$$2u = z_2 - z_1, \quad 2u = z_2 - z_3.$$

Let $W(u, \mathbf{z})$ be a packet Mellin weight of Lemma 7.6. Put $\log_3 T := \log \log \log T$. Uniformly for the retained shifts, the invariant-source specialization $h = k = 1$ of the HLZ diagonal contour with auxiliary factor $H_0 W$ satisfies

$$\boxed{W(0, \mathbf{z})Z_{\mathbf{z},1,1} = \widetilde{Z}_{\mathbf{z},1,1}[H_0 W] + \mathcal{E}_{\text{diag}}(\mathbf{z}; 1, 1),} \quad (40)$$

where, after any fixed number of normalized shift derivatives and any fixed number of the packet derivatives $\partial_\rho, \partial_p$ used below,

$$\mathcal{E}_{\text{diag}} \ll L^{C_W} \left[T^{-2/\log_3 T} (\log T)^C \log_3 T + (\log T)^{-A} \right] \quad (41)$$

for every prescribed fixed $A > 0$. The constant C_W depends only on the fixed shift and packet derivative orders. Thus insertion of the physical packet weight before the three finite shift residues changes the main diagonal by its exact value $W(0, \mathbf{z})$ and costs only the same superlogarithmic/arbitrary-log error class as the unweighted diagonal lemma.

Proof. The packet weight has the form

$$W(u, \mathbf{z}) = \int_0^{C_*} a_{\rho, p, \mathbf{z}, T}(t) e^{Lut} dt$$

(or a finite product of such factors). Lemma 7.6 shows that every fixed $\partial_\rho, \partial_p$ derivative has the same polylogarithmic L^1 bounds. On the strip used in the HLZ contour argument,

$$-\frac{1}{\log_3 T} \leq \Re u \leq \frac{1}{L},$$

its absolute value and every fixed normalized shift/packet derivative are $O(L^{C_W})$: on the right edge this follows from $e^{L\Re(u)t} \leq e^{C_*}$ and the L^1 bounds of Lemma 7.6, while on the left part of the strip the exponential factor only decreases. The auxiliary factor H_0 in the HLZ lemma supplies the required Gaussian decay in $|\Im u|$. Multiplication by W therefore preserves that Gaussian decay, with only the fixed polylogarithmic factor L^{C_W} in front. Thus every vertical and horizontal estimate used in the original contour deformation remains in the same error class after insertion of the packet weight.

The zeros of H_0 at the two moving divisors are unchanged after multiplication by W . Although the HLZ diagonal lemma is stated with the normalization $H(0) = 1$, its contour identity is linear in the auxiliary entire factor. Applying the same proof to $H_0 W$ therefore gives the residue

$$(H_0 W)(0, \mathbf{z}) Z_{\mathbf{z}, 1, 1} = W(0, \mathbf{z}) Z_{\mathbf{z}, 1, 1},$$

without dividing by $W(0, \mathbf{z})$. The same contour deformation crosses only $u = 0$, and the left vertical and truncated horizontal integrals acquire at most the factor L^{C_W} above. This proves the undifferentiated form of (40)–(41).

For the fixed normalized shift derivatives used in the paper, apply Cauchy's formula on circles of radius comparable to $1/L$. A fixed number of such derivatives costs only a fixed power $L^{C_{\text{sh}}}$. Since the logarithmic remainder in the HLZ lemma is available with an arbitrary prescribed exponent, we first invoke that lemma with exponent $A + C_{\text{sh}} + C_W + 2$ and only then differentiate. After absorbing the resulting fixed powers into C_W , the final differentiated estimate has the displayed exponent A . The same argument applies to the fixed $\partial_\rho, \partial_p$ derivatives because Lemma 7.6 gives the required uniform L^1 bounds for the differentiated packet amplitudes. In particular the packet weight is bounded uniformly in $\Im u$ on each vertical line used in the deformation, so it introduces no additional vertical exponential growth. $\square$

Lemma 6.4 (Normalization of the finite HLZ Euler factor). *In the notation of the completed HLZ diagonal, the finite local factor attached to k is*

$$\mathcal{E}_{\alpha, \beta, \gamma}(k) := \prod_{p^{\kappa_p} \parallel k} \frac{\sum_{m \geq 0} f_{\alpha, \gamma}(p^{m+\kappa_p}) p^{-m(1+\beta)}}{\sum_{m \geq 0} f_{\alpha, \gamma}(p^m) p^{-m(1+\beta)}}, \quad (42)$$

where

$$f_{\alpha,\gamma} = \mu * n^{-\alpha} * n^{-\gamma}$$

in Dirichlet-convolution notation. Then

$$\boxed{\mathcal{E}_{0,0,0}(k) = 1.} \quad (43)$$

Moreover, uniformly for $k \leq Y = L^\kappa$ and $|\alpha| + |\beta| + |\gamma| \leq C/L$,

$$\boxed{\mathcal{E}_{\alpha,\beta,\gamma}(k) = 1 + O_C\left(\frac{\log Y}{L}\right),} \quad (44)$$

and every fixed derivative with respect to the normalized variables $L\alpha, L\beta, L\gamma$ is bounded by a fixed power of $\log Y/L$ (with the zeroth derivative bounded uniformly).

Proof. For one prime put $z = p^{-(1+\beta)}$. The local generating series is

$$\sum_{m \geq 0} f_{\alpha,\gamma}(p^m) z^m = \frac{1 - z}{(1 - p^{-\alpha} z)(1 - p^{-\gamma} z)}.$$

At $\alpha = \beta = \gamma = 0$ this is $(1 - z)^{-1}$, so every local coefficient is 1 and the numerator and denominator in (42) coincide. This proves (43).

On the stated shift polydisc one has $|z| \leq p^{-1+o(1)}$, uniformly for $p \leq Y$. Hence the local series and its reciprocal are analytic and uniformly bounded on a fixed slightly larger polydisc. Differentiating a local factor j times in the unnormalized shifts costs at most $O_j((\kappa_p + 1)^j (\log p)^j)$. Summing logarithmic derivatives over $p^{\kappa_p} \parallel k$ and using

$$\sum_{p^{\kappa_p} \parallel k} \kappa_p \log p = \log k \leq \log Y$$

shows that the first logarithmic derivative is $O(\log Y)$ and every fixed higher derivative is bounded by a fixed power of $\log Y$. Passing to the normalized variables divides each derivative by L . The mean-value theorem, together with (43), gives (44) and the asserted derivative bounds. $\square$

Proposition 6.5 (Height versus fibre endpoint). *In the completed three- Z source, every explicit power of the interpolation height X belongs to the functional-equation/archimedean factor, whereas the packet-dependent finite endpoint is produced by the diagonal Mellin factor. More precisely,*

$$X_{\text{arch}} = \frac{t}{2\pi}, \quad \lambda(t) := \frac{\log X_{\text{arch}}}{L} = 1 + O(L^{-1})$$

on a retained dyadic height block, while the diagonal factor

$$T^{2u}(mn)^{-u} = \left(\frac{T^2}{hk\ell^2} \right)^u$$

produces the moving fibre endpoint. After the two Hermitian stationary cutoffs are intersected, the full-fibre scale is the same local height,

$$Y_{\text{fib}} := X_{\text{arch}} = \frac{t}{2\pi}, \quad \frac{\log Y_{\text{fib}}}{L} = \lambda(t) = 1 + O(L^{-1}).$$

Then

$$\ell \leq Y_{\text{fib}} e^{-|v|}, \quad v = w + \log(h/k),$$

so

$$\boxed{\rho(v) = \frac{\log Y_{\text{fib}} - |v|}{L} = 1 - \frac{|v|}{L} + O(L^{-1}).} \quad (45)$$

The symbol $Y_{\text{fib}} \asymp T$ is unrelated to the arithmetic truncation $Y = L^\kappa$ introduced later. This is the raw stationary coordinate: the term $\log(h/k) = \log h - \log k$ is removed later by an exact pair of packet translations in the common positive analysis map, not by a small endpoint approximation. Consequently packet lag changes the finite endpoint ρ but does not replace the explicit archimedean powers by powers of ρ .

Proof. Before the arithmetic master is contracted, the three-face interpolation contains the common monomial

$$X^{-(\alpha+\beta+\gamma)/2} X^{(z_1-z_2+z_3)/2}.$$

Taking the finite Cauchy residues gives the explicit factors

$$1, \quad X^{-\alpha-\beta}, \quad X^{-\beta-\gamma}.$$

These factors are present even after the smooth Mellin cutoff has been suppressed from the displayed fixed-fibre coefficient, so their X is the physical height $X_{\text{arch}} = t/(2\pi)$. By contrast, after (39) the Mellin variable occurs through

$$T^{2u}(mn)^{-u} = (T^2/(hkl^2))^u,$$

which is the source of the diagonal cutoff. The packet dilation of Section 5 changes the two oriented cutoffs to $Y_{\text{fib}}e^{-v}$ and $Y_{\text{fib}}e^v$; their common part is $Y_{\text{fib}}e^{-|v|}$, proving (45).

The distinction survives differentiation. For instance,

$$\partial_\gamma [X^{-q-\gamma} A^\gamma]_{\gamma=0} = X^{-q} \log(A/X),$$

so the X^{-q} and the $-\log X$ term have the same archimedean provenance, whereas the $A = h\ell$ contribution is a fibre variable. Likewise, for $\delta = 2/L$,

$$\left(\frac{kX}{hn_2n_3}\right)^\delta = \left(\frac{k}{h}\right)^\delta X^\delta (n_2n_3)^{-\delta},$$

with

$$X^\delta = e^{2\lambda} = e^2(1 + O(L^{-1})),$$

while only the last factor is integrated over the common fibre. Hence the archimedean and endpoint roles are distinct already at the source level. $\square$

Lemma 6.6 (Exact outer-ratio translation intertwiner). *On the raw stationary coordinate the common-prefix lag is*

$$v = w + \log h - \log k.$$

On the ambient packet Fourier space $L^2(\mathbb{R}, du)$ let

$$(U_h F)(u) := F(u - \log h), \quad (U_k F)(u) := F(u - \log k).$$

Then U_h, U_k are unitary, and the raw pair kernel depending on $w + \log(h/k)$ is exactly the pullback by $U_h^ U_k$ of the universal kernel in w . This identity holds for the full smooth packet profile, including its transition caps; no flat-amplitude approximation is involved. The reserve condition (10) is used only later when the physical stationary band is represented inside the flat plateau for shell analysis.*

Proof. Multiplication of a Dirichlet monomial by h^{-it} translates the packet Fourier coordinate by $\log h$; the adjoint k -orientation gives the analogous translation by $\log k$. Translation on $L^2(\mathbb{R})$ preserves Lebesgue measure and the full translated packet profile, so the relative displacement is exactly $\log h - \log k$ and conjugation by U_h, U_k replaces $w + \log(h/k)$ by w without a cutoff error. These ambient unitaries are part of the common packet analysis map in both A_T^G and $\mathfrak{G}_T$. When the later shell representation is restricted to the physical stationary band, $|\log h|, |\log k| \leq \kappa \log L$ and the chosen reserve keep the translated physical copies strictly inside the constant plateau. $\square$

Proposition 6.7 (The critical- J self diagonal has the same full fibre). *At the pointwise principal diagonal, the primal and dual approximate-functional- equation halves of the critical- J self source glue to the same full finite interval as the completed three- Z diagonal. In particular, if*

$$U := \frac{\log X}{L}, \quad X = \frac{t}{2\pi},$$

and $A := L\alpha$, $B := L\beta$, then for $h = k = 1$ the two diagonal halves are

$$E_{U/2}(A + B)$$

and

$$e^{-U(A+B)} E_{U/2}(-(A + B)),$$

whose sum is exactly

$$\boxed{E_{U/2}(A + B) + e^{-U(A+B)} E_{U/2}(-(A + B)) = E_U(A + B).} \quad (46)$$

The three- Z diagonal with the same local height X has the radial factor

$$\left(\frac{X^2}{hk\ell^2} \right)^u,$$

so it uses the same full logarithmic endpoint. The factors h, k do not create a separate principal endpoint after the common analysis map is fixed: by Lemma 6.6 their logarithms are absorbed by exact packet translations, leaving the universal packet lag. Thus no $O(\log L/L)$ scalar endpoint approximation is used here.

Proof. For the shifted product

$$\zeta(1/2 + it + \alpha) \zeta(1/2 - it + \beta),$$

a smoothed pointwise approximate functional equation has radial conductor mn/X , not mn/X^2 . On the diagonal (39), $mn = hk\ell^2$. With $h = k = 1$, one AFE half therefore has $\ell^2 \lesssim X$, i.e.

$$0 \leq \frac{\log \ell}{L} \leq \frac{U}{2}.$$

Its local continuum factor is $E_{U/2}(a + b)$. The dual half has the opposite shift and the functional-equation factor $e^{-U(a+b)}$; changing variables $v = U - r$ converts its interval to $[U/2, U]$, proving (46).

The completed three- Z master is represented before scalar contraction by a single diagonal Perron integral. After $m = k\ell, n = h\ell$, its Mellin factor is $(X^2/(hk\ell^2))^u$. Taking the same local height $X = t/(2\pi)$ therefore produces the full interval $0 \leq \log \ell/L \leq U$, exactly the glued interval of (46). The smooth Mellin window can be chosen consistently in both representations, so the equality is at the principal feature level, not only after scalar summation. $\square$

For the remainder of the proof fix the stable logarithmic core

$$\boxed{W_T := L - A_0 \log L, \quad X_T := e^{W_T} = \frac{T}{2\pi L^{A_0}},} \quad (47)$$

where the fixed lower bounds on A_0 are imposed in Section 13. This definition belongs to the invariant-source geometry and is independent of the auxiliary divisor comparison below.

Proposition 6.8 (Auxiliary unsummed common-divisor feature). *For comparison with the general twisted formula, let $d, e \leq Y$ be auxiliary outer Dirichlet-polynomial indices, with coefficients c_d, c_e , and let m, n be the two zeta indices. For the invariant contour source Proposition 6.1 specializes this family to $d = e = 1$; the general finite-range identity below is auxiliary. On the stationary diagonal*

$$\boxed{dm = en =: N,} \quad (48)$$

the base principal shifted product has the exact positive factorization

$$\boxed{\sum_N N^{-1-2a} \left| \sum_{\substack{d|N \\ d \leq Y}} c_d d^a \right|^2, \quad a = \frac{p}{L}.} \quad (49)$$

The translated principal master with $\delta = 2/L$, inner shifts $a + \delta, a - \delta$, and reciprocal outer deformation $c_d \mapsto c_d d^{-\delta}$, $c_e \mapsto c_e e^{\delta}$ has exactly the same factor (49). The critical-J self slot acts through the same outer-divisor feature. The factorization is Hilbert-valued as well: if $b_d \in \mathcal{H}$ and U_d are unitaries on $\mathcal{H}$, then the corresponding outer feature is

$$\boxed{\sum_N N^{-1-2a} \left\| \sum_{\substack{d|N \\ d \leq Y}} d^a U_d b_d \right\|_{\mathcal{H}}^2 \geq 0.} \quad (50)$$

Thus the exact packet translations, including the full smooth packet profile, may be retained inside the common positive analysis map. Hence, before scalar contraction, the completed signed principal dependence is radial and the outer arithmetic factor is one common positive congruence.

If $d = gh$, $e = gk$, $(h, k) = 1$, and $m = k\ell$, $n = h\ell$, then

$$\boxed{\ell = \frac{Ng}{de}.} \quad (51)$$

Consequently, if this auxiliary finite family is parametrized by $Y = L^\kappa$ for a fixed $\kappa > 0$, then

$$0 \leq \frac{\log N - \log \ell}{L} = \frac{\log(de/g)}{L} \ll_\kappa \frac{\log L}{L}. \quad (52)$$

The same $O(\log L/L)$ scale controls the p -dependence of the positive outer divisor map. Moreover $\mathcal{W}_a$ itself has an explicit finite Möbius inverse on the outer range: restricting its output to $N \leq Y$,

$$\boxed{c_d d^a = \sum_{e|d} \mu(d/e) (\mathcal{W}_a c)(e), \quad d \leq Y.} \quad (53)$$

Consequently the weighted map $c \mapsto (N^{-1/2-a}(\mathcal{W}_a c)(N))_{N \leq Y}$ is injective and has inverse norm $Y^{1+o(1)}$. The same inverse bound holds for the Hilbert-valued map in (50): if $S_N = \sum_{d|N} d^a U_d b_d$, then

$$d^a U_d b_d = \sum_{e|d} \mu(d/e) S_e,$$

so b_d is recovered by applying the unitary U_d^* and the scalar weight d^{-a} . The deformation of this positive divisor map has entrywise normalized size $O(\log L/L)$. To avoid relying on one-sided factorization in any later comparison, we measure it with the uniform two-sided arithmetic transfer bound of Lemma 9.3; hence

$$\boxed{O\left(L^{-1+10\kappa+o(1)} \log L\right) = o(1), \quad \kappa < \frac{1}{10}.} \quad (54)$$

For the invariant contour source, $d = e = 1$ and hence the divisor-map p -deformation, the endpoint displacement (52), and the outer annulus all vanish identically. For the auxiliary general finite family, the endpoint displacement is pair dependent and extends the stationary endpoint in the common N -coordinate; Proposition 6.9 records its separate sideband decomposition. None of these auxiliary losses is needed for the exact source specialization.

Proof. A base diagonal monomial is

$$c_d \overline{c_e} d^{-s} e^{-(1-s)} m^{-(s+a)} n^{-(1-s+a)}.$$

Using $dm = en = N$, equivalently $m = N/d$ and $n = N/e$, this is exactly

$$N^{-1-2a} c_d d^a \overline{c_e e^a}.$$

Summing first over divisors of N proves (49). More generally, replacing the scalar factors by $x_d = d^a U_d b_d \in \mathcal{H}$ gives

$$\sum_{d,e|N} \langle x_d, x_e \rangle = \left\| \sum_{d|N} x_d \right\|^2,$$

which proves (50). For the translated master the extra scalar factor is

$$d^{-\delta} e^{\delta} m^{-a-\delta} n^{-a+\delta} = N^{-2a} d^a e^a,$$

so the same divisor Gram is obtained identically. The critical- J source multiplies the same shifted product, and Proposition 6.7 shows that its primal and dual halves glue before scalar contraction; therefore its outer analysis map is the same one.

For the inverse statement, write $S_N = (\mathcal{W}_a c)(N)$. Then $S_N = \sum_{d|N} c_d d^a$ for $N \leq Y$, and ordinary Möbius inversion gives (53). Cauchy–Schwarz and the divisor bound $\tau(n) = n^{o(1)}$ give

$$\sum_{d \leq Y} |c_d|^2 \ll Y^{1+o(1)} \sum_{N \leq Y} |S_N|^2 \ll Y^{2+o(1)} \sum_{N \leq Y} N^{-1-2a} |S_N|^2,$$

which is the asserted $Y^{1+o(1)}$ inverse norm.

Finally $N = ghk\ell$ and $de = g^2hk$, proving (51). Since $d, e \leq Y$, one has $de/g \leq Y^2$, giving (52). In particular, the physical cutoff

$$\ell \leq Y_{\text{fib}} e^{-|v|}$$

is equivalent in the common-divisor coordinate to

$$N \leq \frac{de}{g} Y_{\text{fib}} e^{-|v|}. \quad (55)$$

Thus the pair-independent range

$$N \leq Y_{\text{fib}} e^{-|v|} \quad (56)$$

is a genuine common sub-prefix for every arithmetic pair; the difference between (55) and (56) is an outer annulus, not a shortening of the common core. After applying Lemma 6.6, the raw lag v is straightened to the universal packet lag w . Imposing the common core then simply truncates the outer sum in (49):

$$\sum_{N \leq Y_{\text{fib}} e^{-|w|}} N^{-1-2a} \left| \sum_{\substack{d|N \\ d \leq Y}} c_d d^a \right|^2 \geq 0. \quad (57)$$

Thus the passage to the pair-independent core preserves the exact positive arithmetic congruence; only the discarded annulus is treated separately. The same truncation also preserves the coordinates needed for the finite Möbius inverse. Indeed, on this stable core,

$$|w| \leq W_T,$$

while $\log Y_{\text{fib}} = L + O(1)$. Hence uniformly on the retained core

$$Y_{\text{fib}} e^{-|w|} \geq L^{A_0+o(1)}. \quad (58)$$

For this auxiliary comparison one may, after A_0 is fixed, choose any $0 < \kappa < \min\{1/10, A_0 - 1\}$ and set $Y = L^\kappa$. Then the right side exceeds Y for all sufficiently large T . Consequently every coordinate $1 \leq N \leq Y$ used in (53) remains present after the common-core restriction, and the inverse bounds for $\mathcal{W}_a$ and $\widetilde{\mathcal{W}}_a$ remain valid. Moreover $d^a = \exp(p \log d/L)$, so each fixed p -derivative of the positive divisor map costs at most a fixed power of $\log Y/L$. Applying the uniform two-sided transfer estimate (132) gives (54). This optional auxiliary choice is not part of the invariant-source parameter hierarchy. $\square$

Proposition 6.9 (Auxiliary outer common-prefix excess). *For the invariant contour source this contribution is identically zero by Proposition 6.1. For a general auxiliary retained outer pair (d, e) , first restrict to the stable logarithmic band $0 \leq u \leq W_T$. On this band decompose the actual N -range (55) into the common core (56) and the excess annulus*

$$Y_{\text{fib}} e^{-|v|} < N \leq \frac{de}{g} Y_{\text{fib}} e^{-|v|}. \quad (59)$$

Let A_T^{out} denote this stable-band annulus and its adjoint; the part on $W_T < u \leq L$ belongs only to (149). After returning to the physical stationary carrier $\ell = Ng/(de)$, expand the smooth unit-shell stationary symbol in Fourier sidebands and write

$$A_T^{\text{out}} = E_T^{\text{out}} + R_T^{\text{out,hi}}, \quad (60)$$

where E_T^{out} contains the sidebands $|r| \leq R_j^{\text{sb}}$ and $R_T^{\text{out,hi}}$ the complementary sidebands. Then

$$\text{rank } E_T^{\text{out}} \ll TY^{3+o(1)} = o(N_T). \quad (61)$$

The term $R_T^{\text{out,hi}}$ belongs to the direct high-sideband regular class (vii) of Proposition 12.4. No estimate of the annulus width is needed for the low-sideband rank bound.

Proof. Equation (51) maps the annulus back to positive integer carriers ℓ satisfying the original stationary saddle relation $t_* = 2\pi\ell y$. If $y \in [J_j, J_j + 1]$ and $t_* \in [T, 2T]$, then

$$\frac{T}{2\pi(J_j + 1)} \leq \ell \leq \frac{2T}{2\pi J_j},$$

which is precisely the stationary carrier interval used in Section 11, up to the fixed shell buffer already present there. Lemma 5.1 gives a smooth symbol uniformly on the annulus. Its low Fourier sidebands replace ℓ by effective shifts within R_j^{sb} of this stationary interval. Proposition 11.1 therefore places all low-sideband terms in the same shell-edge spaces and gives total geometric rank $O(T)$ per fixed arithmetic multiplicity. Multiplying by the conservative $Y^{3+o(1)}$ outer label count proves (61). The high sidebands are not included in this rank assertion; their coefficient square sum is no larger than that of the full direct family and their Fourier decay is estimated in class (vii). $\square$

For a packet pair r, s , let I_r and I_s be the two physical logarithmic prefix intervals determined by the packet displacement and put

$$J_{rs} := I_r \cap I_s, \quad \rho_{rs}^{(T)} := |J_{rs}|.$$

In the invariant-source specialization there is no outer translation and $N = \ell$. Propositions 6.7 and 6.5 show that the principal full-fibre scale is the local archimedean height $Y_{\text{fib}} = X_{\text{arch}} = \tau/(2\pi)$. Hence

$$\nu_T := \frac{\log Y_{\text{fib}}}{L} = \lambda(\tau) = 1 + O(L^{-1}), \quad \nu_T \geq 1$$

on the dyadic interval. The actual normalized common-prefix endpoint is

$$\boxed{\rho_{rs}^{(T)} = \nu_T - \frac{|w_r - w_s|}{L}}. \quad (62)$$

The frozen global model replaces only the scalar full-fibre length ν_T by 1; this $O(L^{-1})$ endpoint deformation is estimated in Lemma 8.4. In the finite- T local formulas below we write ρ_{rs} for $\rho_{rs}^{(T)}$. Since $d = e = g = 1$ in the invariant source, there is no additional arithmetic endpoint displacement. The intersection $I_r \cap I_s$ is formed at the packet level before the local Perron residue is evaluated. Thus ρ_{rs} is determined by the global packet geometry rather than introduced as an independent positivity parameter.

Before the u -residue and the three shift residues are taken, the packet pair contributes one joint analytic weight on the pair-independent common N -core. Here t denotes the normalized logarithmic N -coordinate after translating that core interval to its local origin; it is not the physical height variable. Thus

$$W_{rs,L}(u, \mathbf{z}) = \int_{J_{rs}} a(t, \mathbf{z}) e^{Lut} dt, \quad (63)$$

where a contains the smooth packet, slow archimedean, and fixed shift factors. For every fixed multi-index ν ,

$$|\partial_{\mathbf{z}}^{\nu}[u^n] W_{rs,L}(u, \mathbf{z})| \ll_{\nu} \frac{(CL)^n (1+n)^C}{n!}. \quad (64)$$

If

$$\mathcal{F}(u, \mathbf{z}) = \sum_{j=-J}^{\infty} F_j(\mathbf{z}) u^j,$$

then

$$\text{Res}_{u=0}(W\mathcal{F}) = \sum_{n=0}^{J-1} [u^n] W F_{-n-1}.$$

This is a finite sum, hence the u -residue and the finite three- Z shift residues commute exactly:

$$\boxed{\text{Res}_{\mathbf{z}} \text{Res}_{u=0}(W\mathcal{F}) = \text{Res}_{u=0} \text{Res}_{\mathbf{z}}(W\mathcal{F})}. \quad (65)$$

The standard auxiliary factor in the completed three- Z master cancels the moving divisors $2u = z_2 - z_1$ and $2u = z_2 - z_3$; multiplication by the holomorphic packet factor (63) does not recreate them.

For the basic local pole feature,

$$\int_{J_{rs}} e^{-pt} dt$$

becomes, after translation of J_{rs} to $[0, \rho_{rs}]$,

$$E_{\rho_{rs}}(p) := \int_0^{\rho_{rs}} e^{-pt} dt.$$

This explains the appearance of E_ρ in the local model without asserting that a pointwise fixed- ρ positive form globalizes to a positive packet matrix.

Proposition 6.10 (Completed principal local profile). *Let $\tau \asymp T$ be the physical height and let τ_b be the reference height of its smooth block, of length $H = T/L^D$. Write*

$$\lambda(\tau) := \frac{\log(\tau/2\pi)}{L}, \quad \mu_b := \frac{2F_b}{L},$$

where F_b is the frozen value of the half-self factor f on that block. Stirling's formula and the mean-value theorem give, uniformly for $|\tau - \tau_b| \ll H$,

$$\mu_b = \lambda(\tau) + O(L^{-D-1}) + O((TL)^{-1}). \quad (66)$$

Let ρ be the pair-independent common N -prefix endpoint. Before taking the normalized curvature coefficient, the completed model principal source has local profile

$$\boxed{\mathcal{G}_{\lambda(\tau), \mu_b, \rho}(p) = e^{\lambda(\tau)p} \left[\mu_b E_\rho(2p) - 2E_\rho(p)^2 + e^{2\lambda(\tau)} E_\rho(p+2) E_\rho(p-2) \right]}. \quad (67)$$

Consequently the frozen profile $\lambda = \mu_b = 1$ is

$$\boxed{G(\rho) = [p^2] e^p \left[E_\rho(2p) - 2E_\rho(p)^2 + e^2 E_\rho(p+2) E_\rho(p-2) \right]}. \quad (68)$$

Proof. Use the completed per-orientation source $H_{\text{mod}} = F_b - 2P + P(s - 2/L)$ from (29). By Proposition 6.1, all auxiliary outer indices are trivial, so the common-product coordinate reduces to $N = \ell$ and there is no outward arithmetic annulus. The common-prefix projection is taken before the local residue, so after translating the normalized logarithmic carrier coordinate every principal fibre integral is over the same interval $0 \leq t \leq \rho$.

The self term is read after the two Hermitian orientations are reassembled. Before freezing, the full J source is $2f = -\chi'/\chi$; on the block under consideration its principal frozen value is $2F_b$. The

local pole extraction has the same common positive L^2 Perron amplitude as the other principal faces, so after this common amplitude is factored out the self coefficient is exactly

$$\frac{2F_b}{L} = \mu_b.$$

Proposition 6.7 gives the full glued fibre and Proposition 6.5 keeps the explicit Hardy shift factor at the physical height. Hence the normalized full- J contribution is

$$\mu_b e^{\lambda(\tau)p} E_\rho(2p).$$

The approximation (66) is used only in the later comparison with the frozen model; no factor 2 is absorbed into the definition of (67). The two unshifted logarithmic-derivative faces lie on the common diagonal (39). Their two one-sided fibre integrals are both $E_\rho(p)$; the coefficient $-2P$ therefore gives

$$-2e^{\lambda(\tau)p} E_\rho(p)^2.$$

Finally the translated source $P(s - 2/L)$ contributes the internal shifts $+2$ and -2 in the two diagonal fibre variables. The explicit functional-equation factor is archimedean, hence $X^{2/L} = e^{2\lambda(\tau)}$ by Proposition 6.5. Its contribution is therefore

$$e^{\lambda(\tau)p} e^{2\lambda(\tau)} E_\rho(p+2) E_\rho(p-2).$$

Adding the three contributions proves (67). Setting $\lambda = \mu_b = 1$ and extracting the coefficient of p^2 gives (68). By Proposition 6.1, no auxiliary outer arithmetic label is present in this invariant-source calculation. $\square$

Lemma 6.11 (The local projector contracts the internal Dirichlet fibre). *Fix a packet pair. In the invariant-source specialization the common shifted-zeta and packet analysis maps are fixed before the internal HLP index is exposed; there is no separate outer arithmetic map. After the common diagonal is imposed, every exact HLP level has a radial Dirichlet series*

$$\mathcal{D}_k(z) = \sum_{M \geq 1} a_{k,M} M^{-z}.$$

The index M is a summation index of the contour coefficient, not a coordinate of the packet Hermitian space. For $k \leq K_0$, the actual packet Mellin weight W enters the scalar residue functional

$$\boxed{\mathcal{L}_{\text{pr}, \leq K_0}^{\text{ex}}[W] = \sum_{0 \leq k \leq K_0} \omega_{k,L} \text{Res}_{u=0}(\mathcal{F}_{k,L}(1+u; p) W(u)).} \quad (69)$$

Thus no vector indexed by M remains in the exact low-level common principal fibre. The additional translated residues in (89) are not part of (69); they remain direct one- χ terms and retain their M -carriers. The levels $k > K_0$ are likewise not acted on by this projector and retain their direct M -carriers as in Lemma 7.5. Replacing the level factors by their leading Laurent jets and extending the resulting model sum over all k gives the model projector (85) and the radial profile of Proposition 6.10. The extension beyond K_0 is $K_{\text{lead}, \infty}^{\text{pkt}} - K_{\text{lead}, \leq K_0}^{\text{pkt}}$; the signed correction required in the exact-minus-model decomposition is its negative, namely (99).

Proof. On the safe spectral right edge the internal Dirichlet sums and the geometric HLP hierarchy are absolutely convergent. After stationary phase the packet cutoff is represented by W . For the exact source, perform the local contour move only for $k \leq K_0$. Lemma 7.8 permits the internal

Dirichlet sum and contour integral to be interchanged levelwise. Taking the $u = 0$ component of (89) and summing over $k \leq K_0$ gives exactly (69); the translated-residue component remains in the direct one- χ decomposition. The levels $k > K_0$ remain on the direct right-edge representation and therefore retain their M -carriers.

For the model comparison, replace the level factors by their leading Laurent jets. The resulting infinite model series is absolutely convergent and Lemma 7.9 gives the self, base, and translated coefficients of Proposition 6.10, hence $\mathcal{G}_{\lambda, \mu_b, \rho}$. Its portion beyond K_0 is the radial model truncation error (99); the exact high-level carriers remain in the direct decomposition of Lemma 7.5. $\square$

7 Exact HLP levels and the smooth packet local replacement

Fix once and for all a spectral right edge

$$\Re s = 1 + \delta, \quad \delta > 0.$$

All Dirichlet expansions in this section are first made on this line. Absolute convergence gives $P(s) = O_\delta(1)$ and $Q(s) = O_\delta(1)$, while $f(s) = \frac{1}{2}L + O_\delta(1)$ on $t \asymp T$. Hence

$$\left| \frac{P(s)}{f(s)} \right| \ll_\delta L^{-1},$$

The coefficient organization is the one appearing in the local-extrema analysis of Hughes–Lugmayer–Pearce–Crump [7], but no RH-dependent estimate from that paper is used here. Indeed, on the present fixed safe line the exact identity

$$\boxed{\frac{Q(s)}{f(s) - P(s)} = \sum_{k \geq 1} f(s)^{-k} P(s)^{k-1} Q(s)} \quad (70)$$

follows directly from the geometric series, and converges absolutely and uniformly because $|P/f| \ll_\delta L^{-1}$. Thus the levelwise expansion used below is unconditional before any contour movement. Since $f'(s) = O(T^{-1})$, its separated f' contribution is also $O_\delta((TL)^{-1})$ on this line. The exact multiplier at level k changes only the Dirichlet coefficient sequence. If the total convolution index is M , the stationary phase remains

$$t_* = 2\pi My, \quad \text{carrier} = e(-My). \quad (71)$$

Thus exact levels do not introduce a new moving support variable. Their packet weights remain in the finite-interval Laplace/Mellin class (63).

7.1 Uniform factorial control of the HLP hierarchy

For $k \geq 0$ define the nonnegative Dirichlet-convolution coefficients Λ_k by

$$P(s)^k = \sum_{n \geq 1} \frac{\Lambda_k(n)}{n^s}, \quad \Lambda_0(1) = 1,$$

and for $k \geq 1$ let α_k be the coefficient sequence of $P^{k-1}Q$.

Lemma 7.1 (Derivative identity and support). *For every $k \geq 1$,*

$$\boxed{\alpha_k(n) = \frac{\log n}{k} \Lambda_k(n), \quad \Lambda_k(n) \neq 0 \implies n \geq 2^k.} \quad (72)$$

Proof. Differentiate the identity for P^k . Since $P' = -Q$,

$$-\sum_n (\log n) \Lambda_k(n) n^{-s} = k P^{k-1} P' = -k P^{k-1} Q = -k \sum_n \alpha_k(n) n^{-s}.$$

Coefficient comparison gives the first identity. Every nonzero term of the k -fold convolution defining Λ_k is a product of integers at least 2, giving the support statement. $\square$

Proposition 7.2 (Uniform factorial square-sum majorant). *There are absolute constants $A, C > 0$ such that, for every $x \geq 2$ and $k \geq 1$,*

$$\sum_{n \leq x} \alpha_k(n)^2 \leq C \frac{A^k}{k!} x (\log(ex))^{2k+1}. \quad (73)$$

Proof. Put

$$S_k(x) := \sum_{n \leq x} \Lambda_k(n)^2.$$

The convolution identity

$$\Lambda_k(n) = \sum_{d|n} \Lambda(d) \Lambda_{k-1}(n/d)$$

and $\sum_{d|n} \Lambda(d) = \log n$ give by weighted Cauchy–Schwarz

$$\Lambda_k(n)^2 \leq (\log n) \sum_{d|n} \Lambda(d) \Lambda_{k-1}(n/d)^2.$$

Hence

$$S_k(x) \leq (\log x) \sum_{d \leq x} \Lambda(d) S_{k-1}(x/d). \quad (74)$$

Chebyshev’s estimate $\psi(y) = \sum_{n \leq y} \Lambda(n) \ll y$ and Stieltjes integration by parts imply, uniformly for $0 \leq v \ll \log(ex)$,

$$\sum_{d \leq x} \frac{\Lambda(d)}{d} \left(\log \frac{ex}{d} \right)^v \ll \frac{(\log(ex))^{v+1}}{v+1}. \quad (75)$$

Indeed, apply partial summation to $d^{-1}(\log(ex/d))^v$ and use $\psi(t) \ll t$.

Inducting in (74), starting from $S_1(x) \ll x \log(ex)$, and applying (75) with $v = 2k - 3$ gives

$$S_k(x) \leq C \frac{A^{k-1}}{(k-1)!} x (\log(ex))^{2k-1}.$$

If $S_k(x) \neq 0$, Lemma 7.1 gives $2^k \leq x$, so the required condition $v \ll \log(ex)$ is automatic. Finally

$$\sum_{n \leq x} \alpha_k(n)^2 \leq \frac{(\log x)^2}{k^2} S_k(x),$$

and enlarging A, C proves (73). $\square$

Lemma 7.3 (Quadratic bound for completed direct carriers). *After the common shifted-zeta factor has been placed in the completed analysis map and the deterministic stationary/shell normalization defining $\mathcal{K}_{j,T}$ has been applied, let $c_m^{\text{dir}}(p)$ denote the remaining normalized coefficient of a direct one- χ carrier. For the ordinary logarithmic-derivative term, the exact HLP hierarchy, and their translated variants, uniformly for $x \leq T$ and $0 \leq q \leq 2$,*

$$\boxed{\sum_{m \leq x} \left| \partial_p^q c_m^{\text{dir}}(p) \right|^2 \ll x L^{C_{\text{dir}}}.} \quad (76)$$

The same bound holds for any subset of the HLP levels, in particular $k > K_0$, for any support restriction of these direct carriers, and uniformly for the translated principal-part subfamilies arising from the locally projected levels $k \leq K_0$ in (82).

Proof. By Proposition 6.1, no nontrivial outer arithmetic map remains. The direct terms considered here are precisely the terms for which Lemma 6.11 leaves the radial carrier M uncontracted: the additional translated residues and the exact levels above K_0 retain their direct carrier. Thus for $k \geq 1$ the changing internal multiplier is exactly $f^{-k} P^{k-1} Q$, with coefficient $f^{-k} \alpha_k(m)$. Since $|f| \asymp L$ and $\log(ex) \ll L$ for $x \leq T$, (73) gives

$$\left(\sum_{m \leq x} |f^{-k} \alpha_k(m)|^2 \right)^{1/2} \ll (xL)^{1/2} \frac{C^k}{\sqrt{k!}}.$$

A fixed p -derivative introduces only a polynomial in k and factors $(\log m)/L = O(1)$. Minkowski's inequality and $\sum_k C^k (1+k)^C / \sqrt{k!} < \infty$ therefore give the same bound after summing any subset of levels. The ordinary $k = 0$ coefficient is controlled by $\sum_{m \leq x} \Lambda(m)^2 \leq (\log x) \psi(x) \ll xL$; translated factors $m^{O(1/L)}$ are uniformly bounded. A support restriction only sets some of the Hilbert-valued coefficient coordinates to zero and therefore cannot increase the square sum. For a fixed level k , each additional translated principal part is obtained by a finite residue functional on a normalized local contour and does not introduce a new carrier index. Cauchy's estimate and the fixed-factor bounds used in (94) give at most $L^C C_1^k (1+k)^C$ beyond the level coefficient bound. Hence Minkowski's inequality reduces the complete translated family to the convergent series

$$\sum_{k \geq 1} \frac{C_2^k (1+k)^C}{\sqrt{k!}} < \infty.$$

Thus the translated principal-part contribution is uniform in K_0 and obeys (76) after enlarging C_{dir} . $\square$

Lemma 7.4 (Uniform freezing of the slow half-self factor). *Let a smooth height block be supported on $|t - \tau_b| \ll H = T/L^D$ on the fixed right edge $\Re s = c > 1$, and let $F_b = f(1/2 + i\tau_b)$ be as in (27). Then*

$$\boxed{|f(c + it) - F_b| \ll L^{-D}, \quad |f(c + it)^{-k} - F_b^{-k}| \ll k L^{-k-D-1} \quad (k \geq 1).} \quad (77)$$

After restricting the HLP hierarchy to $k \leq K_0$ and applying the local principal projector, there is a fixed C_{fr} such that the normalized local kernels obtained from (26) and (28) satisfy, for the derivative orders used below,

$$\boxed{\partial_\rho^r \partial_p^q (K_{\text{var},b} - K_{\text{fr},b}) = O(L^{-D-1+C_{\text{fr}}}), \quad 0 \leq r, q \leq 2.} \quad (78)$$

The same estimate holds for the corresponding diagonal derivative jump.

Proof. Stirling's formula on a fixed vertical line gives $f(c + it) = \frac{1}{2} \log(t/2\pi) + O(T^{-1})$ and $\partial_t f(c + it) = O(T^{-1})$. Moreover $f'(s) = O(T^{-1})$ uniformly on the bounded horizontal segment joining $1/2 + i\tau_b$ to $c + i\tau_b$. Therefore

$$|f(c + it) - F_b| \ll \frac{|t - \tau_b|}{T} + T^{-1} \ll L^{-D}.$$

Since both $f(c + it)$ and F_b have modulus $\asymp L$, the mean-value theorem applied to $z \mapsto z^{-k}$ gives the second estimate in (77).

The normalized self term contains the factor $2f/L$, so its freezing error is $O(L^{-D-1})$. At HLP level k , every occurrence of the slow factor is multiplied by f^{-k} ; after a fixed number of stationary and packet derivatives the loss is bounded by $L^{C_{\text{fr}}}(1+k)^{C_{\text{fr}}}$. Proposition 7.2 therefore bounds the required sum uniformly in K_0 , since

$$\sum_{1 \leq k \leq K_0} \frac{R^k (1+k)^{C_{\text{fr}}+1}}{\sqrt{k!}} \leq \sum_{k \geq 1} \frac{R^k (1+k)^{C_{\text{fr}}+1}}{\sqrt{k!}} < \infty.$$

The finite Cauchy and curvature factors have uniform derivatives on the fixed normalized shift polydisc. This proves (78), including the one-sided diagonal derivatives. $\square$

Lemma 7.5 (High exact HLP levels remain direct sideband terms). *Let $K_0 = L^{1/8}$. Keep the stationary levels $k > K_0$ in their original pointwise- $f(t)$, absolutely convergent safe-line Dirichlet representation; no block freezing or local Perron projector is applied to them. On a retained unit shell put $d_j \asymp T/J_j$ and*

$$R_j^{\text{sb}} := \epsilon_L d_j,$$

with $\epsilon_L = L^{-\sigma}$ fixed in Section 13. Expand the smooth stationary amplitude in shell Fourier modes r . The part $|r| \leq R_j^{\text{sb}}$ and its adjoint define $E_T^{\text{HLP,hi}}$ with

$$\boxed{\text{rank } E_T^{\text{HLP,hi}} \ll T = o(N_T).} \tag{79}$$

Write $R_T^{\text{HLP,hi}}$ for the complementary part $|r| > R_j^{\text{sb}}$; it is disjoint from $E_T^{\text{HLP,hi}}$ and is estimated later as regular class (vii).

Proof. The safe-line geometric expansion converges absolutely. Each level changes only the coefficient of the total index M , and Lemma 7.7 gives carrier $e(-My)$ with $t_* = 2\pi My$. Lemma 5.1 is uniform over the summed exact HLP hierarchy, so its shell amplitude has the same Fourier-sideband decay used in class (vii). The low sidebands replace M by the effective carrier $M - r$ and hence lie in the fixed edge spaces of Proposition 11.1. By Proposition 6.1 there is no additional arithmetic multiplicity, which gives (79). The complementary sidebands are exactly the high-sideband class (vii). $\square$

7.2 Canonical local pole projection and the smooth packet Mellin class

The principal local projection is defined levelwise. Expand the exact resolvent geometrically on the absolute-convergence right edge and let $\mathcal{F}_{k,L}(1+u; p)$ denote one completed retained level after the finite three- Z /Cauchy completion has been formed. Its common principal local singularity is the complete Laurent part at $u = 0$, including the actual multiplicity created by coalescing faces. Define

$$\Pi_0^{(k)} \mathcal{F}_{k,L} := \text{PP}_{u=0} \mathcal{F}_{k,L}. \tag{80}$$

If that completed level has additional translated one- χ poles separated from the common local cluster, let $\Sigma_{\text{tr},k}$ be their finite set. Its cardinality is bounded independently of k : these poles arise only from the fixed three- Z /Cauchy completion, while the HLP level changes the pole multiplicity at the common point $u = 0$. Put

$$\Pi_{\Sigma_{\text{tr},k}} F := \sum_{u_0 \in \Sigma_{\text{tr},k}} \text{PP}_{u=u_0} F. \quad (81)$$

Then the exact level has the mutually exclusive decomposition

$$\boxed{\mathcal{F}_{k,L} = \Pi_0^{(k)} \mathcal{F}_{k,L} + \Pi_{\Sigma_{\text{tr},k}} \mathcal{F}_{k,L} + \mathcal{F}_{k,L}^{\text{hol}}}, \quad (82)$$

where $\mathcal{F}_{k,L}^{\text{hol}}$ is holomorphic throughout the local neighbourhood. Identity (82) is algebraic for every k , but the exact source decomposition applies its local projectors only for $k \leq K_0$. Thus the exact locally projected functional is the finite sum

$$\mathcal{L}_{\text{pr}, \leq K_0}^{\text{ex}}[W] := \sum_{0 \leq k \leq K_0} \omega_{k,L} \text{Res}_{u=0}(\mathcal{F}_{k,L}(1+u; p)W(u)). \quad (83)$$

The levels $k > K_0$ are not projected and remain in the direct sideband decomposition of Lemma 7.5.

For comparison with the positive model, replace the leading Laurent factor at each level and resum the resulting *model* hierarchy over all k . This is not a second extraction of the exact high-level source. Lemma 7.9 then produces the model cluster

$$\boxed{\Sigma_0^{\text{mod}} = \{0, 2/L\}}, \quad (84)$$

with residues $-2, +1$. For the resummed model define

$$\Pi_{\Sigma_0^{\text{mod}}} F := \sum_{u_0 \in \{0, 2/L\}} \text{PP}_{u=u_0} F. \quad (85)$$

Thus the translated model pole belongs to the *model* principal projector, while the algebraic point $u_*(F_b)$ of Lemma 4.2 serves only to identify the same translated scale. The levelwise decomposition therefore separates the common principal cluster, the additional translated poles, and the holomorphic remainder into mutually exclusive contributions.

Lemma 7.6 (Smooth bounded-support Mellin weights). *Let $0 \leq t \leq C_*$ with C_* fixed. Assume that for some fixed derivative order R_* , large enough for the applications below,*

$$\|\partial_t^m \partial_\rho^r \partial_p^j a_{\rho,p,T}\|_{L^1(0,C_*)} \leq L^{A_*+o(1)} \quad (m+r+j \leq R_*).$$

Define

$$W_{L,\rho,p}(u) := \int_0^{C_*} a_{\rho,p,T}(t) e^{Lut} dt = \sum_{n \geq 0} w_n(\rho, p) u^n. \quad (86)$$

Then

$$\boxed{|\partial_\rho^r \partial_p^j w_n(\rho, p)| \leq L^{A_*+o(1)} \frac{(C_* L)^n}{n!}}. \quad (87)$$

The same conclusion holds for a moving endpoint $\int_0^\rho a_{\rho,p,T}(t) e^{Lut} dt$, and the class is stable under every fixed finite product of such weights.

Proof. Expanding the exponential and using Fubini gives

$$w_n(\rho, p) = \frac{L^n}{n!} \int_0^{C_*} a_{\rho, p, T}(t) t^n dt,$$

which proves (87). The one-dimensional $W^{1,1}$ inequality on the fixed interval controls every endpoint value by the corresponding L^1 norm together with one t -derivative. Repeated ρ -differentiation of a moving endpoint therefore produces only finitely many controlled endpoint jets, beginning with

$$a_{\rho, p, T}(\rho) \frac{(L\rho)^n}{n!},$$

and integrals of the same form. Finally, products correspond to convolution of the compactly supported t -amplitudes, so the support remains bounded and the same factorial estimate holds with a larger fixed C_* . $\square$

Lemma 7.7 (Exact HLP levels preserve the packet radial variable). *On the absolute-convergence right edge, multiplication of finitely many Dirichlet series changes only the coefficient of the total Dirichlet index M . After insertion of the physical packet lag and the outer-denominator dilation, the corresponding one- χ term has saddle and phase*

$$t_* = 2\pi My, \quad e(-My).$$

Thus exact HLP levels do not create an additional packet-support coordinate; their packet/common-prefix weights belong to the class of Lemma 7.6.

Proof. If $A_1(s) \cdots A_r(s) = \sum_M c_M M^{-s}$, then after the exact logarithmic translation the s -dependence is $(Me^{-u})^{-s}$ up to a scalar amplitude. The one- χ stationary calculation therefore depends on the single total index M , giving the displayed saddle and phase. A translated factor such as $P(s - 2/L)$ merely multiplies its coefficient at M by $M^{2/L}$ and does not change the stationary variable. $\square$

Lemma 7.8 (Cluster contour and levelwise residues). *Fix a sufficiently large constant $C_* > 4$. Since the completed three- Z /Cauchy construction has only finitely many normalized translated shifts, C_* may be chosen once so that every local translated pole in $\Sigma_{\text{tr}, k}$, uniformly in k , lies inside*

$$\Gamma_L := \{u : |u| = C_*/L\}.$$

For all large T , Γ_L also encloses $u = 0$, the algebraic locator $u_(F_b)$ of Lemma 4.2, and the model point $2/L$, while*

$$\sup_{u \in \Gamma_L} \left| \frac{P(1+u)}{F_b} \right| \leq q < 1 \tag{88}$$

with a fixed q . Hence, for every packet Mellin weight W analytic inside Γ_L , the boundary integral of the bare resummed quotient equals the absolutely convergent sum of the geometric-level contour integrals, each of which has only the pole at $u = 0$. After the fixed completed Cauchy/shift factors are adjoined, termwise contour integration remains valid, but any additional enclosed translated poles are retained explicitly. For $k \leq K_0$ one has

$$\frac{1}{2\pi i} \int_{\Gamma_L} W(u) \mathcal{F}_{k,L}(1+u; p) du = \text{Res}_{u=0}(W \mathcal{F}_{k,L}) + \sum_{u_0 \in \Sigma_{\text{tr}, k}} \text{Res}_{u=u_0}(W \mathcal{F}_{k,L}). \tag{89}$$

The holomorphic complement contributes zero. Exact local projectors are used only for $k \leq K_0$, while the full infinite leading sum is used only on the model side. No resummed exact quotient is transported from the original Perron line through u_ .*

Proof. Uniformly for $|u| \leq r_0$, $P(1+u) = u^{-1} + O(1)$, while $F_b = L/2 + O(1)$. Thus on Γ_L ,

$$|P(1+u)/F_b| \leq 2/C_* + o(1),$$

which proves (88) after increasing C_* . The algebraic locator $u_* = 2/L + O(L^{-2})$ lies inside this circle. On Γ_L the identity

$$\frac{Q}{F_b - P} = \sum_{j \geq 0} \frac{QP^j}{F_b^{j+1}}$$

therefore converges uniformly. Multiplication by the analytic W preserves uniform convergence, so contour integration may be interchanged with the sum. Every bare geometric term on the right is meromorphic inside Γ_L only at $u = 0$. Cauchy's theorem therefore identifies the boundary integral of the bare resummed quotient with the sum of these complete levelwise residues. After the fixed completed factors are restored, uniform convergence on Γ_L still permits termwise integration. Cauchy's theorem then adds exactly the residues of the finitely many enclosed translated poles, giving (89); the remaining part is holomorphic and integrates to zero. This equality is used only after the safe levelwise deformation and is not used to transport the resummed quotient from the original Perron line. The model cluster $(0, 2/L)$ is enclosed by the same circle, so the resummed leading family is represented on the same local contour. $\square$

Lemma 7.9 (Leading HLP jet resums to the translated model cluster). *At the principal Laurent level*

$$P(1+u) \rightsquigarrow u^{-1}, \quad Q(1+u) \rightsquigarrow u^{-2},$$

the complete geometric hierarchy in $Q/(F_b - P)$ resums exactly as

$$\boxed{\sum_{k \geq 1} F_b^{-k} u^{-k-1} = -\frac{1}{u} + \frac{1}{u - F_b^{-1}}.} \quad (90)$$

Consequently the resummed leading Laurent family of the completed source has generating principal part

$$\boxed{-\frac{2}{u} + \frac{1}{u - F_b^{-1}}.} \quad (91)$$

Since $F_b = L/2 + O(1)$,

$$F_b^{-1} = \frac{2}{L} + O(L^{-2}),$$

so (91) is precisely the finite- T precursor of the model principal parts $-2/u + 1/(u - 2/L)$ in (29). The $O(L^{-2})$ displacement of the translated pole is one-log lower after the common L^2 Perron normalization and is included in the exact-model local comparison below.

Proof. The geometric expansion on Γ_L is

$$\frac{Q}{F_b - P} = \sum_{k \geq 1} F_b^{-k} QP^{k-1}.$$

Replacing only the common principal Laurent factors gives the series on the left of (90). Summing it as a geometric series gives

$$\frac{1}{F_b u^2} \frac{1}{1 - (F_b u)^{-1}} = \frac{1}{u(F_b u - 1)} = -\frac{1}{u} + \frac{1}{u - F_b^{-1}}.$$

Adding the principal part $-u^{-1}$ of the separate $-P$ source proves (91). Multiplication by the common shifted-zeta, Cauchy, curvature, and packet factors does not alter this source-level pole identity. $\square$

Now write the independent Perron variable as $z = 1 + u$. Put

$$A(u) := uP(1+u), \quad B(u) := u^2Q(1+u), \quad C(u) := u^2\zeta(1+u)^2.$$

On a fixed disk $|u| \leq r_0$, the Laurent expansions at the zeta pole show that A, B, C are analytic, $A(0) = B(0) = C(0) = 1$, and all fixed derivatives are bounded. For $k \geq 1$ and $0 \leq q \leq 2$, differentiate the completed level q times with respect to the normalized curvature shift p and then set $p = 0$. A derivative falling on a factor shifted by $a = p/L$ contributes one factor L^{-1} and can raise the local pole order by one; derivatives falling on the normalized archimedean or packet factors do not raise the pole order and need not carry this factor. In particular the twice pole-raising contribution is fixed by the exact curvature identity (31):

$$\boxed{L^{-2}Q(1+u)^2P(1+u)^{k-1}\zeta(1+u)^2 = L^{-2}u^{-k-5}A(u)^{k-1}B(u)^2C(u).} \quad (92)$$

The fixed shift residues commute with these derivatives by (65). After the moving shift divisors have been canceled, Leibniz' rule and Cauchy's estimate therefore give, for $k \leq K_0$,

$$\boxed{\partial_p^q \mathcal{F}_{k,L}^{\text{ex}}(1+u; p) \Big|_{p=0} = \sum_{\ell=0}^q L^{-\ell} u^{-k-3-\ell} (H_{k,q,\ell,0} + u\Theta_{k,q,\ell,L}(u)), \quad 0 \leq q \leq 2.} \quad (93)$$

Here ℓ records the number of derivatives that actually raise the local pole order. For every fixed a used below,

$$\boxed{|H_{k,q,\ell,0}| + |\partial_u^a \Theta_{k,q,\ell,L}(u)| \ll L^{o(1)} C_H^k (1+k)^C, \quad 0 \leq \ell \leq q \leq 2.} \quad (94)$$

The separate ordinary face $k = 0$ has fixed pole order and is covered by the same finite estimate after enlarging the constant. The levels $k > K_0$ are not locally projected, by Lemma 7.5. Hence the constants in the local Laurent error are uniform through $k \leq L^{1/8}$; no fixed- k asymptotic is being summed. Package the three required curvature coefficients into the order-two intermediate leading Taylor jet

$$\boxed{\mathcal{F}_{k,L}^{\text{lead},(2)}(1+u; p) := \sum_{q=0}^2 \frac{p^q}{q!} \sum_{\ell=0}^q L^{-\ell} u^{-k-3-\ell} H_{k,q,\ell,0}.} \quad (95)$$

Its derivatives through order two at $p = 0$ are exactly the leading terms in (93). This jet is not resummed from arbitrary k -dependent coefficients: it is obtained by taking the same finite p -derivatives of the absolutely convergent leading geometric family in Lemma 7.9. Hence differentiation and geometric resummation commute, and the translated-pole parameter remains the one given there. No higher p -jet is used in the proof. The level weight obeys

$$\omega_{k,L} \ll C_0^k L^{-k}.$$

For $k \leq L^{1/8}$, write

$$W_{L,\rho,p}(u) = \sum_{n \geq 0} w_n(\rho, p) u^n.$$

The finite-interval Laplace representation gives, for $r, j \leq 2$,

$$|\partial_\rho^r \partial_p^j w_n(\rho, p)| \leq CL^n B^n \frac{(1+n)^C}{n!}. \quad (96)$$

Define the packet-local residue before curvature extraction by

$$\mathcal{L}_{k,L}^{\text{pkt}}(\rho, p) := \omega_{k,L} \text{Res}_{u=0} [\mathcal{F}_{k,L}(1+u; p) W_{L,\rho,p}(u)].$$

All p -derivatives in the estimates below are evaluated at $p = 0$. For a term in $\partial_p^q \mathcal{L}_{k,L}^{\text{pkt}}$ with q_1 derivatives falling on $\mathcal{F}_{k,L}$ and $q - q_1$ on W , decompose the first factor according to (93). Its error part is a finite sum over $0 \leq \ell \leq q_1$ of

$$\omega_{k,L} L^{-\ell} \text{Res}_{u=0} \left[u^{-k-2-\ell} \Theta_{k,q_1,\ell,L}(u) \partial_p^{q-q_1} W_{L,\rho,p}(u) \Big|_{p=0} \right].$$

Expanding $\Theta_{k,q_1,\ell,L} = \sum_j \theta_{k,q_1,\ell,j} u^j$ and using (94) and (96) gives, for each ℓ ,

$$\ll L^{-k-\ell+o(1)} C^k (1+k)^C \sum_{j=0}^{k+1+\ell} r_0^{-j} \frac{(BL)^{k+1+\ell-j} (1+k)^C}{(k+1+\ell-j)!}.$$

Because $\ell \leq q_1 \leq 2$ is fixed and $k/L = o(1)$, the same factorial comparison as above makes the sum uniformly bounded after factoring its $j = 0$ term. Hence, uniformly for $r, q \leq 2$,

$$\boxed{|\partial_\rho^r \partial_p^q \Delta \mathcal{L}_{k,L}^{\text{pkt}}(\rho, 0)| \ll L^{1+o(1)} \frac{C_1^k (1+k)^{C_1}}{(k+1)!}}. \quad (97)$$

Only the levels $k \leq K_0$ enter this local comparison. By Lemma 7.5, the higher exact levels retain their direct one- χ representation: their low sidebands belong to $E_T^{\text{HLP,hi}}$ and their high sidebands belong to regular class (vii).

The normalization is now explicit. For the component indexed by ℓ , the leading jet has pole order $k + 3 + \ell$, so residue extraction uses $w_{k+2+\ell}$. Its natural size is

$$L^{-k-\ell} \frac{L^{k+2+\ell}}{(k+2+\ell)!} = L^2 \frac{1}{(k+2+\ell)!}.$$

Thus every increase of the local pole order is paired with exactly the same power of L^{-1} from the normalized shift. In particular, the twice pole-raising curvature contribution in (92) uses w_{k+4} together with L^{-2} and still has the common L^2 Perron scale. Let $K_{\text{lead}, \leq K_0}^{\text{pkt}}$ be the sum of these leading jets for $k \leq K_0$. After the common positive Perron amplitude L^2 is divided out, (97) gives

$$\partial_\rho^r \partial_p^q (K_{\text{ex}, \leq K_0}^{\text{pkt}} - K_{\text{lead}, \leq K_0}^{\text{pkt}}) = O(L^{-1+o(1)}), \quad r, q \leq 2, \quad (98)$$

Here and in all derivative estimates through (101), the p -derivatives are evaluated at $p = 0$. Let $K_{\text{lead}, \infty}^{\text{pkt}}$ denote the full leading Laurent family before freezing the translated pole. The same calculation, together with (96) and (94), gives

$$\left| \partial_\rho^r \partial_p^q K_{\text{lead}, k}^{\text{pkt}} \right| \ll \frac{C^k (1+k)^C}{(k+2)!}, \quad r, q \leq 2,$$

after the common L^2 normalization. Therefore, for every fixed $A > 0$,

$$\boxed{\partial_\rho^r \partial_p^q (K_{\text{lead}, \leq K_0}^{\text{pkt}} - K_{\text{lead}, \infty}^{\text{pkt}}) = O_A(L^{-A}), \quad r, q \leq 2.} \quad (99)$$

This is a model-side radial error; no fast one- χ carrier remains in it. It remains to compare the full leading family with the fixed translated model. Put

$$c_b := \frac{L}{F_b} = \frac{2}{\mu_b} = 2 + O(L^{-1}).$$

Lemma 7.9 shows that the translated pole of the intermediate leading family is at $u = c_b/L$. After the common N -prefix has been fixed, the resummed leading residue functional is a finite combination of bounded-support Mellin weights and fixed Cauchy factors whose dependence on c_b is analytic for c_b in a fixed neighbourhood of 2; the pole remains a fixed positive normalized distance from the outer local contour. Lemma 7.6 and Cauchy's estimate therefore give a uniform bound for the c_b -derivative after up to two (ρ, p) derivatives. The mean-value theorem gives

$$\partial_\rho^r \partial_p^q (K_{\text{lead}, \infty}^{\text{pkt}} - K_{\text{mod}, \text{loc}}^{\text{pkt}}) = O(L^{-1}), \quad r, q \leq 2, \quad (100)$$

where $K_{\text{mod}, \text{loc}}^{\text{pkt}}$ uses the fixed model parameter $c = 2$, i.e. the pole $u = 2/L$. The three differences telescope exactly:

$$\begin{aligned} K_{\text{ex}, \leq K_0}^{\text{pkt}} - K_{\text{mod}, \text{loc}}^{\text{pkt}} &= (K_{\text{ex}, \leq K_0}^{\text{pkt}} - K_{\text{lead}, \leq K_0}^{\text{pkt}}) \\ &\quad + (K_{\text{lead}, \leq K_0}^{\text{pkt}} - K_{\text{lead}, \infty}^{\text{pkt}}) \\ &\quad + (K_{\text{lead}, \infty}^{\text{pkt}} - K_{\text{mod}, \text{loc}}^{\text{pkt}}). \end{aligned}$$

Hence (98), (99), and (100) give

$$\boxed{\partial_\rho^r \partial_p^q (K_{\text{ex}, \leq K_0}^{\text{pkt}} - K_{\text{mod}, \text{loc}}^{\text{pkt}}) = O(L^{-1+o(1)}), \quad r, q \leq 2.} \quad (101)$$

By Proposition 6.1, the invariant source has no nontrivial arithmetic whitening. Consequently the exact/model local discrepancy has normalized diagonal jump $O(L^{-1+o(1)}) = o(1)$, and its fixed mixed lag derivatives are bounded by a fixed power of L on every retained shell. The shell-local primitive criterion of Lemma 12.3, together with the metric comparison (159) and the choice $A_0 > C_{\text{reg}} + 3$ in (178), therefore gives

$$\boxed{\|\mathfrak{G}_T^{-1/2} R_T^{\text{loc}} \mathfrak{G}_T^{-1/2}\|_{S_2}^2 = o(N_T).} \quad (102)$$

Definition 7.10 (Finite- T principal operator). Let W_T be the stable logarithmic core length fixed in (47), and let C_T denote restriction of the straightened physical logarithmic coordinate to $0 \leq u \leq W_T$. After this restriction use the normalized lag coordinate $r = u/L$ and the unitary output rescaling $g(u) \mapsto L^{1/2}g(Lr)$. Define P_T to be the pullback, through the exact packet and diagonal maps, this core restriction, and this rescaling, of the *model* principal-part projector $\Pi_{\Sigma_0^{\text{mod}}}$ on the pair-independent common carrier core. By Proposition 6.1, the auxiliary arithmetic map specializes to its $d = 1$ coordinate and every outer-ratio translation is the identity for the invariant contour source. Thus P_T retains the finite- T parameters $\lambda(\tau)$ and μ_b , the common-core endpoint, and the actual packet/common-prefix weight of (63) on $0 \leq u \leq W_T$. No stationary Fourier-mode restriction is imposed inside this core. The exact levels $k > K_0$ are not part of P_T and

retain the direct sideband decomposition of Lemma 7.5; only their model leading tail is present through the full resummed projector. The complementary logarithmic tail $W_T < u \leq L$ is assigned by Proposition 10.3. The auxiliary outward arithmetic annulus of Proposition 6.9 is empty in the invariant-source specialization. Slow-parameter freezing and the low-level exact/model discrepancy are excluded from P_T .

Proposition 7.11 (Uniform-twisted packet source decomposition). *After the f' source, horizontal connectors, and nonstationary sector have been separated, restrict the stationary exact HLP/Hardy source to the stable logarithmic band $0 \leq u \leq W_T$. Here “stable” refers only to the logarithmic packet coordinate. By Proposition 6.1 the invariant source has no outward arithmetic annulus. On this stable band the same packet contour matrix has the disjoint decomposition*

$$\boxed{A_T^{\text{stat,ex,stable}} = P_T + E_T^{\text{one-}\chi} + R_T^{\text{src}}.} \quad (103)$$

Here $E_T^{\text{one-}\chi}$ is the direct physical one- χ family formed by the low-sideband parts of the high exact HLP levels, the additional translated principal parts, and the remaining direct stationary sources. The auxiliary outer-excess family is zero for the invariant contour source. Its rank is bounded by Proposition 11.1. Every complementary high-sideband part belongs instead to R_T^{src} , together with the other mutually exclusive regular source classes identified in the proof below. Their relative Hilbert–Schmidt bounds are established in Proposition 12.4.

Proof. Work first on the absolute-convergence right edge and separate the horizontal connectors and the nonstationary sector. The packet/common-prefix weight is kept exactly at this stage; no replacement by the flat plateau is made in the present source decomposition.

Expand the pointwise resolvent (26) geometrically on the safe right edge before any freezing. At level k , multiplication of Dirichlet series only changes the total coefficient index M , and Section 5 gives

$$d_{k,M}(p) a_{k,M,T}(y; p) e(-My), \quad t_* = 2\pi My.$$

Split the hierarchy at $K_0 = L^{1/8}$. The levels $k > K_0$ remain in this original pointwise- $f(t)$ representation. Lemma 7.5 assigns their low sidebands to $E_T^{\text{one-}\chi}$ and their high sidebands to the direct regular source later estimated in class (vii).

For the finite hierarchy $k \leq K_0$, insert the smooth height partition. On each block Lemma 7.4 replaces the pointwise source by the frozen source (28). The partition has bounded overlap, so the freezing contribution is

$$O(L^{-D-1+C_{\text{fr}}}) = o(1)$$

by (176); it belongs only to R_T^{src} . It remains to analyze the frozen low-level source on one block.

For each $k \leq K_0$, apply the canonical decomposition (82) on the closed contour Γ_L of Lemma 7.8. The exact residue identity (89) pairs the complete Laurent operator $\Pi_0^{(k)}$ with the actual joint packet weight of (63) and retains every enclosed translated projector $\Pi_{\Sigma_{\text{tr},k}}$ separately. Each translated term remains a direct stationary one- χ source and is expanded in unit-shell Fourier sidebands: its low sidebands belong to the edge term and its high sidebands to regular class (vii). After these principal parts have been subtracted, the remaining function $\mathcal{F}_{k,L}^{\text{hol}}$ is holomorphic throughout the interior of Γ_L . Hence, for every analytic packet weight W ,

$$\boxed{\frac{1}{2\pi i} \int_{\Gamma_L} W(u) \mathcal{F}_{k,L}^{\text{hol}}(1+u; p) du = 0.} \quad (104)$$

Thus no independent “near-pole analytic remainder” survives the local closed contour. By (65), the same levelwise extraction applies to all three completed Z -faces, and Lemma 7.6 gives the required factorial Mellin bounds. For $k \leq K_0$, the exact stationary level has the mutually exclusive form

$$\mathcal{T}_k^{\text{stat}} = \mathcal{L}_k^{\text{ex}} + \mathcal{E}_k^{\text{tr}} + \mathcal{D}_k^{\text{dir}}.$$

Here $\mathcal{L}_k^{\text{ex}}$ is the complete common local Laurent contribution, $\mathcal{E}_k^{\text{tr}}$ the additional translated principal parts, and $\mathcal{D}_k^{\text{dir}}$ the complementary pole-free right-edge one- χ contribution. The levels $k > K_0$ are not subjected to this local projector and remain in the direct edge representation of Lemma 7.5. Any smooth transition used to join the local contour to the complementary Mellin contour is supported away from the pole cluster and is included in $\mathcal{D}_k^{\text{dir}}$.

After removing the unprojected levels $k > K_0$ by Lemma 7.5, apply the trivial HLZ outer-twist specialization of Proposition 6.1. Its outer indices are all equal to 1 and the auxiliary divisor coordinate is the identity; this notation is independent of the HLP level index k used throughout the present section. Accordingly the outward annulus of Proposition 6.9 is empty. On the resulting common carrier core, replace (93) first by its intermediate leading jet and then replace the resummed pole parameter $c_b = L/F_b$ by the fixed model value 2. The resulting principal family, with the actual finite- T packet weight, defines P_T . Proposition 6.5 identifies the explicit X -powers with $\lambda(\tau)$, while the reassembled self term retains $\mu_b = 2F_b/L$. No nontrivial outer translation or outer-arithmetic factor occurs in the invariant contour source.

The local regular term R_T^{loc} is the sum of the low-level exact/leading discrepancy (98), the model truncation tail (99), and the translated-pole displacement (100); their combined estimate is (102). Only in Section 8, for the purpose of proving a Loewner floor, are the remaining slow parameters of P_T compared with the frozen reference model A_T^G .

The local holomorphic remainder contributes zero by (104). On the complementary Mellin region, the original absolutely convergent Dirichlet expansion is retained term by term, so $\mathcal{D}_k^{\text{dir}}$ keeps the physical carrier $e(-My)$. Its pole-free transition factors remain in the smooth stationary-symbol class of Lemma 5.1. Expanding that unit-shell symbol in Fourier sidebands, the low sidebands lie in the common macroscopic shell-edge range and hence have rank $o(N_T)$, while the high sidebands satisfy part (vii) of Proposition 12.4. No diagonal-jump estimate for the order-one complementary source is used.

Finally Lemma 7.5 sends the low sidebands of $k > K_0$ to the edge matrix and their high sidebands to regular class (vii). The local estimates are summed only over $k \leq K_0$, with the model truncation tail (99). This gives (103). The direct carrier structure of the complementary source is inherited from the original levelwise right-edge Dirichlet expansion; its quantitative high-sideband estimate is deferred to Proposition 12.4. $\square$

8 The completed global model and its spectral lower bound

By Proposition 6.10, the frozen completed local profile is the source-derived function G of (68). Let $C(x) := G(1-x)$ on the retained interior lag interval. If q is mean zero and

$$Q(x) := \int_0^x q(t) dt,$$

then two integrations by parts give the exact primitive identity

$$\langle K_C q, q \rangle = -2C'(0+) \|Q\|_2^2 - \iint C''(|x-y|) Q(x) \overline{Q(y)} dx dy. \quad (105)$$

Define the triangular primitive form

$$Q_\Delta(q) := 2\|Q\|_2^2.$$

After the internal principal contraction of Lemma 6.11 and the trivial outer-twist specialization of Proposition 6.1, the principal analysis has a single packet fibre and no nontrivial packet-coordinate multiplier. On the central flat plateau of ϕ_L , translate the physical Fourier coordinate by its fixed left endpoint u_0 . The common flat analysis is exactly

$$(\mathcal{U}_T^{\text{flat}} v)(u) = L_{\text{pkt}}^{-1/2} \sum_{\nu} e^{i\tau_{\nu} u_0} v_{\nu} e^{i\tau_{\nu} u}. \quad (106)$$

The phase $v_{\nu} \mapsto e^{i\tau_{\nu} u_0} v_{\nu}$ is unitary and commutes with the diagonal multiplier $v_{\nu} \mapsto \tau_{\nu}^{-1} v_{\nu}$. Let C_T be the stable-core restriction from Definition 7.10, put

$$R_T^{\text{core}} := \frac{W_T}{L} = 1 - \frac{A_0 \log L}{L}, \quad \mathcal{U}_T^{\text{core}} := C_T \mathcal{U}_T^{\text{flat}},$$

and normalize the core coordinate by $r = u/L$, so $0 \leq r \leq R_T^{\text{core}}$. Introduce the unitary rescaling of the core output

$$(\widehat{\mathcal{U}}_T^{\text{core}} v)(r) := L^{1/2} (\mathcal{U}_T^{\text{core}} v)(Lr). \quad (107)$$

Define A_T^G as the pullback through $\widehat{\mathcal{U}}_T^{\text{core}}$ of the frozen kernel

$$K_G(r, s) := G(1 - |r - s|),$$

and define $\mathfrak{G}_T$ through the same normalized core map by the triangular primitive form Q_Δ . Thus

$$\mathfrak{G}_T = (\mathcal{W}_T^{\text{glob}})^* \mathcal{W}_T^{\text{glob}}, \quad (108)$$

where $\mathcal{W}_T^{\text{glob}}$ is obtained from $\mathcal{U}_T^{\text{core}}$ by adjoining the primitive on $0 \leq u \leq W_T$ and multiplying the primitive output by $\sqrt{2}/L$. Indeed, after $r = u/L$ this gives exactly

$$Q_\Delta = \frac{2}{L^2} \int_0^{W_T} |R(u)|^2 du.$$

The endpoint mean constraint is imposed only on V_{mean} . Lemma 10.2 compares the shell-local primitive metric with this same global reference metric. Hence A_T^G and $\mathfrak{G}_T$ are pullbacks from one fixed core analysis map and differ only in the final lag form.

The coefficient extraction in (68) gives the closed form

$$G(\rho) = \frac{1}{12} [-14\rho^4 + 32\rho^3 - 24\rho^2 + 6\rho + e^2 ((6\rho^2 - 12\rho + 9) \sinh^2 \rho - 3\rho^2)]. \quad (109)$$

Proposition 8.1 (Explicit spectral floor on the stable core). *On each mean-zero core fibre $0 \leq r \leq R_T^{\text{core}}$, the frozen completed lag form satisfies*

$$K_G \succeq 0.06 Q_\Delta.$$

Consequently

$$A_T^G \succeq 0.06 \mathfrak{G}_T. \quad (110)$$

on the full packet space subject only to the core mean-zero constraint. No additional model spectral deletion is required.

Proof. Extend a core input q by zero from $[0, R_T^{\text{core}}]$ to $[0, 1]$. Its mean remains zero, and both K_G and Q_Δ restrict to the original core forms. It therefore suffices to prove the inequality on $[0, 1]$. Apply (105) to $C(|x - y|) = G(1 - |x - y|)$ and put $g(r) := G''(r)$. Direct differentiation of (109) gives

$$g(r) = -14r^2 + 16r - 4 + \frac{e^2}{2} [4(r - 1) \sinh(2r) + (2r^2 - 4r + 4) \cosh(2r) - 2]. \quad (111)$$

The cusp coefficient is

$$a_G := -2C'(0+) = 2G'(1) = \frac{-2 + 3e^2(\sinh 2 - 2)}{6} > 5.677. \quad (112)$$

For $0 \leq x \leq 4$ and $m + 2 > x$, the positive Taylor series gives the rational bounds

$$\sum_{j=0}^m \frac{x^j}{j!} \leq e^x \leq \sum_{j=0}^m \frac{x^j}{j!} + \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \quad (113)$$

Using $m = 64$ and exact rational arithmetic in the formulas for G', g, g' gives

$$\begin{aligned} g(0.16) &> 0.025, & g(0.1625) &< -0.010, \\ g(0.6145) &< -0.0004, & g(0.6146) &> 0.0012, \\ g'(0.39) &< -0.41, & g'(0.40) &> 0.19, \\ g(0) &< 3.39, & g(0.8375) &> 6.9, \\ g'(0.3854) &> -0.70, & g'(0.5) &> 6.8, \\ g(0.3144) + g(0.6856) &> 0.0019, \\ g(0.3145) + g(0.6855) &< -0.0010, \end{aligned} \quad (114)$$

and

$$\begin{aligned} 2.8385 &< G'(1) < 2.8386, & G'(0) &= \frac{1}{2}, \\ 0.7440 &< G'(a) < 0.7442, & 0.2361 &< G'(b) < 0.2364, \\ G'(0.3144) &< 0.6142, & G'(0.6855) &> 0.2846, \end{aligned} \quad (115)$$

where $a \in (0.16, 0.1625)$ and $b \in (0.6145, 0.6146)$ are the two zeros of g . To see that these are the only zeros, differentiate twice more:

$$G''''(r) = 2 [e^2 ((2r^2 - 4r + 9) \cosh(2r) + (8r - 8) \sinh(2r)) - 14].$$

Writing $x = 1 - r$ and using $\tanh y \leq y$ gives

$$7 + 2x^2 - 8x \tanh(2r) \geq 18x^2 - 16x + 7 \geq \frac{31}{9},$$

so $G''''(r) > 20$ on $[0, 1]$. Thus g is strictly convex and (114) gives exactly the two roots a, b and a unique minimum between 0.39 and 0.40.

The regular primitive kernel has Schur row sum

$$M(x) := \int_0^1 |g(1 - |x - y|)| dy.$$

It is symmetric about $1/2$. For $0 \leq x \leq 1/2$ write

$$M(x) = A + D(x), \quad A := \int_0^1 |g(r)| dr,$$

$$D(x) := \int_{1-x}^1 |g(r)| dr - \int_0^x |g(r)| dr.$$

We determine the maximum of D directly from the sign pattern of g and the strict convexity established above. For $0 \leq x \leq a$, both $g(x)$ and $g(1-x)$ are positive. Since the unique minimum of g lies in $(0.39, 0.40)$, g decreases on $[0, a]$ and increases on $[0.8375, 1]$. Hence (114) gives

$$D'(x) = g(1-x) - g(x) \geq g(0.8375) - g(0) > 0.$$

For $a < x < 1-b$, one has $g(x) < 0 < g(1-x)$ and therefore

$$D'(x) = g(x) + g(1-x) =: h(x).$$

Strict convexity makes g' strictly increasing, so, because $x < 1-x$,

$$h'(x) = g'(x) - g'(1-x) < 0.$$

The last two sign tests in (114) give $h(0.3144) > 0 > h(0.3145)$, hence a unique zero $x_0 \in (0.3144, 0.3145)$. Finally, for $1-b \leq x < 1/2$, both values of g are negative. Put $t = 1/2 - x \in (0, b - 1/2]$. The function

$$H(t) := g(1/2 + t) - g(1/2 - t)$$

satisfies $H(0) = 0$ and, throughout this range,

$$H'(t) = g'(1/2 + t) + g'(1/2 - t) \geq g'(0.5) + g'(0.3854) > 0.$$

Thus $g(1-x) > g(x)$ and $D'(x) = g(x) - g(1-x) < 0$. Consequently x_0 is the unique maximizer of D on $[0, 1/2]$, and symmetry gives the only other maximizer of M at $1 - x_0$. Since

$$A = G'(1) - G'(0) + 2G'(a) - 2G'(b)$$

and

$$D(x_0) = G'(1) - G'(1-x_0) - 2G'(a) + G'(0) + G'(x_0),$$

(115) yields

$$\boxed{A < 3.3548, \quad D(x_0) < 2.1802, \quad \sup_{x \in [0,1]} M(x) < 5.535.} \quad (116)$$

All decimal inequalities above are rigorous rational interval consequences of (113); no floating-point estimate is used in (116).

The primitive identity is therefore

$$Q_G(q) = a_G \|Q\|_2^2 - \iint g(1 - |x - y|) Q(x) \overline{Q(y)} dx dy.$$

Schur's test and (112)–(116) give

$$Q_G(q) > (5.677 - 5.535) \|Q\|_2^2 > 0.142 \|Q\|_2^2 > 0.06 Q_\Delta(q).$$

By Lemma 6.11 the internal Dirichlet index has already been contracted by the principal Laurent functional. By Proposition 6.1, no nontrivial outer coordinate remains in the invariant contour source. Thus A_T^G and $\mathfrak{G}_T$ are pullbacks through the same normalized core map $\widehat{\mathcal{U}}_T^{\text{core}}$ of (107), and congruence preserves the whole-space Loewner inequality. This proves (110) without deleting any model spectral subspace. $\square$

Lemma 8.2 (The whole-space floor survives restriction). *Let V be any subspace of the finite model coordinate space after the common positive pullback, and let P_V be its orthogonal projection. Then*

$$\boxed{P_V A_T^G P_V \succeq 0.06 P_V \mathfrak{G}_T P_V \quad \text{on } V.} \quad (117)$$

Thus every retained spectral restriction inherits the same numerical floor.

Proof. For $v \in V$, one has $P_V v = v$. Proposition 8.1 gives

$$\langle P_V A_T^G P_V v, v \rangle = \langle A_T^G v, v \rangle \geq 0.06 \langle \mathfrak{G}_T v, v \rangle = 0.06 \langle P_V \mathfrak{G}_T P_V v, v \rangle.$$

This is exactly (117). $\square$

Remark 8.3 (Global positivity). Positivity of the finite-interval covariance for each fixed scalar ρ does not by itself imply positivity after replacing ρ by the pairwise lag $\rho(r - s)$. Proposition 8.1 instead applies the primitive identity (105) directly to the whole completed lag kernel, which yields the required global Loewner bound.

For a Hermitian lag form B on the mean-zero fibre define its triangular form norm by

$$\|B\|_{\text{op}, \Delta} := \sup_{\substack{q \neq 0 \\ \int q = 0}} \frac{|\langle Bq, q \rangle|}{Q_{\Delta}(q)}. \quad (118)$$

Lemma 8.4 (Stability under the finite- T principal parameters). *Let $K_{\lambda, \mu, \theta}$ be the mean-zero lag form obtained from (67) after normalized curvature with*

$$\rho(x, y) = 1 + \theta - |x - y|, \quad 0 \leq x, y \leq 1.$$

Uniformly for

$$|\lambda - 1| + |\mu - 1| + \theta \leq \eta, \quad \theta \geq 0,$$

one has

$$\boxed{\|K_{\lambda, \mu, \theta} - K_{1, 1, 0}\|_{\text{op}, \Delta} \ll |\lambda - 1| + |\mu - 1| + \theta.} \quad (119)$$

Proof. After taking $[p^2]$, the profile (67) is analytic in (λ, μ, ρ) on a fixed neighbourhood of $(1, 1, [0, 1])$. For $\theta \geq 0$ sufficiently small, $\rho(x, y)$ remains in the fixed compact interval $[0, 1 + \eta]$. Differentiation with respect to λ , μ , or θ therefore gives kernels whose diagonal derivative jumps and regular second lag derivatives are uniformly bounded. In particular, changing θ changes the cusp coefficient by $O(\theta)$ and the regular primitive kernel by $O(\theta)$ in Schur norm. Applying the primitive identity (165) and the mean-value theorem yields

$$|\langle (K_{\lambda, \mu, \theta} - K_{1, 1, 0})q, q \rangle| \ll (|\lambda - 1| + |\mu - 1| + \theta) Q_{\Delta}(q),$$

which is (119). $\square$

Proposition 8.5 (Finite- T principal comparison). *On V_{mean} of (182),*

$$\boxed{\left\| \mathfrak{G}_T^{-1/2} (P_T - A_T^G) \mathfrak{G}_T^{-1/2} \right\|_{\text{op}} \ll L^{-1} = o(1).} \quad (120)$$

Consequently, for all sufficiently large T ,

$$\boxed{P_T \succeq c_0 \mathfrak{G}_T} \quad (121)$$

with any fixed $c_0 < 0.06$; in particular one may take $c_0 = 0.03$.

Proof. By Proposition 6.1, the invariant contour source has no nontrivial outer translation, divisor deformation, or outer-prefix excess. The finite HLZ Euler factor attached to the trivial outer twist is the empty Euler product and is identically one for all retained shifts. Thus the only finite- T deformation of the principal lag form is archimedean. Uniformly on the retained stable interior,

$$\lambda(\tau) = \frac{\log(\tau/2\pi)}{L} = 1 + O(L^{-1}), \quad \mu_b = \lambda(\tau) + O(L^{-D-1}) + O((TL)^{-1}),$$

and, by (62),

$$\theta_T := \nu_T - 1 = \lambda(\tau) - 1 = O(L^{-1}), \quad \theta_T \geq 0.$$

Lemma 8.4 therefore gives an $O(L^{-1})$ perturbation in the triangular metric. Pulling this form inequality back through the common normalized map $\hat{\mathcal{U}}_T^{\text{core}}$ and using the definition of $\mathfrak{G}_T$ converts it exactly into the relative operator norm in (120). The full smooth packet profile and the finite Cauchy factors are contained in the common analysis and residue maps on both sides of the comparison, so they create no additional error. This proves (120). The exact-model local discrepancy (101) is excluded from this comparison by definition and belongs only to R_T^{tri} . Combining (120) with Proposition 8.1 gives (121); no model spectral subspace is deleted. $\square$

9 Auxiliary finite divisor comparison

The divisor maps in this section are finite auxiliary comparison coordinates. By Proposition 6.1, they are not part of the invariant contour source itself: the actual outer twist is trivial. They are used only to record finite comparison estimates for the general twisted family and are never inserted as an additional source factor or replaced by a scalar normalization.

Lemma 9.1 (Hilbert-valued gcd identity). *For $b_1, \dots, b_Y$ in a Hilbert space $\mathcal{H}$,*

$$\sum_{m,n \leq Y} \frac{(m,n)}{mn} \langle b_m, b_n \rangle = \sum_{d \leq Y} \varphi(d) \left\| \sum_{d|n} \frac{b_n}{n} \right\|^2 \geq 0. \quad (122)$$

Proof. Insert the elementary identity

$$(m,n) = \sum_{d|(m,n)} \varphi(d)$$

and interchange the finite sums. The inner double sum becomes the squared Hilbert norm in (122). $\square$

Let U_n be isometries on $\mathcal{H}$ and define

$$(D_Y b)_d := \sqrt{\varphi(d)} \sum_{d|n} \frac{U_n b_n}{n}, \quad d \leq Y. \quad (123)$$

Thus the auxiliary finite arithmetic Gram is

$$G_{\text{ar}} := D_Y^* D_Y. \quad (124)$$

The isometries U_n may contain the exact shell translations coming from the physical one- χ carrier; no scalarization is involved.

The common outer-divisor feature of Proposition 6.8 has the normalized analysis map

$$(\widetilde{\mathcal{W}}_a c)(N) := N^{-1/2-a} \sum_{\substack{d|N \\ d \leq Y}} c_d d^a. \quad (125)$$

Its full output contains, in particular, the coordinates $1 \leq N \leq Y$. Writing $S_N = N^{1/2+a}(\widetilde{\mathcal{W}}_a c)(N)$, finite Möbius inversion gives

$$c_n n^a = \sum_{k|n} \mu(k) S_{n/k}. \quad (126)$$

For bounded p and $a = p/L$, $n^{\pm a} = L^{o(1)}$ uniformly for $n \leq Y$. Cauchy–Schwarz in (126) therefore gives the conservative inverse estimate

$$\|\widetilde{\mathcal{W}}_a^{-1}\|_{\text{op}, \text{Ran } \widetilde{\mathcal{W}}_a} \leq Y^{1+o(1)}. \quad (127)$$

Thus the new unsummed factorization introduces no larger arithmetic inverse loss than the finite divisor maps below.

Proposition 9.2 (Finite Möbius inverse). *The map D_Y is injective and*

$$\|D_Y^{-1}\|_{\text{op}, \text{Ran } D_Y} \leq Y \tau_{\max}(Y)^{1/2} = Y^{1+o(1)}. \quad (128)$$

The estimate is uniform in the ambient Hilbert space.

Proof. Put

$$c_n := \frac{U_n b_n}{n}, \quad S_d := \frac{(D_Y b)_d}{\sqrt{\varphi(d)}}.$$

Then

$$S_d = \sum_{d|n} c_n.$$

Finite Möbius inversion on the divisibility poset gives

$$c_n = \sum_{k \leq Y/n} \mu(k) S_{nk}. \quad (129)$$

Since U_n is an isometry,

$$\begin{aligned} \|b_n\| &\leq n \sum_{k \leq Y/n} \frac{\|(D_Y b)_{nk}\|}{\sqrt{\varphi(nk)}} \\ &\leq \sqrt{nY} \left(\sum_{k \leq Y/n} \|(D_Y b)_{nk}\|^2 \right)^{1/2}. \end{aligned}$$

Squaring and summing in n yields

$$\sum_{n \leq Y} \|b_n\|^2 \leq Y^2 \tau_{\max}(Y) \|D_Y b\|^2,$$

which proves (128). Injectivity follows either from this bound or directly by descending induction in d . $\square$

Lemma 9.3 (Conservative arithmetic transfer loss). *For an auxiliary finite twisted family with $1 \leq h, k, g \leq Y = L^\kappa$ and fixed $\kappa > 0$, the arithmetic label space has dimension*

$$\boxed{\dim \mathcal{H}_{\text{ar}} \leq Y^{3+o(1)}}. \quad (130)$$

Every retained finite arithmetic pullback used in the small-outer comparison below contains at most three nonunitary divisibility maps in total; each is either the common map (125) or a finite auxiliary map of the form (123). Hence the inverse of one complete arithmetic analysis map on its range has norm at most

$$\boxed{Y^{3+o(1)}}. \quad (131)$$

Moreover, any Hermitian arithmetic-fibre kernel whose entries, after the common positive weights have been factored out, have operator norm at most ε transfers to the divisor-whitened metric with the deliberately coarse bound

$$\boxed{\|G_{\text{ar}}^{-1/2} R_{\text{ar}} G_{\text{ar}}^{-1/2}\|_{\text{op}} \ll \varepsilon Y^{10+o(1)}}. \quad (132)$$

The exponent 10 is only a safety exponent. For the Hilbert–Schmidt norm on the same arithmetic fibre one also has

$$\boxed{\|G_{\text{ar}}^{-1/2} R_{\text{ar}} G_{\text{ar}}^{-1/2}\|_{S_2} \ll \varepsilon Y^{23/2+o(1)}}. \quad (133)$$

by (130). In particular a polylogarithmic entrywise mixed-derivative bound remains polylogarithmic after arithmetic whitening.

Proof. There are at most Y^3 triples (g, h, k) , with divisor refinements contributing only $Y^{o(1)}$. Each nonunitary divisibility inverse has norm $Y^{1+o(1)}$ by (127) or Proposition 9.2; three factors give (131).

For the two-sided estimate, first take the crude operator norm of a normalized arithmetic matrix of entry size at most ε . By (130) it is at most $\varepsilon Y^{3+o(1)}$. Whitening on the left and right costs at most one complete inverse factor (131) on each side, hence at most $Y^{6+o(1)}$. Thus the total loss is $\varepsilon Y^{9+o(1)}$, which is bounded by the safer displayed $\varepsilon Y^{10+o(1)}$ for all large Y . Perturbations that factor through the common analysis map, or arise from a one-sided deformation of that map, actually cost less; throughout the proof we are free to use the uniform worst-case bound (132). Finally, on an arithmetic fibre of dimension at most $Y^{3+o(1)}$,

$$\|B\|_{S_2} \leq (\dim \mathcal{H}_{\text{ar}})^{1/2} \|B\|_{\text{op}},$$

so (132) implies (133). □

Lemma 9.4 (Operator-valued positive lift). *If $(H_{mn}) \succeq 0$ and U_n are unitary, then for arbitrary scalars a_n ,*

$$\sum_{m,n} \overline{a_m} a_n H_{mn} U_m^* U_n \succeq 0.$$

Proof. Choose a Gram representation $H_{mn} = \sum_j \overline{x_{j,m}} x_{j,n}$ and set $X_j = \sum_n a_n x_{j,n} U_n$. The displayed operator is $\sum_j X_j^* X_j$. □

10 Stationary-shell frame and spectral compression

The shell transform is built before the stationary restriction. This order is important: the full transform is exactly Parseval, and only the explicitly omitted nonstationary modes create a defect.

10.1 Critical packet lattice, translation reserve, and Fourier trim

Put

$$R_{\text{pkt}} := \frac{C_{\text{pkt}}}{2} \log L, \quad L_{\text{pkt}} = L + 2R_{\text{pkt}}.$$

Use the full flat packet Fourier interval

$$0 \leq u \leq L_{\text{pkt}},$$

whose central physical zeta band is $R_{\text{pkt}} \leq u \leq R_{\text{pkt}} + L$. By Proposition 6.1, the invariant contour source has no nontrivial outer-denominator translation. The logarithmic reserve is used only for the packet-dimension slack and the smooth transition caps; it does not encode an arithmetic truncation.

For exact lattice centres

$$\tau_k = \tau_0 + \frac{2\pi k}{L_{\text{pkt}}}$$

put

$$S_v(u) := L_{\text{pkt}}^{-1/2} \sum_k v_k e^{i\tau_k u}. \quad (134)$$

Fourier orthogonality gives the exact isometry

$$\boxed{\int_0^{L_{\text{pkt}}} |S_v(u)|^2 du = \|v\|_2^2.} \quad (135)$$

Recall W_T, X_T from (47). Let C_T be the stable-core restriction of Definition 7.10. On packet coefficients its positive Gram is

$$\langle P_W v, v \rangle := \int_{R_{\text{pkt}}}^{R_{\text{pkt}} + W_T} |S_v(u)|^2 du = \|C_T S_v\|_2^2.$$

This is the same core restriction used in P_T , A_T^G , and $\mathfrak{G}_T$; it is not an additional packet-coordinate deletion.

Lemma 10.1 (Stable-core packet frame). *For every admissible packet-coordinate compression there is a subspace V_{core} such that*

$$\boxed{\text{codim } V_{\text{core}} \ll T \log L = o(N_T), \quad \|C_T S_v\|_2^2 \geq c_{\text{core}} \|v\|_2^2 \quad (v \in V_{\text{core}}),} \quad (136)$$

with a fixed $c_{\text{core}} > 0$. The corresponding bound for the diagonal multiplier $v_\nu \mapsto \tau_\nu^{-1} v_\nu$ holds with the right side multiplied by T^{-2} .

Proof. For exact lattice centres, (135) is Parseval on an interval of length L_{pkt} . Its complement to the stable core has length

$$L_{\text{pkt}} - W_T = O(\log L).$$

If D_{tail} is the Gram of restriction to that complement, then each coordinate vector contributes $O(\log L/L)$ to its trace. Hence on a packet fibre of dimension $O(TL)$,

$$\text{tr } D_{\text{tail}} \ll T \log L.$$

Removing the spectral subspace of D_{tail} for eigenvalues at least $1/4$ costs $O(T \log L)$ dimensions and leaves the core Gram at least $3/4$ of the full flat Gram. Denote this good packet subspace by V_0 .

By Proposition 6.1 there is only the identity outer translation. For the primitive estimate intersect V_0 with $D_\tau^{-1}V_0$, where $D_\tau v = (\tau_\nu^{-1}v_\nu)_\nu$. Since D_τ is invertible and $T \leq \tau_\nu \leq 2T + O(1)$, this preimage has the same codimension as V_0 , while

$$\|D_\tau v\|_2^2 \asymp T^{-2}\|v\|_2^2.$$

Thus the intersection has codimension $O(T \log L)$ and gives both lower bounds in (136). Moving every centre by at most T^{-100} changes these finite Grams by $o(1)$ in operator norm, by the estimate in the proof of Lemma 3.3; the fixed lower bound therefore persists. $\square$

10.2 Logarithmic unitary and exact local Parseval

On the retained physical band write

$$\tilde{u} := u - R_{\text{pkt}}, \quad x = e^{\tilde{u}},$$

and define

$$F_v(x) := x^{-1/2} S_v(R_{\text{pkt}} + \log x), \quad 1 \leq x \leq X_T. \quad (137)$$

Since $dx = x d\tilde{u} = x du$,

$$\boxed{\int_1^{X_T} |F_v(x)|^2 dx = \int_{R_{\text{pkt}}}^{R_{\text{pkt}} + W_T} |S_v(u)|^2 du = \langle P_W v, v \rangle.} \quad (138)$$

We use an explicit fixed-overlap partition by length-one windows. Choose $\vartheta \in C_c^\infty((-1/2, 1/2))$ which is positive on $[-1/4, 1/4]$, put $c_j = j/2$, and define

$$D(x) := \left(\sum_{j \in \mathbb{Z}} \vartheta(x - c_j)^2 \right)^{1/2}, \quad \psi_j(x) := \frac{\vartheta(x - c_j)}{D(x)}.$$

Because every real number lies within distance $1/4$ of some c_j , the $1/2$ -periodic function D is smooth and bounded above and below by positive constants. With

$$I_j = [J_j, J_j + 1], \quad J_j := c_j - \frac{1}{2},$$

we therefore have

$$\boxed{\text{supp } \psi_j \subset I_j, \quad \sum_{j \in \mathbb{Z}} \psi_j(x)^2 = 1, \quad \|\psi_j^{(r)}\|_\infty \ll_r 1.} \quad (139)$$

Extend F_v by zero outside $[1, X_T]$ and retain shells with $\psi_j \not\equiv 0$ on $(1, X_T)$. The $O(1)$ shells meeting the two core endpoints remain in the Parseval decomposition; their endpoint jets are treated by Proposition 11.2.

Lemma 10.2 (Global triangular metric and shell-local primitives). *Let g be Hilbert-valued on $[0, W_T]$ with $\int_0^{W_T} g = 0$, and put*

$$R(u) := \int_0^u g(v) dv, \quad f(x) := x^{-1/2} g(\log x).$$

Then $R(0) = R(W_T) = 0$. Extend f and $\mathcal{R}(x) := R(\log x)$ by zero outside $[1, X_T]$. After the normalization $r = u/L$, the flat triangular primitive form is

$$\mathcal{Q}_{\text{glob}}(g) := \frac{2}{L^2} \int_0^{W_T} \|R(u)\|^2 du = \frac{2}{L^2} \int_1^{X_T} \left\| \int_1^x y^{-1/2} f(y) dy \right\|^2 \frac{dx}{x}. \quad (140)$$

Let $\mathcal{J}_T$ denote these retained shells. For $j \in \mathcal{J}_T$ define

$$q_j(x) := \psi_j(x) x^{-1/2} f(x), \quad Q_j(x) := \int_{J_j}^x q_j(y) dy,$$

and

$$\mathcal{Q}_{\text{loc}}(f) := \frac{2}{L^2} \sum_{j \in \mathcal{J}_T} \int_{I_j} \|Q_j(x)\|^2 \frac{dx}{x}. \quad (141)$$

Then there is an absolute constant C_Δ such that

$$\boxed{\mathcal{Q}_{\text{loc}}(f) \leq C_\Delta \mathcal{Q}_{\text{glob}}(g)}. \quad (142)$$

The estimate is Hilbert-valued. If, in addition, $\int_{I_j} q_j = 0$ for every shell and output component, then Q_j vanishes at both endpoints and may be used in the two-variable primitive identity of Lemma 12.3.

Proof. The zero extension of $\mathcal{R}$ is absolutely continuous because $R(0) = R(W_T) = 0$, and $\mathcal{R}' = x^{-1/2} f$ almost everywhere. Thus

$$Q_j(x) = \psi_j(x) \mathcal{R}(x) - \int_{J_j}^x \psi_j'(y) \mathcal{R}(y) dy,$$

because ψ_j vanishes at the shell endpoints. The windows have uniformly bounded first derivative and bounded overlap. On the physical part of a length-one shell, $x^{-1} \asymp (1 + J_j)^{-1}$, so Cauchy–Schwarz gives

$$\int_{I_j} \|Q_j(x)\|^2 \frac{dx}{x} \ll \int_{I_j} \|\mathcal{R}(x)\|^2 \frac{dx}{x}.$$

Summing in j proves (142) and is unchanged for Hilbert-valued functions. The final assertion follows directly from the definition of Q_j . $\square$

Define

$$Q_{j,n}(v) := \int_{I_j} \psi_j(x) F_v(x) e^{-2\pi i n x} dx. \quad (143)$$

Because each shell has length one, the integer exponentials are an orthonormal basis and Parseval gives

$$\sum_{n \in \mathbb{Z}} |Q_{j,n}(v)|^2 = \|\psi_j F_v\|_2^2.$$

Summing in j ,

$$\boxed{\|\mathcal{Q}_{\text{full}} v\|_2^2 = \langle P_W v, v \rangle}. \quad (144)$$

No stationary approximation has entered this identity.

10.3 Stationary band and nonstationary leakage

For one packet centre τ and one shell Fourier mode n , the phase is

$$\Phi_{\tau,n}(x) = \tau \log x - 2\pi n x, \quad \Phi'_{\tau,n}(x) = \frac{\tau}{x} - 2\pi n.$$

For $T \leq \tau \leq 2T$ and $x \in I_j$, a stationary point can occur only for

$$\mathcal{I}_j := \left[\frac{T}{2\pi(J_j + 1)}, \frac{2T}{2\pi J_j} \right]. \quad (145)$$

Choose the shell margin

$$R_j := L^B \left(1 + \frac{\sqrt{T}}{J_j} \right) \quad (146)$$

and let $\mathcal{I}_j^+$ contain the integers at distance at most R_j from $\mathcal{I}_j$. Let $\mathbf{Q}_{\text{shell}}$ retain only (j, n) with $n \in \mathcal{I}_j^+$ and let $\mathbf{Q}_{\text{out}}$ be its complement.

Fix a packet basis vector e_k . On a nonboundary shell put $D = \text{dist}(n, \mathcal{I}_j) \geq R_j$. Then

$$|\Phi'| \geq 2\pi D, \quad |\Phi^{(r)}| \ll_r T/J_j^r, \quad |A_{j,k}^{(r)}| \ll_r L_{\text{pkt}}^{-1/2} J_j^{-1/2} L^{Cr} J_j^{-r}, \quad \frac{T/J_j^2}{D^2} \ll L^{-2B}.$$

Repeated integration by parts gives

$$\boxed{|Q_{j,n}(e_k)| \ll_N L_{\text{pkt}}^{-1/2} J_j^{-1/2} L^{C_N} (1 + D)^{-N}.} \quad (147)$$

Thus the squared nonboundary contribution is

$$\ll_N \frac{L^{2C_N}}{L_{\text{pkt}}} \sum_j J_j^{-1} R_j^{1-2N}.$$

For a boundary shell, split instead at the physical endpoint $b \in \{1, X_T\}$. Repeating the same integration by parts N times gives a finite endpoint sum plus an interior remainder. The endpoint sum is a linear combination of

$$\boxed{\partial_x^r (\psi_j(x) F_v(x)) \Big|_{x=b}, \quad 0 \leq r < N,} \quad (148)$$

with scalar coefficients depending on the Fourier mode and the phase; the interior remainder satisfies the same arbitrary-order bound as (147). Thus the physical endpoint contributes only finitely many packet jet functionals, while every nonstationary interior term remains in the regular integration-by-parts class.

The shell enlargement itself has only $o(N_T)$ extra rows. Indeed

$$\begin{aligned} \sum_j R_j &\ll X_T L^B + \sqrt{T} L^B \log X_T \\ &\ll T L^{B-A_0} + \sqrt{T} L^{B+1} = o(N_T) \end{aligned}$$

when $A_0 > B + 3$.

Proposition 10.3 (Terminal logarithmic tail). *Let $X_{\text{full}} := e^L = T/(2\pi)$. On the complementary logarithmic band*

$$W_T < u \leq L, \quad X_T < x \leq X_{\text{full}},$$

the vertical right-edge source and its adjoint decompose as

$$A_T^{\text{tail}} = E_T^{\text{tail}} + E_T^{\text{bdry,tail}} + R_T^{\text{tail}}, \quad (149)$$

where $E_T^{\text{bdry,tail}}$ is the tail-side subfamily of the split endpoint jets at $x = X_T$ and is included, exactly once, in the global boundary matrix E_T^{bdry} of Proposition 11.2. Moreover

$$\text{rank } E_T^{\text{tail}} \ll T \log L = o(N_T), \quad (150)$$

and, on $V_{\text{core}} \cap V_{\text{mean}}$,

$$\|\mathfrak{G}_T^{-1/2} R_T^{\text{tail}} \mathfrak{G}_T^{-1/2}\|_{S_2}^2 = o(N_T). \quad (151)$$

The same estimates include the core–tail cross terms.

Proof. Use the same unit-shell Fourier transform on $X_T < x \leq X_{\text{full}}$, but put only the integers at distance at most 1 from $\mathcal{I}_j$ into the tail stationary range. Its total number of rows is

$$\sum_{X_T \ll J_j \ll X_{\text{full}}} \left(1 + \frac{T}{J_j}\right) \ll T + T \log \frac{X_{\text{full}}}{X_T} \ll T \log L.$$

By Proposition 6.1 there is no arithmetic fibre multiplicity. Every stationary tail term and every core–tail cross term therefore factors through a space of dimension $O(T \log L)$, which proves the rank bound.

For the complementary modes let $D = \text{dist}(n, \mathcal{I}_j) > 1$. Since $J_j \geq X_T = T/(2\pi L^{A_0})$,

$$\frac{T}{J_j^2} \ll \frac{L^{2A_0}}{T}.$$

The integration-by-parts argument of (147), now with the fixed tail margin 1, gives for every fixed N

$$|Q_{j,n}(e_k)| \ll_N L_{\text{pkt}}^{-1/2} J_j^{-1/2} \left(\frac{L^{C_N}}{J_j}\right)^N (1+D)^{-N}.$$

Squaring this estimate and summing first in n gives a bounded $\sum_{D>1} (1+D)^{-2N}$. Since the retained unit shells satisfy $J_j \geq X_T \asymp T L^{-A_0}$,

$$\sum_{J_j \geq X_T} J_j^{-1-2N} \ll X_T^{-2N}.$$

After summing over the $O(TL)$ packet coordinates, with no additional arithmetic multiplicity by Proposition 6.1, we obtain the sufficient absolute estimate

$$\|R_T^{\text{tail}}\|_{S_2}^2 \ll T^{1-2N} L^{C_N, A_0}. \quad (152)$$

The same estimate holds for the nonstationary core–tail cross terms, since the core analysis map has fixed operator norm up to a power of L . Lemma 12.1 gives on $V_{\text{core}} \cap V_{\text{mean}}$

$$\|\mathfrak{G}_T^{-1/2}\|_{\text{op}}^2 \ll T^2 L^{C_G}.$$

Consequently

$$\|\mathfrak{G}_T^{-1/2} R_T^{\text{tail}} \mathfrak{G}_T^{-1/2}\|_{S_2}^2 \ll T^{5-2N} L^{C_N, A_0 + 2C_G} = o(N_T),$$

because the fixed order N chosen in Section 13 is in particular larger than 3. Endpoint jets at the physical endpoint $x = X_{\text{full}}$ factor through finitely many evaluation functionals and are included in E_T^{tail} . The tail-side split jets at $x = X_T$ are precisely $E_T^{\text{bdry, tail}}$ and belong only to the global boundary matrix E_T^{bdry} of Proposition 11.2. This proves (149) and (151). $\square$

11 Macroscopic one- χ shifts have low edge rank

For a direct stationary one- χ source on a unit shell write

$$\sum_m W_{j,m} a_{j,m}(y; p) e(-my), \quad a_{j,m} = \sum_r \hat{a}_{j,m}(r; p) e(ry).$$

Here $m \in \mathcal{I}_j$. The sideband r has effective carrier $q = m - r$, since $e(ry)e(-my) = e(-qy)$. If P_j projects onto $\mathcal{I}_j^+$ and $(S_q b)_n = b_{n+q}$, its compression is $P_j S_q P_j$. Only $|r| \leq R_j^{\text{sb}}$ enters the edge matrix; the complement is regular class (vii).

Write

$$A_j := \frac{T}{2\pi(J_j + 1)}, \quad B_j := \frac{2T}{2\pi J_j}.$$

If $m \in \mathcal{I}_j$, then $P_j S_m P_j$ can have nonzero range only on the common low-edge interval

$$A_j - R_j \leq n \leq B_j + R_j - A_j.$$

Thus all forward shifts share one edge space E_j^- with

$$\dim E_j^- \ll B_j - 2A_j + 2R_j + 1 \ll \frac{T}{J_j^2} + R_j + 1. \quad (153)$$

The adjoints lie in a high-edge space E_j^+ of the same dimension. Therefore, for arbitrary scalar weights $W_{j,m}$,

$$\text{rank} \left(\sum_{m \in \mathcal{I}_j} W_{j,m} P_j S_m P_j + \text{adjoint} \right) \ll \frac{T}{J_j^2} + R_j + 1. \quad (154)$$

The same conclusion holds with coefficients acting on an auxiliary common fibre only when they preserve this shell-coordinate edge range; no such extra fibre is present in the invariant source by Proposition 6.1. For the parameter $\epsilon_L = L^{-\sigma}$ fixed in Section 13, the low Fourier sidebands of a smooth stationary symbol replace m by the effective shift $m - r$ with $|r| \leq \epsilon_L d_j$, where $d_j \asymp T/J_j$. Enlarging $E_j^\pm$ by this amount therefore changes the right side of (154) only by $O(\epsilon_L d_j)$. Now

$$\sum_j \frac{T}{J_j^2} \ll T, \quad \sum_j R_j \ll T L^{B-A_0} + \sqrt{T} L^{B+1} = o(T),$$

and

$$\sum_j \epsilon_L d_j \ll \epsilon_L T \log X_T \ll T L^{1-\sigma}.$$

Thus the first two geometric edge contributions are $O(T)$. The low-sideband enlargement contributes $T L^{1-\sigma} = o(T)$ once $\sigma > 2$. Hence

Proposition 11.1 (Uniform rank bound for low-sideband stationary shifts). *Let E_T^{shift} be any finite sum of the low-sideband stationary compressed shifts described above, with base carrier indices in the intervals $\mathcal{I}_j$ and sidebands $|r| \leq R_j^{\text{sb}}$. Then*

$$\boxed{\text{rank } E_T^{\text{shift}} \ll T = o(N_T).} \quad (155)$$

The estimate depends only on the geometric range of the retained low-sideband shifts and is independent of their coefficients or signs. It also holds for norm-convergent sums of such shifts, since every partial sum has range in the same fixed finite-dimensional edge space. In particular it applies to the low-sideband part of each additional translated projector $\Pi_{\Sigma_{\text{tr},k}}$: by Lemma 7.7, taking its Mellin residue changes only the radial coefficient and leaves the same total integer carrier M ; the smooth stationary amplitude is then split at R_j^{sb} exactly as above. The complementary sidebands are not part of this rank assertion.

Proposition 11.2 (Stable-core endpoint terms have finite rank). *Let E_T^{bdry} be the sum of the endpoint terms produced by the fixed N integrations by parts on the $O(1)$ shell windows meeting $b \in \{1, X_T\}$, including both one-sided contributions at the split $x = X_T$. Denote the contribution from the tail side of this split by $E_T^{\text{bdry,tail}}$, as in (149). Then*

$$\boxed{\text{rank } E_T^{\text{bdry}} \ll L^{C_{\text{bdry}}} = o(N_T)} \quad (156)$$

for a fixed exponent C_{bdry} .

Proof. For each boundary shell, fixed local-source label α , endpoint b , and $0 \leq r < N$, let

$$(J_T v)_{b,r,\alpha} := \partial_x^r (\psi_j(x) F_{v,\alpha}(x)) \big|_{x=b}.$$

Equation (148) shows that every endpoint contribution with one boundary leg factors as $J_T^* B Q$ for some output operator B and some companion analysis Q ; the reflected term factors as $Q^* B^* J_T$. Hence each Hermitian pair has rank at most $2 \text{rank } J_T$. There are two physical endpoints, $O(1)$ boundary shells, and the derivative order N is fixed. The fixed local-source labels contribute at most $L^{C_{\text{bdry}}}$ coordinates. Thus

$$\text{rank } J_T \ll L^{C_{\text{bdry}}} = o(N_T).$$

□

Define E_T^{edge} to be the sum of the low-sideband centered stationary shift family, the high exact HLP low-sideband matrix $E_T^{\text{HLP,hi}}$, the low-sideband parts of the additional translated-pole terms, the boundary-shell matrix E_T^{bdry} , and the terminal logarithmic tail matrix E_T^{tail} . The auxiliary outer-excess term is zero by Proposition 6.1; the horizontal packet-centre collars are handled only by V_{end} . Lemma 7.5 and Propositions 11.1, 11.2, and 10.3 give

$$\boxed{\text{rank } E_T^{\text{edge}} \ll T \log L = o(N_T).} \quad (157)$$

12 Reference metric and regular remainder

The final reference metric $\mathfrak{G}_T$ is the global flat-lag metric defined in (108). For the regular-remainder estimates it is convenient to use a shell-local auxiliary metric. Let $\mathcal{W}_T^{\text{loc}}$ be the composition of

- (i) packet synthesis followed by the logarithmic unitary and the fixed bounded-overlap shell windows, with the input weight $x^{-1/2}$ appearing in (180);
- (ii) the invariant-source specialization of Proposition 6.1; the outer translation and divisor maps are the identity and introduce no additional arithmetic fibre;
- (iii) the local primitive $q_j \mapsto Q_j$ of Lemma 10.2, with the endpoint mean constraint imposed only after restriction to V_{mean} ;
- (iv) multiplication of the primitive output by $\sqrt{2}L^{-1}x^{-1/2}$, so that its ordinary $L^2(dx)$ norm is exactly the weighted norm in (141), together with the remaining positive weights common to both normalizations.

Put

$$\mathfrak{G}_T^{\text{loc}} := (\mathcal{W}_T^{\text{loc}})^* \mathcal{W}_T^{\text{loc}}. \quad (158)$$

The stationary Fourier restriction is not part of this reference map; it is an orthogonal compression applied to the source operators and can only decrease their Hilbert–Schmidt norms. The signed coefficient $G(\rho)$ is part of neither reference metric. By Proposition 6.1 there is no nontrivial outer curvature deformation $d^{p/L}$ in the invariant source.

Lemma 10.2, applied to the common packet/core map, gives on V_{mean}

$$\mathfrak{G}_T^{\text{loc}} \preceq C_{\Delta} \mathfrak{G}_T. \quad (159)$$

Both forms are positive definite on every finite packet-coordinate compression. Indeed, if $\mathcal{W}_T^{\text{loc}} v = 0$, then every local primitive Q_j vanishes; differentiation gives $q_j = 0$. Since $\sum_j \psi_j^2 = 1$, the logarithmic-unitary output is zero on the open physical interval. Hence the underlying packet synthesis

$$u \mapsto \sum_{\nu} v_{\nu} e^{i\tau_{\nu} u}$$

vanishes on a nonempty open interval. It is a finite exponential polynomial; analytic uniqueness (equivalently the Vandermonde argument of Lemma 3.2) gives $v = 0$. The global map is injective by the same argument: vanishing of its primitive makes the packet synthesis zero on the global interval. The mean-zero conditions are needed for the endpoint identities, not for injectivity.

Lemma 12.1 (Polynomial floor for the core reference metric). *On $V_{\text{core}} \cap \ker \mathcal{M}_T^{\text{glob}}$ there is a fixed exponent C_G such that*

$$\mathfrak{G}_T \succeq T^{-2} L^{-C_G} I. \quad (160)$$

Proof. By (106), after translating the flat core to $0 \leq u \leq W_T$ the global reference input is exactly

$$g(u) = L_{\text{pkt}}^{-1/2} \sum_{\nu} \tilde{v}_{\nu} e^{i\tau_{\nu} u}, \quad \tilde{v}_{\nu} = e^{i\tau_{\nu} u_0} v_{\nu}.$$

The phase $v \mapsto \tilde{v}$ is unitary and commutes with $D_{\tau} v = (\tau_{\nu}^{-1} v_{\nu})_{\nu}$. Hence the two inequalities in Lemma 10.1, after translating the integration interval, apply directly to this synthesis and to its D_{τ} multiple.

Write

$$R(u) = \int_0^u g(s) ds = a + L_{\text{pkt}}^{-1/2} \sum_{\nu} \frac{\tilde{v}_{\nu}}{i\tau_{\nu}} e^{i\tau_{\nu} u}.$$

Since $T \leq \tau_\nu \leq 2T + O(1)$, Lemma 10.1 gives

$$\left\| L_{\text{pkt}}^{-1/2} \sum_{\nu} \frac{\tilde{v}_\nu}{i\tau_\nu} e^{i\tau_\nu u} \right\|_{L^2(0, W_T)}^2 \gg T^{-2} \|v\|_2^2.$$

The constant a cannot cancel this oscillatory norm. Indeed $|\int_0^{W_T} e^{i\tau_\nu u} du| \leq 2/|\tau_\nu|$; using $d_T = O(TL)$ and $L_{\text{pkt}} \asymp W_T \asymp L$, the squared norm of the projection of the oscillatory sum onto the constants is $O(T^{-3})\|v\|_2^2$. Hence

$$\|R\|_{L^2(0, W_T)}^2 \gg T^{-2} \|v\|_2^2.$$

Inserting the factor $2L^{-2}$ in the normalized triangular primitive gives (160), after enlarging the fixed exponent C_G . $\square$

The final perturbation will use only fixed- T compactness, but (160) is used for the direct terminal-tail estimate. All p -dependence excluded above belongs only to the principal comparison (120).

The factor used in the shell normalization is fixed by the change from the normalized flat lag coordinate to the physical logarithmic coordinate. If $r = u/L$ and $\widehat{q}(r) = L^{1/2}g(Lr)$, then for every lag profile H ,

$$\iint H(1 - |r - s|) \widehat{q}(r) \overline{\widehat{q}(s)} dr ds = \frac{1}{L} \iint H\left(1 - \frac{|u - v|}{L}\right) g(u) \overline{g(v)} du dv. \quad (161)$$

After $x = e^u$, the logarithmic unitary and the input factor $x^{-1/2}$ in (180) cancel the measure Jacobians. Thus every normalized local lag term carries the same external factor L^{-1} ; all remaining height, stationary, and fixed local-source factors are contained in its smooth operator-valued amplitude.

Lemma 12.2 (Shell transport of the triangular normalization). *Let $I_j = [J_j, J_j + 1]$, $x = J_j + t$, $y = J_j + s$, and let*

$$\rho_j(t, s) = \nu - \frac{|\log x - \log y|}{L}, \quad |\nu - 1| \ll L^{-1}.$$

Suppose a Hilbert-valued shell kernel is a finite sum of terms

$$D_j(t, s) = \frac{1}{L} a_j(t, s) H(\rho_j(t, s)), \quad (162)$$

where, for integers $a, b \geq 0$ with $a + b \leq 2$,

$$\|\partial_t^a \partial_s^b a_j(t, s)\| \ll L^C J_j^{-a-b},$$

and H, H', H'' are bounded by a fixed power of L on the relevant compact interval. Then the diagonal derivative jump satisfies

$$\boxed{J_{D_j}(t) = \omega_j(t) a_j(t, t) H'(\nu), \quad \omega_j(t) = \frac{2}{L^2(J_j + t)},} \quad (163)$$

with the convention for J_D in Lemma 12.3. Moreover the regular mixed derivative obeys

$$\boxed{\left\| \omega_j(t)^{-1/2} D_{j,ts}^{\text{reg}}(t, s) \omega_j(s)^{-1/2} \right\| \ll L^{C'}} \quad (164)$$

for a fixed C' . The same conclusions hold for finite sums with finite-dimensional operator-valued amplitudes a_j .

Proof. Off the diagonal,

$$\partial_t \rho_j = -\frac{\operatorname{sgn}(t-s)}{Lx}, \quad \partial_s \rho_j = \frac{\operatorname{sgn}(t-s)}{Ly}.$$

The derivatives of the smooth amplitude a_j have no diagonal jump. The two one-sided values of $\partial_t H(\rho_j)$ therefore differ by $2H'(\nu)/(Lx)$, and the external factor L^{-1} in (162) gives (163).

For the regular mixed derivative, Leibniz' rule produces terms with two amplitude derivatives, one amplitude and one lag derivative, or two lag derivatives. On a length-one shell $x, y \asymp J_j$; hence they are bounded, respectively, by fixed polylogarithmic multiples of

$$L^{-1}J_j^{-2}, \quad L^{-2}J_j^{-2}, \quad L^{-3}J_j^{-2}.$$

Since

$$\omega_j(t)^{1/2}\omega_j(s)^{1/2} \asymp L^{-2}J_j^{-1},$$

division by the reference weights leaves at most a fixed factor $L^{C'}$ (indeed the first class costs only $O(L/J_j)$). This proves (164). The argument is unchanged for finite-dimensional operator coefficients after taking operator or Hilbert–Schmidt norms. $\square$

Lemma 12.3 (Two-variable primitive remainder criterion). *Let $I = [0, 1]$ and let $q = Q'$ with $Q(0) = Q(1) = 0$. Let $D(x, y)$ be a Hermitian kernel which is continuous across the diagonal, C^1 on each open triangle, and whose regular mixed derivative is in $L^2(I^2)$. Define the diagonal jump*

$$J_D(x) := \partial_x D(x, x+) - \partial_x D(x, x-).$$

Then, distributionally,

$$\partial_x \partial_y D = J_D(x) \delta_{x=y} + D_{xy}^{\operatorname{reg}},$$

and two integrations by parts give

$$\boxed{\iint D(x, y) q(x) \overline{q(y)} dx dy = \int J_D(x) |Q(x)|^2 dx + \iint D_{xy}^{\operatorname{reg}}(x, y) Q(x) \overline{Q(y)} dx dy.} \quad (165)$$

For the global application, let $D_{j,T}$ be the corresponding operator-valued kernel on the finite output fibre of the j th retained unit shell. All common positive output weights are already included in $\mathcal{W}_T^{\operatorname{loc}}$; by Proposition 6.1 there is no additional outer arithmetic metric. In the local coordinate $0 \leq t \leq 1$ put

$$\omega_j(t) := \frac{2}{L^2(J_j + t)}. \quad (166)$$

By (141), $\omega_j(t)$ is the exact scalar primitive reference density on this output fibre. Define the normalized jump and regular mixed kernel by

$$\mathcal{J}_{j,T}(t) := \omega_j(t)^{-1} J_{D_{j,T}}(t),$$

$$\mathcal{K}_{j,T}(t, s) := \omega_j(t)^{-1/2} D_{j,T,ts}^{\operatorname{reg}}(t, s) \omega_j(s)^{-1/2}.$$

If

$$\sup_{j,x} \|\mathcal{J}_{j,T}(x)\|_{\operatorname{op}} = \delta_T = o(1) \quad (167)$$

and

$$M_T := \sum_{j \in \mathcal{J}_T} \iint \|\mathcal{K}_{j,T}(x, y)\|_{S_2}^2 dx dy = o(N_T), \quad (168)$$

then every finite retained compression of total dimension $O(N_T)$ satisfies

$$\boxed{\|\mathfrak{G}_T^{-1/2} R_T \mathfrak{G}_T^{-1/2}\|_{S_2}^2 \ll \delta_T^2 N_T + M_T = o(N_T).} \quad (169)$$

In particular, a uniform per-shell estimate

$$\iint \|\mathcal{K}_{j,T}(x, y)\|_{S_2}^2 dx dy \ll L^C$$

for fixed C implies (168) whenever $A_0 > C + 2$, since $\#\mathcal{J}_T = O(X_T)$ and $X_T = T/(2\pi L^{A_0})$.

Proof. The endpoint conditions kill all boundary terms in the two integrations by parts, proving (165). On the j th local primitive space the exact reference quadratic form is

$$\int_0^1 \omega_j(t) \|Q(t)\|^2 dt.$$

Conjugating by multiplication with $\omega_j(t)^{1/2}$ turns the diagonal term in (165) into the multiplication operator $M_{\mathcal{J}_{j,T}}$ and the regular term into the integral operator $T_{\mathcal{K}_{j,T}}$. On the orthogonal direct sum of shell outputs put

$$B_{\text{jump}} := \bigoplus_j M_{\mathcal{J}_{j,T}}, \quad B_{\text{reg}} := \bigoplus_j T_{\mathcal{K}_{j,T}}.$$

Then

$$\|B_{\text{jump}}\|_{\text{op}} \leq \delta_T, \quad \|B_{\text{reg}}\|_{S_2}^2 \leq \sum_j \iint \|\mathcal{K}_{j,T}(t, s)\|_{S_2}^2 dt ds = M_T.$$

The multiplication operator B_{jump} need not itself be Hilbert–Schmidt on the continuous shell-output space; finite dimensionality enters only after pullback. Put $G_{\text{loc}} := \mathfrak{G}_T^{\text{loc}}$ on the retained compression, $W := \mathcal{W}_T^{\text{loc}}$, and

$$V := W G_{\text{loc}}^{-1/2}.$$

Then V is an isometry from the retained packet space, of dimension $d_T = O(N_T)$, into $\text{Ran } W$. Therefore

$$\|V^* B_{\text{jump}} V\|_{S_2}^2 \leq d_T \|B_{\text{jump}}\|_{\text{op}}^2 \ll \delta_T^2 N_T,$$

whereas the ideal property of the Hilbert–Schmidt norm gives

$$\|V^* B_{\text{reg}} V\|_{S_2} \leq \|B_{\text{reg}}\|_{S_2} \leq M_T^{1/2}.$$

Since all common output weights are already contained in $\mathcal{W}_T^{\text{loc}}$, the pulled-back remainder is $V^*(B_{\text{jump}} + B_{\text{reg}})V$, and we obtain

$$\|G_{\text{loc}}^{-1/2} R_T G_{\text{loc}}^{-1/2}\|_{S_2}^2 \ll \delta_T^2 N_T + M_T.$$

Finally (159) gives $G_{\text{loc}} \preceq C_\Delta \mathfrak{G}_T$. With

$$A := G_{\text{loc}}^{1/2} \mathfrak{G}_T^{-1/2}$$

one has $\|A\|_{\text{op}}^2 \leq C_\Delta$ and

$$\mathfrak{G}_T^{-1/2} R_T \mathfrak{G}_T^{-1/2} = A^* (G_{\text{loc}}^{-1/2} R_T G_{\text{loc}}^{-1/2}) A.$$

Thus global whitening costs only the fixed factor C_Δ , proving (169). $\square$

Proposition 12.4 (Relative Hilbert–Schmidt bound for the regular remainder). *After the complete principal cluster has been extracted and every additional nonprincipal translated pole has been separated from it and subjected to the same direct sideband split, the regular remainder is a sum of mutually exclusive pieces. After quadratic-form compression to $\mathcal{P}_T^{\text{ret}}$, it satisfies*

$$\left\| (\mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}})^{-1/2} (R_T^{\text{tri}}|_{\mathcal{P}_T^{\text{ret}}}) (\mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}})^{-1/2} \right\|_{S_2}^2 = o(N_T). \quad (170)$$

Proof. We treat the primitive-controlled source classes with Lemma 12.3; the boundary nonstationary and high-sideband classes are estimated directly in relative Hilbert–Schmidt norm below. Throughout, “normalized jump” and “normalized mixed kernel” mean the quantities $\mathcal{J}_{j,T}$ and $\mathcal{K}_{j,T}$ defined there, including the exact shell density (166). The common factor L^{-1} is fixed by (161). By Proposition 6.1 there is no arithmetic whitening in the invariant source. Lemma 12.2 converts the previously established lag-profile derivative bounds into these shell-normalized quantities. All remaining fixed losses are powers of L and are included in the exponents chosen in Section 13.

(i) *Exact/model local discrepancy.* The matrix R_T^{loc} is the low-level exact/model discrepancy defined by (101); it includes the model truncation tail (99). Its shell-local primitive normalization and relative estimate are exactly (102). The exact levels $k > K_0$ are not part of class (i): Lemma 7.5 assigns their low sidebands to E_T^{edge} and their high sidebands to class (vii).

(ii) *Low-level freezing of the slow half-self factor.* This class contains only the locally projected hierarchy $k \leq K_0$; the exact levels $k > K_0$ are never frozen and are owned by Lemma 7.5. By Lemma 7.4, the normalized jump and mixed kernel satisfy

$$\delta_T \ll L^{-D-1+C_{\text{fr}}} = o(1), \quad \|\mathcal{K}_{\text{freeze}}\|_{L^2(S_2)} \ll L^{-D-1+C_{\text{fr}}} = o(1).$$

The bounded overlap of the smooth height partition preserves these estimates. Thus (169) applies to the complete block-freezing error.

(iii) *Hardy-gauge f' source.* This class is exactly $R_{f'}^{\text{src}}$ from (34), including the curvature, logarithmic-derivative, and quadratic f' terms; none remains in the principal source. On the high strip,

$$f'(s) = O(T^{-1}), \quad |f(s) - P(s)| \asymp L$$

on the safe right edge. The remaining factors are the same completed shifted zeta/Cauchy factors already present in the principal master. A fixed number of stationary, packet and curvature derivatives therefore costs only L^C , while every monomial in (34) retains at least one factor T^{-1} . Thus

$$\|\mathcal{J}_{f'}\|_{\text{op}} \ll T^{-1}L^C = o(1), \quad \|\mathcal{K}_{f'}\|_{L^2(S_2)} \ll T^{-1}L^C.$$

(iv) *Horizontal connectors.* By Lemma 3.6, after the endpoint-collar deletion already included in V_{end} we choose its arbitrary decay exponent as $A = M_* + C_{\text{hor}}$, where C_{hor} dominates the fixed polynomial losses from the finite pullbacks and mixed derivatives. Hence

$$\mathcal{J}_{\text{hor}} = 0, \quad \|\mathcal{K}_{\text{hor}}\|_{L^2(S_2)} \ll T^{-M_*}L^C.$$

No further decay order is chosen later.

(v) *First stationary and Stirling corrections to the lag-profile terms.* The finite- T principal deformations of λ and the common endpoint remain inside P_T and are compared with A_T^G only in (120); the exact/model Laurent discrepancy is already class (i). What remains here consists only of the first omitted stationary-phase and Stirling coefficients attached to the same local lag profiles.

By the stationary expansion in Lemma 5.1 and Stirling's formula, each carries an additional factor $T^{-1}L^C$ after the fixed derivatives. Lemma 12.2 therefore gives

$$\|\mathcal{J}_{\text{corr}}\|_{\text{op}} \ll T^{-1}L^C = o(1), \quad \|\mathcal{K}_{\text{corr}}\|_{L^2(S_2)} \ll T^{-1}L^C.$$

The order-one complementary pole-free direct source is not included in this class; it is treated by the sideband decomposition in (vii).

(vi) *Nonstationary and smooth height-partition terms.* Differentiate first. On every retained nonboundary shell $j \in \mathcal{J}_T$, use the integration-by-parts order N and buffer exponent B fixed in Section 13. On a retained boundary shell meeting $\{1, X_T\}$, first remove the finite endpoint sum (148); Proposition 11.2 assigns those terms only to E_T^{edge} . The remaining interior integral obeys the same repeated integration-by-parts estimate. By (147), $R_j \geq L^B$, and the bounded-overlap shell sum, the squared leakage before the fixed relative-normalization loss is

$$\ll N_T L^{2C_N - B(2N-1) + o(1)}.$$

After the primitive normalization, whose squared fixed loss is at most $L^{C_{\text{reg}}}$, the choice of B and M_* in Section 13 gives

$$\boxed{\mathcal{J}_{\text{nonstat}} = 0, \quad \sum_j \iint \|\mathcal{K}_{j,\text{nonstat}}(x, y)\|_{S_2}^2 dx dy = o(N_T).} \quad (171)$$

For the slow height decomposition choose a smooth partition $\sum_b \omega_b(t) = 1$ on scale $H = T/L^D$ and insert it in the linear contour integral before stationary phase. The leading pieces sum exactly to the original amplitude. Every differentiated partition factor satisfies

$$\partial_t^r \omega_b(t) = O_r(H^{-r}) = O_r(T^{-r} L^{Dr}),$$

so the non-leading partition terms have zero principal cusp and satisfy the same regular criterion. No block-boundary packet deletion is introduced.

(vii) *High sidebands of the direct one- χ source.* This class contains the high sidebands of every stable direct one- χ family: exact HLP levels $k > K_0$, the pole-free $\mathcal{D}_k^{\text{dir}}$ and transition factors, and additional translated principal parts. The auxiliary outer-excess family is zero for the invariant source. Lemmas 5.1 and 7.3 cover all these families. On $I_j = [J_j, J_j + 1]$ write

$$a_{j,m}(y; p) = \sum_{r \in \mathbb{Z}} \widehat{a}_{j,m}(r; p) e(ry).$$

Uniform fixed-order stationary phase gives, for every chosen N ,

$$|\partial_p^s \widehat{a}_{j,m}(r; p)| \ll_{N,s} L^{C_{N,s}} (1 + |r|)^{-N}.$$

Put $d_j \asymp T/J_j$ and $R_j^{\text{sb}} := \epsilon_L d_j$ with $\epsilon_L = L^{-\sigma}$. The choice $A_0 > \sigma + 3$ gives $R_j^{\text{sb}} \geq L^3$ on every retained shell. Place $|r| \leq R_j^{\text{sb}}$ in the common low-sideband edge range.

For the high sidebands choose Fourier decay order $M_{\text{sb}} + 4$. Let $c_{j,m}$ be the normalized coefficient on the finite output fibre after the fixed mixed-kernel derivatives. Since the stationary interval has upper endpoint $B_j \asymp d_j$, Lemma 7.3 gives

$$\sum_{m \in \mathcal{I}_j} \|c_{j,m}\|_{S_2}^2 \ll d_j L^{C_{\text{sb}}}. \quad (172)$$

For the effective Fourier shift $q = m - r$ put

$$H_{q,m} := \widehat{a}_{j,m}(m - q; p) \mathbf{1}_{|q-m| > R_j^{\text{sb}}}.$$

The Fourier decay gives both row and column Schur bounds

$$\sup_m \sum_q |H_{q,m}| + \sup_q \sum_m |H_{q,m}| \ll L^{C_{\text{sb}}} (R_j^{\text{sb}})^{-M_{\text{sb}}-3}.$$

Hence the Hilbert-space-valued Schur estimate implies

$$\sum_q \left\| \sum_m H_{q,m} c_{j,m} \right\|_{S_2}^2 \ll L^{C_{\text{sb}}} d_j (R_j^{\text{sb}})^{-2M_{\text{sb}}-6}.$$

Distinct compressed shifts $P_j S_q P_j$ occupy distinct matrix diagonals and are Hilbert–Schmidt orthogonal; moreover $\|P_j S_q P_j\|_{S_2}^2 \ll d_j$. Therefore

$$\|\mathcal{K}_{j,\text{hi}}\|_{S_2}^2 \ll L^{C_{\text{sb}}} d_j^2 (R_j^{\text{sb}})^{-2M_{\text{sb}}-6},$$

which, since $d_j, R_j^{\text{sb}} \geq 1$, implies the deliberately weaker bound

$$\boxed{\|\mathcal{K}_{j,\text{hi}}\|_{S_2}^2 \ll L^{C_{\text{sb}}} d_j^3 (R_j^{\text{sb}})^{-M_{\text{sb}}}}. \quad (173)$$

Thus, since $d_j \asymp T/J_j$,

$$\begin{aligned} \sum_j \|\mathcal{K}_{j,\text{hi}}\|_{S_2}^2 &\ll L^{C_{\text{sb}}+\sigma M_{\text{sb}}} T^{3-M_{\text{sb}}} \sum_{J_j \leq X_T} J_j^{M_{\text{sb}}-3} \\ &\ll T L^{C_{\text{sb}}+\sigma M_{\text{sb}}-A_0(M_{\text{sb}}-2)}. \end{aligned}$$

By the prior choice (178), the logarithmic exponent is less than -3 , so the right side is $o(TL) = o(N_T)$. No derivative order is chosen at this stage. This piece has no diagonal jump because the local symbol is C^∞ .

(viii) *Local holomorphic complement.* After the common and translated principal parts are removed, the local Mellin source is holomorphic inside the closed contour Γ_L . Its local contour contribution therefore vanishes identically by (104). The smooth transition to the complementary pole-free contour belongs to the direct one- χ source whose high sidebands are class (vii), so this step contributes no additional remainder.

(ix) *HLZ diagonalization error.* Choose the arbitrary logarithmic exponent $A = A_{\text{diag}}$ in (41). Under the trivial outer-twist specialization of Proposition 6.1, the superlogarithmic term $L^{O(1)} T^{-2/\log_3 T}$ is smaller than every fixed negative power of L . The error in Lemma 6.3 remains in the common lag-profile channel: after the diagonal Mellin sum has been taken it has no independent stationary carrier index. Hence its fixed (ρ, p) derivatives produce a scalar smooth lag-profile error on each fixed local-source/refinement component. The number of such components is only polylogarithmic, and its Hilbert–Schmidt square-root loss is included in C_{diag} by definition. Consequently the packet-stable lemma gives

$$\varepsilon_{\text{diag}} \ll L^{-A_{\text{diag}}+C_{\text{diag}}+o(1)} = o(1),$$

and Lemma 12.2 yields

$$\delta_T \ll \varepsilon_{\text{diag}}, \quad \|\mathcal{K}_{j,\text{diag}}\|_{L^2(S_2)} \ll \varepsilon_{\text{diag}}$$

on one shell. In particular there is no factor d_j : such a factor belongs to the direct one- χ carrier families of class (vii), not to the completed diagonal error. Consequently

$$\sum_{j \in \mathcal{J}_T} \|\mathcal{K}_{j,\text{diag}}\|_{L^2(S_2)}^2 \ll X_T L^{-2A_{\text{diag}}+2C_{\text{diag}}+o(1)} = o(N_T)$$

by $A_{\text{diag}} > C_{\text{diag}} + 10$. The error has no extracted local pole, so Lemma 12.3 applies with these bounds.

(x) *Terminal logarithmic tail.* The nonstationary part of the complementary band $W_T < u \leq L$ is exactly R_T^{tail} of Proposition 10.3, and already satisfies the required relative Hilbert–Schmidt bound. Its stationary and endpoint parts belong only to E_T^{edge} .

Class (i) has the relative bound (102). For (ii)–(v) and the smooth height-partition corrections in (vi), the squared relative Hilbert–Schmidt contribution is $L^{C_{\text{reg}}+o(1)}$ per shell, so the total is

$$\ll X_T L^{C_{\text{reg}}+o(1)} = T L^{C_{\text{reg}}-A_0+o(1)} = o(N_T)$$

by (178). The nonstationary interior of (vi) is accounted for only by (171); its endpoint part is not a regular term at all and belongs to E_T^{bdry} by Proposition 11.2. Class (vii) has (173) and its global sum, while (viii) vanishes. The primitive-controlled jump bounds are uniformly $o(1)$, so Lemma 12.3 applies to those classes. Adding the separately controlled classes (i), the nonstationary part of (vi), (vii), (ix), and (x) gives squared Hilbert–Schmidt norm $o(N_T)$, proving (170). $\square$

Proposition 12.5 (Finite- T decomposition of the contour form). *For the same normalized contour matrix A_T of (7), the exact source evaluation has the identity*

$$\boxed{A_T = P_T + E_T^{\text{edge}} + R_T^{\text{tri}}}, \quad (174)$$

where, after quadratic-form compression to $\mathcal{P}_T^{\text{ret}}$ of (186),

$$\begin{aligned} P_T|_{\mathcal{P}_T^{\text{ret}}} &\succeq c_0 \mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}}, \\ \text{rank}(E_T^{\text{edge}}|_{\mathcal{P}_T^{\text{ret}}}) &\leq \text{rank } E_T^{\text{edge}} = o(N_T), \\ \left\| (\mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}})^{-1/2} (R_T^{\text{tri}}|_{\mathcal{P}_T^{\text{ret}}}) (\mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}})^{-1/2} \right\|_{S_2}^2 &= o(N_T). \end{aligned} \quad (175)$$

Here P_T is Definition 7.10, E_T^{edge} is the edge matrix of (157), and R_T^{tri} is the regular remainder of Proposition 12.4. Their disjointness is fixed first by the level split at K_0 : for $k \leq K_0$ use (82), while Lemma 7.5 splits $k > K_0$ into its low- and high-sideband parts. The signed passage from the truncated leading family to the full model is (99). Independently, the geometric source is split at the stable logarithmic core and its terminal complement. By Proposition 6.1, there is no additional outer arithmetic annulus in the invariant source.

Proof. Start from the exact same-form identity (6) and the right-edge split (22). Proposition 6.1 first fixes all auxiliary HLZ outer indices at their trivial values, so no outer arithmetic annulus or outer-translation source is present. On the stable logarithmic band, Proposition 7.11 gives the principal term, the low-sideband edge pieces of every direct one- χ family, and the complementary high-sideband/regular stationary pieces. The separated f' , horizontal, and nonstationary pieces are classes (iii), (iv), and (vi) of Proposition 12.4. Finally split $0 \leq u \leq L$ at W_T ; the complementary terminal band is (149). These source classes are disjoint and exhaust the contour source, proving (174).

The three estimates are respectively (121), (157), and (170). $\square$

13 Reduced spectral space and parameter hierarchy

Choose the parameters in the following order. First fix any constant $C_{\text{pkt}} > 0$, which determines the logarithmic packet reserve (10) before any stationary-buffer or symbol order is chosen. By Proposition 6.1, no arithmetic range parameter is required.

Fix once and for all the finite derivative orders required for two curvature derivatives and two primitive mixed derivatives. Let C_{fr} be the freezing exponent in Lemma 7.4, and choose a fixed height partition exponent

$$D > C_{\text{fr}} + 5, \quad H := T/L^D. \quad (176)$$

Let C_{mean} dominate the fixed local-source/refinement multiplicity. Let C_{reg} dominate the logarithmic exponents of the *squared* per-shell relative Hilbert–Schmidt contributions of the fixed lag-profile, freezing, f' , and stationary/Stirling correction classes, as well as the squared fixed loss in the shell primitive normalization. It does not include the nonstationary integration order or the high-sideband Fourier order, whose exponents C_N and C_{sb} are introduced only after those orders are chosen below. Let C_{diag} dominate all fixed packet, curvature, shell, and fixed local-source/refinement-fibre losses (including the square root of that polylogarithmic fibre dimension in Hilbert–Schmidt norm) multiplying the arbitrary-log term in (41), and fix

$$A_{\text{diag}} > C_{\text{diag}} + 10. \quad (177)$$

Choose

$$\sigma > 2, \quad \epsilon_L = L^{-\sigma}.$$

The direct high-sideband estimate below is obtained with the explicit shell power d_j^3 ; no additional shell-dimension exponent is assumed. Next fix $M_* > 2C_{\text{reg}} + 5$. We use the same numerical target as a logarithmic saving exponent for the stable-core nonstationary leakage and, after increasing it if necessary, as a power-saving target for the horizontal connectors. Choose the integration-by-parts order N so that

$$2N - 1 > M_* + 5,$$

and then choose the stationary buffer exponent B so that

$$B(2N - 1) > 2C_N + M_*.$$

Independently choose one finite high-sideband order

$$M_{\text{sb}} > 5,$$

and let C_{sb} be the corresponding fixed logarithmic exponent in (173). Finally choose A_0 so large that

$$A_0 > \max \left\{ B + 3, \sigma + 3, C_{\text{mean}} + 3, C_{\text{reg}} + 3, \frac{C_{\text{sb}} + \sigma M_{\text{sb}} + 3}{M_{\text{sb}} - 2} \right\}. \quad (178)$$

With W_T, X_T defined by (47), these derivative orders and exponents are fixed once and for all before taking the limit $T \rightarrow \infty$.

Lemma 13.1 (Global and local mean constraints). *Let $g(u; v)$, $0 \leq u \leq W_T$, denote the single packet-fibre output of (106) immediately before the global primitive. Define the global mean map*

$$\mathcal{M}_T^{\text{glob}} v := \int_0^{W_T} g(u; v) du. \quad (179)$$

For the shell-local remainder analysis, let $f_\alpha(x;v)$, $1 \leq x \leq X_T$, denote the finite-dimensional output components immediately before the local primitives, with α collecting the fixed local-source and bounded-overlap refinement labels. On $I_j = [J_j, J_j + 1]$ set

$$q_{j,\alpha}(t;v) := \psi_j(J_j + t)(J_j + t)^{-1/2} f_\alpha(J_j + t;v), \quad 0 \leq t \leq 1, \quad (180)$$

and define the local mean map

$$\mathcal{M}_T^{\text{loc}} v := \left(\int_0^1 q_{j,\alpha}(t;v) dt \right)_{j,\alpha}. \quad (181)$$

Finally put

$$\boxed{V_{\text{mean}} := \ker \mathcal{M}_T^{\text{glob}} \cap \ker \mathcal{M}_T^{\text{loc}}.} \quad (182)$$

Then

$$\boxed{\text{codim } V_{\text{mean}} \ll (1 + X_T) L^{C_{\text{mean}}} = o(N_T).} \quad (183)$$

For every $v \in V_{\text{mean}}$, the global primitive

$$R(u;v) := \int_0^u g(s;v) ds$$

satisfies $R(0;v) = R(W_T;v) = 0$, while every local primitive

$$Q_{j,\alpha}(t;v) := \int_0^t q_{j,\alpha}(s;v) ds$$

satisfies $Q_{j,\alpha}(0;v) = Q_{j,\alpha}(1;v) = 0$. Thus the global model primitive identity and the shell-local remainder identity hold on the same retained subspace.

Proof. Both mean maps are linear. The global mean map has one output component. The fixed local-source/refinement labels contribute at most $L^{C_{\text{mean}}}$ components per retained shell, and there are $O(X_T)$ retained unit shells. Thus the target of $\mathcal{M}_T^{\text{loc}}$ has $O(X_T L^{C_{\text{mean}}})$ coordinates, and the codimension of the intersection of the two kernels is bounded by $1 + O(X_T L^{C_{\text{mean}}})$. The choice $A_0 > C_{\text{mean}} + 3$ in (178) makes this $o(TL) = o(N_T)$. The endpoint identities for R and $Q_{j,\alpha}$ are exactly the two kernel conditions in the definition of V_{mean} . $\square$

All edge-rank, shell-row, and regular-remainder estimates above were proved on the full admissible near-lattice packet family. Passing to the stationary-node-sized subfamily selected in Lemma 15.1 is a coordinate compression. Rank and absolute Hilbert–Schmidt norm cannot increase, and the uniform output-fibre estimates used for the relative bounds may be rerun on the compressed packet family with the same asymptotics.

It remains only to impose the horizontal endpoint restriction. Let P_{end} be the coordinate projection which removes the packet centres in the two collars of Lemma 3.6, and set

$$V_{\text{end}} := \text{Ran } P_{\text{end}}. \quad (184)$$

Then

$$\boxed{\text{codim } V_{\text{end}} = O(T) = o(N_T).} \quad (185)$$

This is a genuine packet-coordinate restriction, so Lemma 3.5 applies after intersection with V_{end} . The stationary buffer rows are not deleted. Their contribution is accounted for by the R_j enlargement in the edge-rank estimate and by the complementary nonstationary remainder bounds.

Define the retained subspace by

$$\boxed{\mathcal{P}_T^{\text{ret}} = V_{\text{core}} \cap V_{\text{mean}} \cap V_{\text{end}}.} \quad (186)$$

By Lemma 10.1, (183), and (185),

$$\boxed{\text{codim } \mathcal{P}_T^{\text{ret}} = o(N_T).} \quad (187)$$

The frozen model requires no further spectral deletion by Proposition 8.1. Stationary shell rows, the smooth height partition, and the low-sideband ranges are not additional subspace restrictions. The auxiliary outer-excess family is absent for the invariant source by Proposition 6.1.

14 Finite-dimensional inertia reduction

Lemma 14.1 (Rank plus relative Hilbert–Schmidt min–max). *Let V be finite dimensional. Let A, P, E, R, G be Hermitian on V , with $G \succ 0$, and suppose*

$$A = P + E + R, \quad P \succeq c_0 G, \quad \text{rank } E \leq r,$$

with

$$s := \|G^{-1/2} R G^{-1/2}\|_{S_2}^2.$$

Then there is a subspace $V_{\text{good}} \subset V$ with

$$\boxed{\text{codim}_V V_{\text{good}} \leq r + 16c_0^{-2}s, \quad A|_{V_{\text{good}}} \succeq \frac{3c_0}{4} G|_{V_{\text{good}}}.} \quad (188)$$

Consequently

$$\boxed{n_-(A|_V) \leq r + 16c_0^{-2}s.} \quad (189)$$

If V has codimension d in the ambient space, then

$$\boxed{n_-(A) \leq d + r + 16c_0^{-2}s.} \quad (190)$$

Proof. Whiten by G :

$$\tilde{P} = G^{-1/2} P G^{-1/2}, \quad \tilde{E} = G^{-1/2} E G^{-1/2}, \quad \tilde{R} = G^{-1/2} R G^{-1/2}.$$

Let $S_E = \text{Ran } \tilde{E}$. Since $\tilde{E}$ is Hermitian, $S_E^\perp = \ker \tilde{E}$, and $\dim S_E \leq r$. Let S_R be the spectral subspace of $\tilde{R}$ for $|\lambda| \geq c_0/4$. Then

$$\frac{c_0^2}{16} \dim S_R \leq \|\tilde{R}\|_{S_2}^2 = s,$$

so $\dim S_R \leq 16c_0^{-2}s$. Put

$$\tilde{V}_{\text{good}} := (S_E + S_R)^\perp, \quad V_{\text{good}} := G^{-1/2} \tilde{V}_{\text{good}}.$$

On this space,

$$\tilde{P} + \tilde{E} + \tilde{R} \succeq c_0 I - \frac{c_0}{4} I = \frac{3c_0}{4} I.$$

This proves (188). Since A is positive on V_{good} , every negative subspace of $A|_V$ meets V_{good} trivially; hence its dimension is at most $\text{codim}_V V_{\text{good}} \leq r + 16c_0^{-2}s$, proving (189). If V has ambient codimension d , the dimension inequality for the intersection of a maximal negative subspace with V gives (190). $\square$

15 Multiplicity perturbation and proof of the main theorem

Lemma 15.1 (Generic affine perturbation at fixed height). *Fix T , the good horizontal contours, and the admissible T^{-100} intervals for the packet centres. Given any real neighbourhood U_0 of $(0, 0)$, there are parameters $(\eta, \varepsilon) \in U_0$, chosen compatibly with the Hardy conjugation symmetry, such that*

$$H(z) = F(z) + \eta z + \varepsilon$$

has only simple zeros and simple stationary zeros in Ω_T , and H, H' have no common zero. Choose one packet centre per stationary node so that Lemma 3.2 gives an evaluation isomorphism. On this packet space define V_{core} by Lemma 10.1, V_{mean} by (182), V_{end} by (184), and $\mathcal{P}_T^{\text{ret}}$ by (186). Then (187) holds. Write

$$\mathfrak{G}_T^{\text{ret}} := \mathfrak{G}_T|_{\mathcal{P}_T^{\text{ret}}}.$$

Proposition 12.5 is therefore applied to the original function F on exactly the same finite packet space on which the zero-side evaluation of H is performed. Moreover the perturbation can be chosen so small that, on this selected $\mathcal{P}_T^{\text{ret}}$,

$$\left\| (\mathfrak{G}_T^{\text{ret}})^{-1/2} (A_H - A_F) (\mathfrak{G}_T^{\text{ret}})^{-1/2} \right\|_{\text{op}} < \frac{c_0}{4}. \quad (191)$$

Proof. Fix T . The chosen boundary contains no zero of F or F' . Intersecting U_0 with a sufficiently small convex real neighbourhood of the origin gives $U \subset U_0$ on which H and H' remain nonvanishing there for every point of the line segment from $(0, 0)$ to $(\eta, \varepsilon) \in U$. Hence homotopy invariance of the argument principle gives

$$N_{\text{good}}(H') = N_{\text{good}}(F') =: m \quad ((\eta, \varepsilon) \in U).$$

There are only finitely many admissible m -coordinate subsets, and each centre ranges over a compact T^{-100} interval. The corresponding reference Grams are continuous and positive definite by Section 12; hence

$$\gamma_T := \inf_{\text{admissible centres}} \lambda_{\min}(\mathfrak{G}_T) > 0.$$

Restriction to $\mathcal{P}_T^{\text{ret}}$ can only increase the least eigenvalue. Uniform continuity of the contour integrand on the same compact centre set allows U to be shrunk so that

$$\sup_{(\eta, \varepsilon) \in U} \sup_{\text{admissible centres}} \|A_H - A_F\|_{\text{op}} < \frac{c_0 \gamma_T}{4}. \quad (192)$$

No asymptotic lower bound for γ_T is required.

It remains to impose genericity inside this fixed neighbourhood. Since $H' = F' + \eta$ and $H'' = F''$, a multiple stationary zero would satisfy

$$F''(z) = 0, \quad \eta = -F'(z).$$

The zeros of F'' in the fixed rectangle are finite, so only finitely many real η are forbidden. Choose a nonzero real η in U outside this set. The stationary zeros $z_1, \dots, z_m$ are then simple. For this fixed η , a common zero of H, H' at z_j would force

$$\varepsilon = -F(z_j) - \eta z_j.$$

Only finitely many real ε are therefore forbidden. Choose a real ε in U outside them. Then H has only simple zeros, H' has only simple zeros, and H, H' have no common zero. Real parameters preserve $H(\bar{z}) = \overline{H(z)}$.

Lemma 3.4 now supplies at least m admissible packet intervals after the endpoint collars are removed. Choose m distinct intervals and then, by Lemma 3.2, centres for which the evaluation map is an isomorphism. Lemma 3.5 preserves all preceding source, frame, rank, and relative-remainder estimates on this subfamily. Finally (192) and $\lambda_{\min}(\mathfrak{G}_T^{\text{ret}}) \geq \gamma_T$ give (191). The right-edge source evaluation remains that of the original function F ; H is used only for the zero-coordinate representation and this finite-matrix comparison. $\square$

Lemma 15.2 (Off-axis zero count for the generic perturbation). *Let $D(H)$ be the number of nonreal conjugate zero pairs of the straightened generic H in the chosen good rectangle Ω_T . Then*

$$\boxed{D(H) = B(H) + P(H') + o(N_T)}. \quad (193)$$

Proof. Choose the perturbation in Lemma 15.1 within the boundary neighbourhood fixed in its proof. Homotopy invariance of the argument principle gives exact equality of the zero counts of F, H and of F', H' inside the good rectangle:

$$N_{\text{good}}(H) = N_{\text{good}}(F), \quad N_{\text{good}}(H') = N_{\text{good}}(F').$$

By the good-rectangle form (17), equivalently its (F', F) version, it follows that

$$N_{\text{good}}(H') - N_{\text{good}}(H) = o(N_T). \quad (194)$$

The packet endpoint collar defining V_{end} is a coordinate restriction and does not enter this zero count. Because the perturbation preserves the real symmetry,

$$N_{\text{good}}(H) = R_{\text{good}}(H) + 2D(H), \quad N_{\text{good}}(H') = R_{\text{good}}(H') + 2P(H').$$

Applying Lemma 3.7 on the real interval $[T_1, T_2]$ gives

$$R_{\text{good}}(H') = R_{\text{good}}(H) + 2B(H) + O(1).$$

Subtracting the two good-rectangle count identities and using (194) yields

$$2\{B(H) + P(H') - D(H)\} = o(N_T),$$

which proves (193). $\square$

Proof of Theorem 1.1. Fix the good rectangle, and let $D_{\text{good}}(F)$ denote the number, with multiplicity, of reflected nonreal zero pairs of F in its interior. Around each distinct nonreal zero of F choose a closed disk, disjoint from the real axis, the other chosen disks, and the boundary. There are finitely many such disks at fixed T . By compactness of their boundaries there is a real neighbourhood U_{disk} of $(0, 0)$ such that

$$|\eta z + \varepsilon| < |F(z)|$$

on every disk boundary whenever $(\eta, \varepsilon) \in U_{\text{disk}}$.

Choose H and the evaluation-isomorphic packet subfamily by Lemma 15.1, with $U_0 = U_{\text{disk}}$. This fixes the packet space, $\mathfrak{G}_T$, and $\mathcal{P}_T^{\text{ret}}$ before any inertia comparison, and gives (191). Proposition 12.5, applied to the original Hardy-gauge matrix A_F on this packet space, gives on $\mathcal{P}_T^{\text{ret}}$

$$A_F = P_T + E_T^{\text{edge}} + R_T^{\text{tri}}.$$

Put $\Delta_T := A_H - A_F$. By (191),

$$P_T + \Delta_T \succeq \frac{3c_0}{4} \mathfrak{G}_T \quad \text{on } \mathcal{P}_T^{\text{ret}}.$$

Hence, after compression to $\mathcal{P}_T^{\text{ret}}$,

$$A_H|_{\mathcal{P}_T^{\text{ret}}} = (P_T + \Delta_T)|_{\mathcal{P}_T^{\text{ret}}} + E_T^{\text{edge}}|_{\mathcal{P}_T^{\text{ret}}} + R_T^{\text{tri}}|_{\mathcal{P}_T^{\text{ret}}}.$$

Set

$$r_T := \text{rank}(E_T^{\text{edge}}|_{\mathcal{P}_T^{\text{ret}}}), \quad s_T := \left\| (\mathfrak{G}_T^{\text{ret}})^{-1/2} (R_T^{\text{tri}}|_{\mathcal{P}_T^{\text{ret}}}) (\mathfrak{G}_T^{\text{ret}})^{-1/2} \right\|_{S_2}^2.$$

By (157), (170), and (187), one has $r_T = o(N_T)$, $s_T = o(N_T)$, and $\text{codim } \mathcal{P}_T^{\text{ret}} = o(N_T)$. Apply the ambient-codimension conclusion (190) of Lemma 14.1 with the lemma's floor parameter replaced by

$$c_* := \frac{3c_0}{4} > 0.$$

The resulting fixed constant $16c_*^{-2}$ does not affect the $o(N_T)$ bounds, and therefore

$$\boxed{n_-(A_H) = o(N_T)}. \tag{195}$$

The zero-coordinate pullback is exact for H , so (14) and (195) give

$$B(H) + P(H') = o(N_T).$$

Lemma 15.2 then gives $D(H) = o(N_T)$. By the defining choice $(\eta, \varepsilon) \in U_{\text{disk}}$, Rouché's theorem preserves, with multiplicity, the number of zeros in every chosen disk; all these disks are disjoint from the real axis. Consequently

$$D_{\text{good}}(F) \leq D(H).$$

Let $D_T(F)$ denote the number, with multiplicity, of reflected nonreal zero pairs of F with height in $[T, 2T]$. Returning from the good rectangle to this dyadic interval changes the zero count only in two strips of length at most one. By Lemma 3.8,

$$D_T(F) \leq D(H) + O(\log T) = o(N_T).$$

Because $c > 1$, the symmetric rectangle (1) contains the whole critical strip; at these heights ζ has no nontrivial zeros on the complementary half-planes. Thus every noncritical zero in $[T, 2T]$, apart from the endpoint adjustment already counted above, belongs to exactly one reflected pair. Hence

$$N_T = N_{0,T} + 2D_T(F) + o(N_T),$$

so

$$\frac{N_{0,T}}{N_T} = 1 - o(1).$$

Dyadic summation gives $N_0(T)/N(T) \rightarrow 1$. □

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